

DISORDERED SYSTEMS AND (SUBCRITICAL) POLYNOMIAL CHAOS WITH HEAVY-TAIL DISORDER

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ABSTRACT. We study discrete statistical mechanics systems perturbed by a random environment without a finite second moment. Specifically, we consider a random environment whose tail distribution satisfies $\mathbb{P}[\omega > x] \sim x^{-\gamma}$ as $x \rightarrow +\infty$ for some $\gamma \in (1, 2)$. Inspired by the seminal work of Caravenna, Sun and Zygouras [CSZ16a], we adopt a general framework that encompasses as key examples both the disordered pinning model and the long-range directed polymer model. We provide some *subcriticality* condition under which we prove that the discrete disordered system possesses a non-trivial scaling limit. We also interpret the subcriticality condition in terms of a generalized Harris criterion without second moment, which gives a prediction for *disorder relevance* depending on the parameters of the system. Our analysis relies on the study of multilinear polynomials of independent heavy-tailed random variables known as *polynomial chaos* and their continuous analogue, given by multiple integrals with respect to a γ -stable Lévy white noise. We develop precise and flexible moments estimates adapted to the heavy-tailed setting.

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1. INTRODUCTION AND PRESENTATION OF THE RESULTS

In this article we study statistical mechanics models defined on a lattice which are perturbed by an external random field. Our work focuses on the study of the scaling limit of such disordered system, in the so-called intermediate disorder regime. The approach we follow is in great part inspired by the work of Caravenna, Sun and Zygouras in [CSZ16a]. Their work focuses on the case where the perturbation lies in the basin of attraction of a Gaussian white noise and their techniques strongly rely on $\mathbb{L}^2$ computations.

We will instead focus on an environment which lies in the domain of attraction of a (γ -stable) Lévy white noise. In that case, $\mathbb{L}^2$ computations are not available and we need to develop flexible $\mathbb{L}^p$ estimates for $p \in (1, 2)$. We obtain a general criterion for *subcriticality* of disordered systems, and we prove that such systems admit a non-trivial scaling limit. This is a generalization of the results presented in [BL21; BL22; BCL23] which focused on a specific model known as the directed polymer in a random environment. In the spirit of [CSZ16a, Section 1.3], it can also be seen as a criterion for *disorder relevance* in the case of a heavy-tail disorder, a question considered for instance in [LS17] in the

case of the disordered pinning model. We then apply our general results to several statistical mechanics models such as the disordered pinning model or the long-range directed polymer model.

In contrast to [CSZ16a], which proves a general Lindeberg principle for polynomial chaos (Theorem 2.6 therein) as one of its main contributions, our approach does not rely on such a principle. Rather, we exploit the Poisson structure of heavy-tail random variables. This will be developed extensively in Section 2.

1.1. Setting and first notations. Our work deals with disordered systems, which are made of two ingredients: a (homogeneous) *reference model*, perturbed by a random environment, also called *disorder*. We now introduce the general framework we work with; we give a few examples that will serve as a common thread.

1.1.1. About the reference model. Let us first introduce the homogeneous model which will be perturbed later on. One should think about it as the law of a rescaled object that is very well understood.

Let $\Omega \subset \mathbb{R}^D$ be a *bounded* open set, with D the dimension of the ambient space. Consider for every $\delta > 0$ a discretization Ω_δ of Ω . Let d_{eff} be the effective dimension of this discretization, that is the unique real number d_{eff} such that for any test function f on Ω ,

$$(1.1) \quad \delta^{d_{\text{eff}}} \sum_{x \in \Omega_\delta} f(x) \xrightarrow{\delta \downarrow 0} \int_{\Omega} f(x) dx.$$

Note that d_{eff} could be different from the ambient dimension D (see the examples below in Section 3). Suppose that a reference probability measure $\mathbb{P}_\delta^{\text{ref}}$ is given on $\{0, 1\}^{\Omega_\delta}$, which describes a field $(\sigma_x)_{x \in \Omega_\delta}$. The law $\mathbb{P}_\delta^{\text{ref}}$ is referred to as the *homogeneous model*.

Example 1 (Renewal processes and pinning model). Let $\tau = (\tau_k)_{k \geq 0}$ be a renewal process on $\mathbb{N}$ with inter-arrival law $\mathbb{P}(\tau_1 = n) \sim c_0 n^{-(1+\alpha)}$ as $n \rightarrow \infty$, for some $\alpha \in (0, 1)$. In other words $\tau_0 = 0$ and the $(\tau_i - \tau_{i-1})_{i \geq 1}$ are i.i.d. $\mathbb{N}$ -valued random variables; with a small abuse of notation, we also interpret the renewal process τ as a subset of $\mathbb{N}$. Then, the dimension is $D = 1$ and we take $\Omega = (0, 1)$. We let $\Omega_\delta = (\delta\mathbb{Z}) \cap \Omega$ where we have set $\delta = \frac{1}{N}$, with $N \in \mathbb{N}$ the length of the system. Here, the effective dimension of the discretization is $d_{\text{eff}} = 1 = D$.

For $x \in \Omega_\delta$, we can define $\sigma_x = \mathbb{1}_{\{xN \in \tau\}}$ and consider $\mathbb{P}_\delta^{\text{ref}}$ the law of the random field $(\mathbb{1}_{\tau_\delta}(x))_{x \in \Omega_\delta}$ where $\tau_\delta = (\delta\tau) \cap \Omega$. Then, the law of $\mathbb{P}_\delta^{\text{ref}}$ admits a scaling limit as $\delta \downarrow 0$, which is the law of the α -stable regenerative set (see e.g. [Gia07, App. A.5.4] and Section 3.1 below).

Although we assume that the domain Ω is bounded (this assumption plays an important role, see the comments after Theorem 2.3), our results can be extended to several models defined on an unbounded domain such as random walks.

Example 2 (Long-range random walk). Let $(S_n)_{n \geq 1}$ be a random walk on $\mathbb{Z}^d$, $d \geq 1$, with i.i.d. increments in the (strict) domain of attraction of an α -stable law with $\alpha \in (0, 2]$; in other words, assume that $(n^{-1/\alpha} S_n)_{n \geq 0}$ converges in distribution towards some d -dimensional α -stable law. In that case, we take $D = 1 + d$ and $\Omega = (0, 1) \times \mathbb{R}^d$. Then, for $N \in \mathbb{N}$ large, let the mesh size be $\delta = \frac{1}{N}$ and consider the discretization $\Omega_\delta = ((\delta\mathbb{Z}) \times (\delta^{1/\alpha}\mathbb{Z}^d)) \cap \Omega$. Here the effective dimension of the discretization is $d_{\text{eff}} = 1 + \frac{d}{\alpha}$ and differs from $D = 1 + d$.

Then, for $x = (t, u) \in \Omega_\delta$, define $\sigma_x = \mathbb{1}_{\{S_{tN} = uN^{1/\alpha}\}}$ so that the random walk trajectory in time-space up to time N can be interpreted through its occupation field in $\{0, 1\}^{\Omega_\delta}$. Thus, we define $\mathbb{P}_\delta^{\text{ref}}$ as the law of the random field $(\mathbb{1}_{\mathcal{A}_\delta}(x))_{x \in \Omega_\delta}$ where $\mathcal{A}_\delta = \{(\frac{n}{N}, \frac{S_n}{N^{1/\alpha}})\}_{1 \leq n \leq N}$. The law $\mathbb{P}_\delta^{\text{ref}}$ admits a scaling limit as $\delta \downarrow 0$, which is the law of the random field $(\mathbb{1}_{\mathbf{A}}(x))_{x \in \Omega}$ where $\mathbf{A} = \{(t, X_t)\}_{t \geq 0}$ with $(X_t)_{t \geq 0}$ an α -stable Lévy process (a Brownian motion when $\alpha = 2$).

1.1.2. Introducing disorder. Let $\omega = (\omega_x)_{x \in \Omega_\delta}$ be i.i.d. *centered* random variables such that $\omega_x > -1$ almost surely (the reason for this choice will be clear a few lines below). Their law will be denoted $\mathbb{P}$ and expectation with respect to ω will be written $\mathbb{E}$. Depending on the context, ω will either stand for a generic random variable which has the same law as ω_x or the full environment $(\omega_x)_{x \in \Omega_\delta}$, and

will be referred to as *disorder*. We insist on the fact that the randomness of ω is not the same as the randomness of the homogeneous model σ , and in fact we assume that ω and σ are independent.

Given $\beta \in [0, 1]$ (the disorder intensity) and a *quenched* realization of ω , we define the disordered probability measure $P_{\Omega_\delta}^{\omega, \beta}$ for the field $\sigma = (\sigma_x)_{x \in \Omega_\delta}$ as a Gibbs modification of P_δ^{ref} as follows:

$$(1.2) \quad P_{\Omega_\delta}^{\omega, \beta}(d\sigma) := \frac{1}{Z_{\Omega_\delta}^{\omega, \beta}} \prod_{x \in \Omega_\delta} (1 + \beta \omega_x \sigma_x) P_\delta^{\text{ref}}(d\sigma).$$

The normalizing constant $Z_{\Omega_\delta}^{\omega, \beta}$ that makes $P_{\Omega_\delta}^{\omega, \beta}(d\sigma)$ a probability measure is called the partition function, and is defined by

$$(1.3) \quad Z_{\Omega_\delta}^{\omega, \beta} := E_\delta^{\text{ref}} \left[\prod_{x \in \Omega_\delta} (1 + \beta \omega_x \sigma_x) \right].$$

We will refer to $P_{\Omega_\delta}^{\omega, \beta}$ as the *disordered system*. The only reason we require $\beta \leq 1$ is to guarantee that the density of $P_{\Omega_\delta}^{\omega, \beta}$ with respect to P_δ^{ref} is positive (recall we assume that $\omega > -1$ a.s.).

Example 3 (Disordered pinning model). Coming back to Example 1, the disordered version of the model is known as the *disordered pinning model on a defect line*. We come back to this example more in depth in Section 3.1 below; see [Gia07; Gia11] for comprehensive references.

Example 4 (Long-range directed polymer). As far as Example 2 is concerned, the disordered system based on (long-range) random walks is known as the (long-range) directed polymer model. We come back to this example more in depth in Section 3.2 below; we refer to [Com17] for a comprehensive overview of the directed polymer model.

Remark 1.1 (Product vs. exponential weights). *Readers familiar with disordered systems might be slightly surprised by our definition of the disordered probability measure. Usually the Gibbs weight is indeed written in the exponential form $\exp(\beta \sum_{x \in \Omega_\delta} \tilde{\omega}_x \sigma_x)$. First of all, let us stress that, for fixed β , this is the same up to the change of variable $\omega = \frac{1}{\beta}(e^{\beta \tilde{\omega}} - 1)$. There are several reasons for which we make this choice. The most obvious one is certainly that the product form allows for a more direct polynomial chaos expansion¹, as presented in Section 1.2 below: in particular, the relevant variables in the expansion are then the ω 's and not the $\tilde{\omega}$'s. Another reason to consider the product form is that, when considering heavy-tailed random variables, the exponential form gives an overwhelming importance to the high values of $\tilde{\omega}$ and the tail of $e^{\beta \tilde{\omega}}$ changes with β ; in particular, after rescaling, ω possesses a scaling limit whereas $e^{\beta \tilde{\omega}}$ never does. This is why [LS17] and then [BL21] considered a product form random perturbation (see in particular Remark 1.2 in [BL21] for more motivations of this choice), which is in fact most natural when investigating the relation between disordered systems and stochastic PDEs, see Sections 3.1.3-3.2.3 below.*

Remark 1.2 (State space of the field σ). *The specific choice of the state space for the random field σ is not restrictive. Our results remain valid for continuous state spaces, provided the random variables $(\sigma_x)_{x \in \Omega_\delta}$ are almost surely bounded from below. This flexibility is a consequence of working directly with the product form of the Gibbs weight; product and exponential weights are equivalent only for binary random variables σ , as explained in the previous remark. We choose the state space $\{0, 1\}$ here primarily because it is the natural setting for examples 3,4. This stands in contrast to the framework of [CSZ16a] where extending the analysis to non-binary random fields requires significant efforts (see [LMS25]).*

The question we address in this article is that of the scaling limit of the disordered system in the limit $\delta \downarrow 0$, choosing $\beta = \beta_\delta \downarrow 0$ in some appropriate scaling window, known as the intermediate disorder regime.

¹Note that this product form is originally used in [HH85] who introduced the directed polymer model, and it also appears explicitly in the seminal work [AKQ14], see Equation (4), as an intermediary step when considering the intermediate disorder scaling limit of the directed polymer model.

1.2. Intermediate disorder in the $\mathbb{L}^2$ case: a quick overview. For pedagogical purposes, before introducing our main results on heavy-tail random variables and Lévy white noises, let us briefly present and comment the case where the random variables ω have a finite $\mathbb{L}^2$ moment. This section reproduces heuristics from Section 1.2. of [CSZ16a] and allows us to introduce some important notation and ideas, in particular the notion of *subcritical polynomial chaos*. Without loss of generality we assume in this section that $\mathbb{E}[\omega^2] = 1$.

The starting point of the discussion is the polynomial chaos expansion of the partition function. It is helpful for heuristically identifying what we mean by *subcritical regime*, what the assumptions on the homogeneous system are, and what the scaling limit of the disordered system should be. Let us focus for now on the partition function $Z_{\Omega_\delta}^{\omega, \beta}$ in (1.3). Expanding the product $\prod_{x \in \Omega_\delta} (1 + \beta \omega_x \sigma_x)$ and taking the expectation yields

$$(1.4) \quad Z_{\Omega_\delta}^{\omega, \beta} = 1 + \sum_{k \geq 1} \beta^k \sum_{\{x_1, \dots, x_k\} \subset \Omega_\delta} \mathbb{E}_\delta^{\text{ref}} \left[\prod_{j=1}^k \sigma_{x_j} \right] \prod_{j=1}^k \omega_{x_j}.$$

In this expansion, note that the k -point correlation functions of the homogeneous system appear. Since the homogeneous system has a scaling limit, it is natural (and satisfied for the systems presented in the above examples) to assume that the correlation functions themselves have a scaling limit. More precisely, define for $x_1, \dots, x_k \in \Omega_\delta$ and $\lambda \in (0, +\infty)$,

$$(1.5) \quad \psi_{\Omega_\delta}^{(k)}(x_1, \dots, x_k) := (\delta^{-\lambda})^k \mathbb{E}_\delta^{\text{ref}} \left[\prod_{j=1}^k \sigma_{x_j} \right],$$

with by convention $\psi_{\Omega_\delta}^{(k)}(x_1, \dots, x_k) = 0$ on the diagonals, *i.e.* if the x_i are not distinct. Then, we assume that the scaling exponent λ is such that, for every $k \geq 1$,

$$(1.6) \quad \psi_{\Omega_\delta}^{(k)}(x_1, \dots, x_k) \xrightarrow{\delta \downarrow 0} \psi_\Omega^{(k)}(x_1, \dots, x_k),$$

for some non-trivial function $\psi_\Omega^{(k)}$, interpreted as the k -point correlation function of the scaling limit of the reference model; we also set $\psi_\Omega^{(k)} = 0$ on the diagonals. For the time being, we are voluntarily remaining evasive about the precise meaning of the convergence (1.6).

Example 5 (Pinning model). For the pinning model of Example 1, we have that $\psi_{\Omega_\delta}^{(k)}(x_1, \dots, x_k) = \prod_{i=1}^k \mathbb{P}((x_i - x_{i-1})N \in \tau)$ for any $0 < x_1 < \dots < x_k < 1$, with by convention $x_0 = 0$. Then, we have that (1.6) holds with $\lambda = 1 - \alpha$ thanks to [Don97], and $\psi_\Omega^{(k)}(x_1, \dots, x_k) = \prod_{i=1}^k (x_i - x_{i-1})^{\alpha-1}$.

Example 6 (Long-range directed polymers). For the long-range directed polymer of Example 2, we have that $\psi_{\Omega_\delta}^{(k)}((t_1, u_1), \dots, (t_k, u_k)) = \prod_{i=1}^k \mathbb{P}(S_{(t_i - t_{i-1})N} = (u_i - u_{i-1})N^{1/\gamma})$ for any $0 < t_1 < \dots < t_k < 1$, with by convention $t_0 = 0, x_0 = 0$. Then, by the local limit theorem, we have that (1.6) holds with $\lambda = \frac{d}{\alpha}$, and $\psi_\Omega^{(k)}(x_1, \dots, x_k) = \prod_{i=1}^k \rho_{t_i - t_{i-1}}(u_i - u_{i-1})$ where $\rho_t(u)$ is the transition density of the limiting α -stable Lévy process $(X_t)_{t \geq 0}$.

With the definition (1.5) and using the symmetry of the k -point correlation functions, this allows us to rewrite the partition function as

$$(1.7) \quad Z_{\Omega_\delta}^{\omega, \beta} = 1 + \sum_{k \geq 1} \frac{1}{k!} (\beta \delta^\lambda)^k \sum_{x_1, \dots, x_k \in \Omega_\delta} \psi_{\Omega_\delta}^{(k)}(x_1, \dots, x_k) \prod_{j=1}^k \omega_{x_j}.$$

We can now compute the $\mathbb{L}^2$ moment of $Z_{\Omega_\delta}^{\omega, \beta}$. Observe that in 1.7, the terms $\prod_{i=1}^k \omega_{x_i}$ and $\prod_{j=1}^l \omega_{y_j}$ are orthogonal in $\mathbb{L}^2$ except if the sets $\{x_1, \dots, x_k\}$ and $\{y_1, \dots, y_l\}$ coincide. Therefore,

$$\mathbb{E}[(Z_{\Omega_\delta}^{\omega, \beta})^2] = 1 + \sum_{k \geq 1} \frac{1}{k!} (\beta \delta^\lambda)^{2k} \sum_{x_1, \dots, x_k \in \Omega_\delta} \psi_{\Omega_\delta}^{(k)}(x_1, \dots, x_k)^2.$$

Then, at least heuristically, for every $k \geq 0$,

$$\sum_{x_1, \dots, x_k \in \Omega_\delta} \psi_{\Omega_\delta}^{(k)}(x_1, \dots, x_k)^2 \approx \delta^{-k d_{\text{eff}}} \int_{\Omega^k} \psi_{\Omega}^{(k)}(x_1, \dots, x_k)^2 dx_1 \dots dx_k,$$

recalling that d_{eff} is the effective dimension of the discretization Ω_δ , see (1.1). This yields

$$(1.8) \quad \mathbb{E}[(Z_{\Omega_\delta}^{\omega, \beta})^2] \approx 1 + \sum_{k \geq 1} \frac{1}{k!} \left(\beta \delta^{\lambda - d_{\text{eff}}/2} \right)^{2k} \int_{\Omega^k} \psi_{\Omega}^{(k)}(x_1, \dots, x_k)^2 dx_1 \dots dx_k.$$

Now, there are two cases to distinguish:

- (i) If $d_{\text{eff}}/2 - \lambda < 0$. For β small enough, one may show that $Z_{\Omega_\delta}^{\omega, \beta}$ has a uniformly bounded $\mathbb{L}^2$ moment. In that case, any weak disorder limit ($\beta = \beta_\delta \downarrow 0$) is trivial and *disorder is irrelevant* (in the sense of [CSZ16a, §1.3]). We also dub this case *supercritical*, since the terms in the chaos expansion (1.8) grow with k ; it also corresponds to a case where the integrals $\int_{\Omega^k} \psi_{\Omega}^{(k)}(x_1, \dots, x_k)^2 dx_1 \dots dx_k$ are divergent.
- (ii) If $d_{\text{eff}}/2 - \lambda > 0$. In particular, for a fixed $\beta > 0$, the $\mathbb{L}^2$ moment of $Z_{\Omega_\delta}^{\omega, \beta}$ diverges. One may hope to scale $\beta = \beta_\delta \downarrow 0$ appropriately to obtain a non-trivial scaling limit and *disorder is relevant* (in the sense of [CSZ16a, §1.3]). We also dub this case *subcritical*, since the terms in the chaos expansion (1.8) vanish with k ; it also corresponds to a case where the integrals $\int_{\Omega^k} \psi_{\Omega}^{(k)}(x_1, \dots, x_k)^2 dx_1 \dots dx_k$ are convergent.

This disjunction of cases is linked to the *Harris criterion* for disorder relevance; we will come back to this in Section 1.5 below. We do not mention the marginal case where $\lambda = \frac{d_{\text{eff}}}{2}$ because this is not the focus of the present article. Instead, we refer to [CSZ17] for an extensive discussion, or [Zyg24; CSZ25] for an overview of the recent progress in the marginally relevant case.

Assume that we are in the subcritical (disorder relevant) case, *i.e.* $\lambda < d_{\text{eff}}/2$. From (1.8), we may identify the correct *intermediate disorder regime*. Indeed, if one scales $\beta = \beta_\delta \downarrow 0$ as $\delta \downarrow 0$, in such a way that

$$(1.9) \quad \lim_{\delta \downarrow 0} \beta_\delta \delta^{d_{\text{eff}}/2 - \lambda} = \hat{\beta} \in [0, \infty),$$

then we obtain that $\mathbb{E}[(Z_{\Omega_\delta}^{\omega, \beta})^2] \approx 1 + \sum_{k \geq 1} \frac{\hat{\beta}^{2k}}{k!} \int_{\Omega^k} \psi_{\Omega}^{(k)}(x_1, \dots, x_k)^2 dx_1 \dots dx_k$. Hence, a very natural assumption on the k -point correlation functions is that

$$(1.10) \quad \sum_{k \geq 1} \frac{\hat{\beta}^{2k}}{k!} \int_{\Omega^k} \psi_{\Omega}^{(k)}(x_1, \dots, x_k)^2 dx_1 \dots dx_k < +\infty.$$

This is the main assumption (written in a slightly less general set up) of [CSZ16a, Theorem 2.3], whose conclusion is that $Z_{\Omega_\delta}^{\omega, \beta}$ converges in distribution as $\delta \rightarrow 0$. We postpone the precise description of the limit to Section 2, where we will compare this with the results we obtain.

1.3. The case of heavy-tail disorder: subcriticality and disorder relevance. From now on, we focus on the case of an environment ω whose tail distribution has a power-law decay. More precisely, assume that $\mathbb{E}[\omega] = 0$ and that there exist $\gamma \in (1, 2)$ and a constant C_0 such that

$$(1.11) \quad \mathbb{P}[\omega > t] \sim C_0 t^{-\gamma}, \quad \text{as } t \rightarrow +\infty.$$

By the so-called layer-cake representation, this implies that ω has a finite expectation and an infinite variance. We could consider a slightly more general class of environments, allowing for slowly-varying functions in the tail of ω . The general setting will be presented in Section 2.2 (see (2.11)). In this introduction, for pedagogical purposes, we restrict ourselves to the pure-power tail (1.11).

Let us now identify the subcritical regime and the intermediate disorder window as we did in the $\mathbb{L}^2$ case. We keep the notation of the rescaled correlation function introduced in (1.5). We can no longer

calculate the $\mathbb{L}^2$ moment of the partition, instead we will use the scaling limit of the environment. Let us define the discrete noise

$$(1.12) \quad \zeta_\delta := \delta^{d_{\text{eff}}/\gamma} \sum_{x \in \Omega_\delta} \omega_x \delta_x.$$

Under assumption (1.11), we have that, as $\delta \downarrow 0$, ζ_δ converges (in some specific Sobolev space) to a γ -stable Lévy white noise ζ , which is a random distribution which is the equivalent of Gaussian white noise in a non- $\mathbb{L}^2$ setting. We will give a more precise description of it below, see Section 2.1; we also refer to Appendix A.2 for details.

With the above notation, we can rewrite our partition function as follows (recall (1.7)):

$$(1.13) \quad \begin{aligned} Z_{\Omega_\delta}^{\omega, \beta} &= 1 + \sum_{k \geq 1} \frac{1}{k!} \left(\beta \delta^{\lambda - d_{\text{eff}}/\gamma} \right)^k \sum_{x_1, \dots, x_k \in \Omega_\delta} \psi_{\Omega_\delta}^{(k)}(x_1, \dots, x_k) \prod_{j=1}^k (\delta^{d_{\text{eff}}/\gamma} \omega_{x_j}) \\ &= 1 + \sum_{k \geq 1} \frac{1}{k!} \left(\beta \delta^{\lambda - d_{\text{eff}}/\gamma} \right)^k \int_{\Omega^k} \bar{\psi}_{\Omega_\delta}^{(k)}(x_1, \dots, x_k) \prod_{j=1}^k \zeta_\delta(dx_j), \end{aligned}$$

where $\bar{\psi}_{\Omega_\delta}^{(k)}$ denotes the piecewise constant extension of $\psi_{\Omega_\delta}^{(k)}$ to Ω^k . Under this form, it is now clear that *subcriticality* or *supercriticality* (that can also be interpreted as disorder *relevance* or *irrelevance*) depends on whether $\lambda > d_{\text{eff}}/\gamma$ or $\lambda < d_{\text{eff}}/\gamma$. Furthermore, in analogy with (1.9) in the $\mathbb{L}^2$ case, the intermediate disorder regime now corresponds to

$$(1.14) \quad \lim_{\delta \downarrow 0} \beta_\delta \delta^{d_{\text{eff}}/\gamma - \lambda} = \hat{\beta} \in [0, \infty).$$

Overall, the limit of the rescaled partition function clearly should be the expression (1.13) with $\psi_{\Omega_\delta}^{(k)}$ and ζ_δ replaced by their respective limits $\psi_\Omega^{(k)}$ and ζ . However, without even mentioning summability issues, the meaning of the iterated integrals appearing in the sum is not clear: indeed ζ lives in some negative Sobolev space and the correlation functions could be too singular.

1.4. Main results in a nutshell. Let us now briefly present our main results to give an overview of the scope of the paper. We purposely remain evasive on some definitions since they would require the introduction of further notation: for the full statements of our results, we refer to Section 2 below.

First of all, we show that, under appropriate (subcritical) integrability condition on the continuum k -point correlation functions, the candidate scaling limit is well defined.

Theorem 1.3 (Well-posedness of the continuous partition function). *Let ζ be a γ -stable Lévy white noise with $\gamma \in (1, 2)$. Assume that there is some $q \in (\gamma, 2]$ such that for every $C \geq 0$,*

$$(1.15) \quad \sum_{k \geq 1} C^k \|\psi^{(k)}\|_q < +\infty,$$

where $\|\cdot\|_q$ denotes the symmetric L^q -norm (see its definition (2.4) below). Then, the continuum partition function

$$(1.16) \quad \mathcal{Z}_\Omega^{\zeta, \hat{\beta}} = 1 + \sum_{k \geq 1} \frac{\hat{\beta}^k}{k!} \int_{\Omega^k} \psi^{(k)}(x_1, \dots, x_k) \prod_{j=1}^k \zeta(dx_j)$$

is a well-defined (and non-degenerate) ζ -measurable random variable in $\mathbb{L}^p$ for every $p \in [0, \gamma)$.

Our second result is that, under an appropriate *subcritical* condition, the limit of the partition function in the intermediate disorder regime (1.14) is the random variable constructed in Theorem 1.3.

Theorem 1.4 (Intermediate disorder limit). *Assume that the noise ω is centered and has a tail distribution given by (1.11) with $\gamma \in (1, 2)$. Assume also that there exists some $q \in (\gamma, 2]$ such that the correlation functions $(\psi_\delta^{(k)})_{k \geq 1}$ satisfy the following:*

(i) For every $k \in \mathbb{N}$ there exists some continuous k -point correlation function $\psi^{(k)}$ such that

$$\lim_{\delta \downarrow 0} \|\bar{\psi}_\delta^{(k)} - \psi^{(k)}\|_q = 0;$$

(ii) The following uniform integrability condition is verified: for every $C > 0$,

$$\limsup_{\delta \downarrow 0} \sum_{k \geq 0} C^k \|\bar{\psi}_\delta^{(k)}\|_q < +\infty \quad \text{and} \quad \lim_{M \rightarrow +\infty} \limsup_{\delta \downarrow 0} \sum_{k > M} C^k \|\bar{\psi}_\delta^{(k)}\|_q = 0.$$

Then, under the subcritical condition $\lambda < \frac{d_{\text{eff}}}{\gamma}$, in the intermediate disorder regime of (1.14), i.e. if $\lim_{\delta \downarrow 0} \beta_\delta \delta^{d_{\text{eff}}/\gamma - \lambda} = \hat{\beta} \in [0, \infty)$, we have the following convergence in distribution:

$$Z_{\Omega_\delta}^{\omega, \beta_\delta} \xrightarrow{(d)} \mathcal{Z}_{\Omega}^{\zeta, \hat{\beta}},$$

where $\mathcal{Z}_{\Omega}^{\zeta, \hat{\beta}}$ is defined in (1.16), with $\zeta = \lim_{\delta \downarrow 0} \zeta_\delta$ a γ -stable Lévy white noise.

In particular, these results can be applied to the disordered pinning model and the directed polymer model. We develop in detail these two examples in Section 3, but let us quickly summarize the results we obtain. In particular, in the subcritical regime $\lambda < \frac{d_{\text{eff}}}{\gamma}$, additionally to the convergence of the disordered partition function, we establish the convergence of the disordered Gibbs measures towards their continuum counterparts. They are the analogue of the results of [BL21; BL22] for the directed polymer model.

1.4.1. For the pinning model. For the disordered pinning model of Example 1 and 3, we prove the following. We refer to Theorem 3.1-3.3 for precise (and slightly more general) statements.

Theorem 1.5. Assume that $\alpha > 1 - \frac{1}{\gamma}$ in Example 1 and that $\beta_\delta \downarrow 0$ with $\lim_{\delta \downarrow 0} \beta_\delta \delta^{\alpha-1+\frac{1}{\gamma}} = \hat{\beta} \in (0, \infty)$. Then, we have the following convergence in distribution for the disordered partition function

$$Z_{\Omega_\delta}^{\omega, \beta_\delta} \xrightarrow[\delta \downarrow 0]{(d)} \mathcal{Z}_{\Omega}^{\zeta, \hat{\beta}},$$

where $\mathcal{Z}_{\Omega}^{\zeta, \hat{\beta}}$ a ζ -measurable non-degenerate random variable expressed as a polynomial chaos (1.16), with $\zeta = \lim_{\delta \downarrow 0} \zeta_\delta$ a γ -stable Lévy white noise on $\Omega = (0, 1)$.

Additionally, there exists a (disordered) continuum measure $\mathbf{Q}_0^{\zeta, \hat{\beta}}$ on the space $\mathcal{C}_\infty$ of closed subsets of $(0, 1)$ (with the Matheron topology) such that we have the following convergence in distribution²:

$$\mathbf{P}_{\Omega_\delta}^{\omega, \beta_\delta} \xrightarrow[\delta \downarrow 0]{(d)} \mathbf{Q}_0^{\zeta, \hat{\beta}}.$$

The disordered measure $\mathbf{Q}_0^{\zeta, \hat{\beta}}$ is referred to as the *continuum directed pinning model with γ -stable Lévy noise*. In particular, Theorem 1.5 establishes the conjecture of [BL22, §2.5.3-(B)] also mentioned in [BL21, §2.2] (note that the roles of α and γ are reversed in [BL21; BL22]). Our proofs build on recent results from [FL], where the random measure $\mathbf{Q}_0^{\zeta, \hat{\beta}}$ is also studied.

1.4.2. For the (long-range) directed polymer model. For the long-range directed polymer model of Example 2 and 4, we prove the following. We refer to Theorems 3.5-3.6 for precise statements.

Theorem 1.6. Assume that $\frac{\alpha}{d} > \gamma - 1$ in Example 2 and that $\beta_\delta \downarrow 0$ with $\lim_{\delta \downarrow 0} \beta_\delta \delta^{\frac{1}{\alpha\gamma}(\alpha-d(\gamma-1))} = \hat{\beta} \in (0, \infty)$. Then, we have the following convergence in distribution for the disordered partition function

$$Z_{\Omega_\delta}^{\omega, \beta_\delta} \xrightarrow[\delta \downarrow 0]{(d)} \mathcal{Z}_{\Omega}^{\zeta, \hat{\beta}},$$

where $\mathcal{Z}_{\Omega}^{\zeta, \hat{\beta}}$ a ζ -measurable non-degenerate random variable expressed as a polynomial chaos (1.16), with $\zeta = \lim_{\delta \downarrow 0} \zeta_\delta$ a γ -stable Lévy white noise on $\Omega = (0, 1) \times \mathbb{R}^d$.

²on the set $\mathcal{M}_1(\mathcal{C}_\infty)$ of probability distributions on $\mathcal{C}_\infty$ endowed with the weak topology.

Additionally, there exists a (disordered) continuum measure $\mathbf{Q}^{\zeta, \hat{\beta}}$ on the space $\mathcal{D}([0, 1], \mathbb{R}^d)$ of càdlàg functions with values in $\mathbb{R}^d$ (with the Skorokhod topology) such that we have the following convergence in distribution³:

$$\mathbf{P}_{\Omega_\delta}^{\omega, \beta} \xrightarrow[\delta \downarrow 0]{(d)} \mathbf{Q}^{\zeta, \hat{\beta}}.$$

The disordered measure $\mathbf{Q}^{\zeta, \hat{\beta}}$ is referred to as the *continuum long-range polymer model with γ -stable Lévy noise*. In particular, Theorem 1.6 establishes the conjecture of [BL22, §2.5.3-(A)] also mentioned in [BL21, §2.2] (again, the roles of α and γ are reversed in [BL21; BL22]).

1.5. Disorder relevance and Harris criterion without second moment. As mentioned above, the disjunctions of cases we have obtained are linked to the Harris criterion introduced in [Har74]. Disorder is said to be *irrelevant* if for small enough disorder intensity β the disordered and homogeneous system share the same features; disorder is said to be *relevant* if the disordered system drastically differs from the homogeneous one, for any disorder intensity $\beta > 0$. Harris criterion [Har74] states that, for random environments with a finite second moment, disorder relevance/irrelevance can be predicted through the *correlation length exponent* ν of the homogeneous system; let us stress that the k -point correlation exponent λ should be related to ν by the relation $\nu = \frac{1}{d_{\text{eff}} - \lambda}$ (see [CSZ16a, Section 1.3] for a discussion).

More precisely, Harris' predictions state that, in the $\mathbb{L}^2$ case, disorder should be irrelevant if $d_{\text{eff}}\nu > 2$ (i.e. if $\lambda < \frac{d_{\text{eff}}}{2}$) and relevant if $d_{\text{eff}}\nu < 2$ (i.e. if $\lambda < \frac{d_{\text{eff}}}{2}$); the remaining case $\nu d_{\text{eff}} = 2$ is dubbed marginal and disorder relevance should depend on finer properties of the model. This corresponds exactly to the dichotomy of supercriticality/subcriticality described above in the $\mathbb{L}^2$ case: in particular, disorder is relevant if the disordered system admits a non-trivial, i.e. disordered, scaling limit.

Harris criterion has been devised for i.i.d. disorder with a finite variance and it does not hold anymore in the heavy-tailed case (1.11). In that case, we recall that subcriticality corresponds to $\lambda < \frac{d_{\text{eff}}}{\gamma}$ and supercriticality to $\lambda > \frac{d_{\text{eff}}}{\gamma}$ (see (1.13) and the discussion that follows). We therefore arrive at the following criterion:

$$(1.17) \quad \text{disorder is } \begin{cases} \text{relevant} & \text{if } d_{\text{eff}}\nu < \frac{\gamma}{\gamma - 1} & (\lambda < \frac{d_{\text{eff}}}{\gamma}), \\ \text{irrelevant} & \text{if } d_{\text{eff}}\nu > \frac{\gamma}{\gamma - 1} & (\lambda > \frac{d_{\text{eff}}}{\gamma}). \end{cases}$$

This criterion has been established in the case of the disordered pinning model of Example 3 in [LS17] (see also [Lac17] for a marginal case) or for the directed polymer model of Example 4 (with the simple random walk, so in particular $\alpha = 2$) in [Viv21]. As far as the construction of a disordered scaling limit is concerned, only the case of the directed polymer has been treated (again with the simple random walk) in [BL21], under the subcritical condition $\lambda < 1 + \frac{d}{2}$.

Our main result put the criterion (1.17) to rigorous ground in a very broad context. We refer to Section 3 below for a more thorough discussion on the different examples.

2. MAIN RESULTS: POLYNOMIAL CHAOS WITH HEAVY-TAIL DISORDER

In this section, we properly state our main results, introducing all the notations we need. Notice that once stated as in (1.13)-(1.16), the problem we deal with is more general and addresses the question of the scaling limit of polynomial chaos with asymptotically Lévy noise under some *subcriticality* condition. We therefore start by defining in Section 2.1 continuum polynomial chaos with γ -stable Lévy white noise under a subcriticality condition. We then state in Section 2.2 the convergence of discrete polynomial chaos with heavy-tailed random variables to their continuous counterpart. Then, in Section 2.3, we come back to our original motivation and state our results in the context of scaling limit of disordered statistical mechanics systems: we give conditions for the convergence of the partition function and of the disordered Gibbs measure, and we also discuss further properties.

2.1. Continuum polynomial chaos with Lévy white noise.

³on the set $\mathcal{M}_1(\mathcal{D}([0, 1], \mathbb{R}^d))$ of probability distributions on $\mathcal{D}([0, 1], \mathbb{R}^d)$ endowed with the weak topology.

2.1.1. Lévy noise and truncated approximation of continuum polynomial chaos. First of all, let us give a precise definition of a Lévy white noise ζ on Ω . Let Λ be a Poisson point process on $\Omega \times \mathbb{R}$, of intensity $\mu(dx dz) = dx \lambda(dz)$ where $\lambda(dz)$ is the so-called Lévy measure. Even if the construction is more general, we focus on the γ -stable case, which corresponds to taking $\lambda(dz) = (c_+ \mathbb{1}_{z>0} + c_- \mathbb{1}_{z<0}) \gamma|z|^{-1-\gamma} dz$ where the constants c_+, c_- tune the asymmetry of the noise and verify $c_+, c_- \geq 0$ and $c_+ + c_- = 1$.

For $a > 0$, define the truncated noise

$$(2.1) \quad \zeta^{(a)} := \sum_{(x,z) \in \Lambda} z \mathbb{1}_{|z|>a} \delta_x - \kappa(a) dx,$$

with $\kappa(a) = \int_{\mathbb{R}} z \mathbb{1}_{|z|>a} \lambda(dz)$. When $a \downarrow 0$, $\zeta^{(a)}$ converges almost surely in any negative local Sobolev space $H_{\text{loc}}^{-s}(\Omega)$ with $s > \frac{D}{2}$ to a random distribution denoted ζ called γ -stable Lévy white noise. We refer to Theorem A.1 in appendix for a statement.

Consider a general family of functions $(\psi^{(k)})_{k \geq 0}$ such that $\psi^{(k)}$ is defined on Ω^k , is symmetric and vanishes on the diagonals. By convention, $\psi^{(0)}$ is a constant. The goal is to properly define the following *Lévy chaos expansion* (in analogy with *Wiener chaos expansion* in the Gaussian case), i.e. the continuous chaos expansion with respect to a γ -stable Lévy white noise ζ :

$$(2.2) \quad \Phi = \sum_{k \geq 0} \frac{1}{k!} \int_{\Omega^k} \psi^{(k)}(x_1, \dots, x_k) \prod_{j=1}^k \zeta(dx_j),$$

Remark 2.1. Note that we do not require any positivity or monotonicity (on the noise or the functions ψ) which were key features of the previous work of Berger and Lacoïn [BL21; BL22].

The way to define (2.2) is via truncation (analogously to what is done in [BL22]). We introduce, for $a > 0$, the truncated chaos

$$(2.3) \quad \Phi^{(a)} := \sum_{k \geq 0} \frac{1}{k!} \int_{x_1, \dots, x_k \in \Omega} \psi^{(k)}(x_1, \dots, x_k) \prod_{j=1}^k \zeta^{(a)}(dx_j).$$

Let $\mathcal{F} = (\mathcal{F}_a)_{a>0}$ be the filtration, where $\mathcal{F}_a = \sigma\{\Lambda \cap (\Omega \times \mathbb{R} \setminus (-a, a))\}$. We will see below that $(\Phi^{(a)})_{a>0}$ is a time-reversed martingale with respect to the filtration $\mathcal{F}$. Therefore, if the martingale is uniformly integrable, it converges almost surely and in L^1 . Our first main result is to derive conditions for this to hold.

2.1.2. Well-posedness of Lévy chaos expansion. Before we state our result, let us introduce symmetric norms. Define for $q > 1$, $k \in \mathbb{N}$ and $\mathbf{f} : \Omega^k \rightarrow \mathbb{R}$ a symmetric function, the *symmetric L^q norm*:

$$(2.4) \quad \|\mathbf{f}\|_q = \left(\frac{1}{k!} \int_{\Omega^k} |\mathbf{f}(x_1, \dots, x_k)|^q \prod_{j=1}^k dx_j \right)^{1/q}.$$

The space of symmetric functions on Ω^k with finite symmetric L^q norm is denoted $L_s^q(\Omega^k)$.

Remark 2.2. It may seem surprising to include a factorial $k!$ in the definition of the L^q norm but this is a fairly common issue encountered in Malliavin calculus. Indeed, consider the case where the white noise ζ is Gaussian and $\Omega = (0, 1)$. Then if $\mathbf{f}$ is an L^2 symmetric function of $(0, 1)^k$, the random variable

$$(2.5) \quad \mathbf{X}(\mathbf{f}) := \frac{1}{k!} \int_{(0,1)^k} \mathbf{f}(x_1, \dots, x_k) \prod_{j=1}^k \zeta(dx_j) = \int_{0 < t_1 < \dots < t_k < 1} \mathbf{f}(t_1, \dots, t_k) \prod_{j=1}^k \zeta(dt_j)$$

is well-defined and

$$\mathbb{E}[\mathbf{X}(\mathbf{f})^2] = \int_{0 < t_1 < \dots < t_k < 1} \mathbf{f}(t_1, \dots, t_k)^2 \prod_{j=1}^k dx_j = \frac{1}{k!} \int_{(0,1)^k} \mathbf{f}(x_1, \dots, x_k)^2 \prod_{j=1}^k dx_j.$$

Therefore $\|\mathbf{X}(\mathbf{f})\|_2 = \|\mathbf{f}\|_2$ in the sense of symmetric norms, and this shows that the condition $\mathbf{f} \in L_s^2$ is actually necessary for $\mathbf{X}(\mathbf{f})$ to be well-defined.

The question of finding a necessary and sufficient condition for the multiple integral $\mathbf{X}(\mathbf{f})$ in (2.5) to be well-defined when ζ is a γ -stable Lévy noise is known to be more delicate than in the Gaussian setting. For instance, it is known that the single integral $\int_{\Omega} \mathbf{f}(x) \zeta(dx)$ is well-defined if and only if $\int_{\Omega} |\mathbf{f}(x)|^{\gamma} dx < +\infty$ but this does not extend to multiple integrals (see for example [RW86]). We will show that a sufficient condition for the k -multiple integral $\mathbf{X}(\mathbf{f})$ to be well-defined is $\mathbf{f} \in L_s^q(\Omega^k)$ for some $q > \gamma$ but this is known to not be optimal (see for example [Sur85]).

Theorem 2.3 (Well-posedness of γ -stable Lévy chaos). *Let ζ be a γ -stable Lévy white noise with $\gamma \in (1, 2)$. Assume that there exist $q \in (\gamma, 2]$ such that family of functions $(\psi^{(k)})_{k \geq 0}$ verifies the following: for every $C > 0$,*

$$(2.6) \quad \sum_{k \geq 0} C^k \|\psi^{(k)}\|_q < \infty.$$

Then $\Phi^{(a)}$ is well-defined for every $a \in (0, 1]$ and $(\Phi^{(a)})_{a \in (0, 1]}$ is a time-reversed martingale with respect to the filtration $\mathcal{F} = (\mathcal{F}_a)_{a > 0}$.

Additionally, for every $p \in (0, \gamma)$, $(\Phi^{(a)})_{a \in (0, 1]}$ is bounded in $\mathbb{L}^p$ and thus admits a limit $\Phi := \lim_{a \downarrow 0} \Phi^{(a)}$ a.s. and in $\mathbb{L}^p$. Furthermore, for every $p \in (0, \gamma)$, there exist a constant $C_0 = C_0(p, q, \gamma, \Omega)$ such that

$$(2.7) \quad \|\Phi\|_p \leq \sum_{k \geq 0} (C_0)^k \|\psi^{(k)}\|_q.$$

Let us stress that it is crucial here to have some room between γ and the borders of the interval $[1, 2]$: indeed, both limit cases $\gamma = 1$ and $\gamma = 2$ would require logarithmic corrections. Additionally, the bounds on the moments of Φ strongly rely on the fact that the domain Ω is bounded. A convincing way to understand why, is the fact that the L^q norm does not dominate the L^p norm if Ω is unbounded.

Remark 2.4. *It may be useful in some application to know how the constant $C_0 = C_0(p, q, \gamma, \Omega)$ appearing in Theorem 2.3-(2.7) depends on p, q, γ and Ω . One can deduce from the proof that, for $p \in (1, \gamma)$,*

$$(2.8) \quad C_0(p, q, \gamma, \Omega) = \frac{C_1}{(p-1)} \max\{(\gamma-p)^{-1/p}, (q-\gamma)^{-1/q}\} (|\Omega| \vee 1)^{\frac{1}{p}-\frac{1}{q}},$$

where the constant C_1 is universal.

2.2. Scaling limit of discrete polynomial chaos. Let us now go back to our initial problem of the scaling limit of the discrete polynomial chaos expansion of the partition function (1.13). We will now allow for a slightly more general setting.

2.2.1. Discretization of Ω and of $L_s^q(\Omega)$. First, let us give a precise definition of what we mean when we say that Ω_{δ} is a discretization of Ω . For every $\delta \in (0, 1)$, Ω_{δ} is a finite subset of Ω . Suppose that to each Ω_{δ} is associated a *tesselation* $\mathcal{C}_{\delta}$ of Ω , that is a map $\mathcal{C}_{\delta} : \Omega_{\delta} \rightarrow \mathcal{B}(\Omega)$ (with $\mathcal{B}(\Omega)$ the Borel subsets of Ω) that verifies the following conditions:

- (i) the “cells” $(\mathcal{C}_{\delta}(x))_{x \in \Omega_{\delta}}$ form a partition of Ω ;
- (ii) for every $x \in \Omega_{\delta}$, $x \in \mathcal{C}_{\delta}(x)$;
- (iii) the volume of the cells $\mathcal{C}_{\delta}(x)$ does not depend on the point $x \in \Omega_{\delta}$ and is denoted v_{δ} .

We assume that $v_{\delta} \rightarrow 0$ as $\delta \downarrow 0$, meaning that the discrete sets Ω_{δ} asymptotically approximate Ω . Note that in Section 1, we had assumed v_{δ} had the specific form $\delta^{d_{\text{eff}}}$. This is actually not necessary and, in particular, relaxing it will allow us to consider more general random walks for the (long-range) directed polymer model (see Section 3.2 below).

Then, for every $\delta \in (0, 1)$, any symmetric function $f : \Omega_{\delta}^k \rightarrow \mathbb{R}$ which vanishes on the diagonals can be extended into a piecewise constant function $\bar{f} : \Omega^k \rightarrow \mathbb{R}$ by assigning value

$$(2.9) \quad \bar{f}(y_1, \dots, y_k) = f(x_1, \dots, x_k),$$

for every $(y_1, \dots, y_k) \in \mathcal{C}_\delta(x_1) \times \dots \times \mathcal{C}_\delta(x_k)$. Note that $\bar{f}$ is symmetric and vanishes on the diagonals and that its symmetric L^q norm is given by

$$(2.10) \quad \|\bar{f}\|_q = \left(\frac{(v_\delta)^k}{k!} \sum_{x_1, \dots, x_k \in \Omega_\delta} |f(x_1, \dots, x_k)|^q \right)^{1/q}.$$

For every $\delta \in (0, 1)$, let $(\omega_x)_{x \in \Omega_\delta}$ be i.i.d. *centered* random variables in the domain of attraction of a γ -stable law with $\gamma \in (1, 2)$. In particular (see [Fel66, p. XVII.5]), there exists a slowly-varying function φ^4 such that for every $t > 0$,

$$(2.11) \quad \mathbb{P}[|\omega| > t] = \frac{\varphi(t)}{t^\gamma},$$

and the following limit exists $c_+ = \lim_{t \rightarrow +\infty} \frac{\mathbb{P}[|\omega| \geq t]}{\mathbb{P}[|\omega| \geq t]}$ and $c_- = 1 - c_+$. Note that this is slightly more general than the pure power-law tail introduced in (1.11), and also removes the condition that $\omega > -1$ a.s. (we do not require any positivity assumption in the section).

Let us define $V_\delta \rightarrow +\infty$ up to asymptotic equivalence by

$$(2.12) \quad \mathbb{P}[|\omega| > V_\delta] \sim v_\delta \quad \text{as } \delta \downarrow 0.$$

One can actually show that there is a slowly-varying function $\hat{\varphi}$ such that $V_\delta = v_\delta^{-1/\gamma} \hat{\varphi}(v_\delta)$, see [BGT87, §1.5.7]. (Note that in Section 1 we had $V_\delta \sim C_0^{-1/\gamma} \delta^{-d_{\text{eff}}/\gamma}$.) The definition (1.12) of the discrete noise ζ_δ has to be adapted and becomes

$$(2.13) \quad \zeta_\delta := \frac{1}{V_\delta} \sum_{x \in \Omega_\delta} \omega_x \delta_x.$$

For any $\delta \in (0, 1)$, we let $(\psi_\delta^{(k)})_{k \in \mathbb{N}}$ be functions defined on Ω_δ^k , symmetric and vanishing on the diagonals. We then consider the following *discrete* polynomial chaos associated with $(\psi_\delta^{(k)})_{k \in \mathbb{N}}$, with disorder ω :

$$(2.14) \quad \Phi_\delta = \sum_{k \in \mathbb{N}} \frac{1}{k!} \sum_{x_1, \dots, x_k \in \Omega_\delta} \psi_\delta^{(k)}(x_1, \dots, x_k) \prod_{j=1}^k \frac{\omega_{x_j}}{V_\delta} = \sum_{k \in \mathbb{N}} \frac{1}{k!} \int_{\Omega^k} \bar{\psi}_\delta^{(k)}(x_1, \dots, x_k) \prod_{j=1}^k \zeta_\delta(dx_j),$$

where we recall that $\bar{\psi}_\delta$ is the piecewise constant extension of ψ_δ .

Our second main theorem shows that, under relevant conditions on the functions $(\psi_\delta^{(k)})_{k \in \mathbb{N}}$, the discrete polynomial chaos Φ_δ converges in law as $\delta \rightarrow 0$ to its continuous counterpart.

Theorem 2.5. *Assume that (2.11) holds with $\gamma \in (1, 2)$ and let V_δ be defined by (2.12). Assume that there exist $q \in (\gamma, 2]$ such that:*

(i) *For every $k \in \mathbb{N}$, there exists some $\psi^{(k)} \in L_s^q(\Omega^k)$ such that $\lim_{\delta \downarrow 0} \|\bar{\psi}_\delta^{(k)} - \psi^{(k)}\|_q = 0$;*

(ii) *The following summability conditions are satisfied: for every $C \geq 0$,*

$$\limsup_{\delta \downarrow 0} \sum_{k \geq 0} C^k \|\bar{\psi}_\delta^{(k)}\|_q < +\infty \quad \text{and} \quad \lim_{M \rightarrow +\infty} \limsup_{\delta \downarrow 0} \sum_{k > M} C^k \|\bar{\psi}_\delta^{(k)}\|_q = 0.$$

Then we have the following joint convergence in distribution in $H_{\text{loc}}^{-s}(\Omega) \times \mathbb{R}$, $s > \frac{D}{2}$

$$(\zeta_\delta, \Phi_\delta) \xrightarrow{(d)} (\zeta, \Phi) \quad \text{as } \delta \downarrow 0.$$

Furthermore, for every $1 < p < \gamma$ and $\delta \in (0, 1)$, for the same constant $C_0 = C_0(p, q, \gamma, \Omega)$ appearing in Theorem 2.3-(2.7), we have $\|\Phi_\delta\|_p \leq \sum_{k \in \mathbb{N}} (C_0)^k \|\bar{\psi}_\delta^{(k)}\|_q$ and $\|\Phi_\delta\|_p \rightarrow \|\Phi\|_p$ as $\delta \downarrow 0$.

⁴A function φ is said to be slowly-varying if $\lim_{t \rightarrow \infty} \frac{\varphi(at)}{\varphi(t)} = 1$ for any $a > 0$, see [BGT87].

2.3. Going back to disordered statistical mechanics models. Let us note that the setting of the introduction, *i.e.* partition functions of statistical mechanics disordered systems, fits that of Section 2.2 above: the only specificity is the fact that $c_- = 0$ and $c_+ = 1$ (recall $\omega > -1$) and also that the functions $\psi_\delta^{(k)}$ are non-negative.

We have also slightly generalized the setting of Section 1 since we can now consider disorder satisfying (2.11) instead of the pure power-law (1.11). In addition, we can consider a more general rescaling of the correlations functions: let J_δ be a normalization factor and, for every $k \geq 1$ and $x_1, \dots, x_k \in \Omega_\delta$, define

$$(2.15) \quad \psi_{\Omega_\delta}^{(k)}(x_1, \dots, x_k) := (J_\delta)^k \mathbb{E}_\delta^{\text{ref}} \left[\prod_{j=1}^k \sigma_{x_j} \right].$$

From (1.7), we can now rewrite the partition function as a polynomial chaos:

$$(2.16) \quad Z_{\Omega_\delta}^{\omega, \beta} = 1 + \sum_{k \geq 1} \left(\frac{\beta V_\delta}{J_\delta} \right)^k \sum_{x_1, \dots, x_k \in \Omega_\delta} \psi_{\Omega_\delta}^{(k)}(x_1, \dots, x_k) \prod_{j=1}^k \frac{\omega_{x_j}}{V_\delta}.$$

Then, the condition for *subcriticality* of polynomial chaos can be translated as having $\lim_{\delta \downarrow 0} \frac{V_\delta}{J_\delta} = +\infty$; the criterion (1.17) becomes, in this case, that disorder should be *relevant* when $\lim_{\delta \downarrow 0} \frac{V_\delta}{J_\delta} = +\infty$. We now present several results under this subcriticality condition.

2.3.1. Intermediate disorder for the partition function. An immediate consequence of Theorems 2.3 and 2.5 above is that under adequate conditions on the rescaled correlation functions (2.15), the partition function $Z_{\Omega_\delta}^{\omega, \beta}$ converges in law as $\delta \rightarrow 0$, in the intermediate disorder scaling limit $\beta_\delta V_\delta / J_\delta \rightarrow \hat{\beta} \in (0, \infty)$.

Theorem 2.6 (Scaling limit of the partition function). *Assume that (2.11) holds with $\gamma \in (1, 2)$ and let V_δ be defined by (2.12). Assume that there exist $q \in (\gamma, 2]$ such that the correlations functions $\psi_{\Omega_\delta}^{(k)}$ in (2.15) satisfy the hypotheses of Theorem 2.5. Then, in the intermediate disorder regime $\lim_{\delta \downarrow 0} \beta_\delta \frac{V_\delta}{J_\delta} = \hat{\beta} \in [0, \infty)$, we have the following joint convergence in distribution:*

$$(\zeta_\delta, Z_{\Omega_\delta}^{\omega, \beta_\delta}) \xrightarrow{\delta \rightarrow 0} (\zeta, \mathcal{Z}_{\Omega}^{\zeta, \hat{\beta}}).$$

Here, $\mathcal{Z}_{\Omega}^{\zeta, \hat{\beta}}$ is the continuum partition function defined in (1.16), which is well-defined thanks to Theorem 2.3.

2.3.2. Intermediate disorder for the disordered Gibbs measure. Another very natural result is the convergence of the disordered probability $\mathbb{P}_{\Omega_\delta}^{\omega, \beta}$ defined in (1.2). For such a result to hold, it makes sense to require that the homogeneous model itself converges in a stronger sense than only through correlation functions.

Let E be a Polish space such that for every $\delta > 0$, finite fields $\{\sigma_x\}_{x \in \Omega_\delta}$ can be viewed as elements of E . (The pinning model and long-range polymers are such examples, we refer to Section 3 for further details.) Denote $\mathcal{M}_1(E)$ the topological space of probability measures on E endowed with the topology of convergence in distribution. For any function G defined on E , $k \geq 1$ and $x_1, \dots, x_k \in \Omega_\delta$, denote

$$(2.17) \quad \psi_{\Omega_\delta}^{(k)}(x_1, \dots, x_k; G) := (J_\delta)^k \mathbb{E}_\delta^{\text{ref}} \left[G(\sigma) \prod_{j=1}^k \sigma_{x_j} \right].$$

We denote by $\mathcal{C}_b(E)$ the set of bounded, continuous functions $G : E \rightarrow \mathbb{R}$. The following result shows the convergence of the disordered model in the intermediate disorder regime at the level of measures, and in particular construct a *continuum disordered model*.

Theorem 2.7 (Scaling limit of the Gibbs measure). *Assume that the conditions of Theorem 2.6 hold. If furthermore,*

- (a) $\mathbb{P}_{\Omega_\delta}^{\text{ref}}$ converges in $\mathcal{M}_1(E)$ to a probability distribution $\mathbf{P}_\Omega$;
- (b) for every function $G \in \mathcal{C}_b(E)$ and every $k \in \mathbb{N}^*$, $\overline{\psi}_{\Omega_\delta}^{(k)}(\cdot, G)$ converges in $L_s^q(\Omega^k)$;

(c) for any $\hat{\beta} \geq 0$, the continuum partition function $Z_{\Omega}^{\zeta, \hat{\beta}}$ is almost surely strictly positive. Then we have the following joint convergence in distribution in $H_{\text{loc}}^{-s}(\Omega) \times \mathcal{M}_1(E)$, $s > \frac{D}{2}$,

$$(\zeta_{\delta}, P_{\Omega_{\delta}}^{\omega, \beta_{\delta}}) \xrightarrow{(d)} (\zeta, P_{\Omega}^{\zeta, \hat{\beta}}) \quad \text{as } \delta \downarrow 0,$$

where $P_{\Omega}^{\zeta, \hat{\beta}}$ is the disordered version of P_{Ω} ; part of the statement is that the measure $P_{\Omega}^{\zeta, \hat{\beta}}$ is well-defined.

Assumptions (a) and (b) are very natural. One could hope that Assumption (c) is actually a consequence of the convergence of $Z_{\Omega_{\delta}}^{\omega, \beta_{\delta}}$ to $Z_{\Omega}^{\zeta, \hat{\beta}}$. Unfortunately, we did not find any way to prove this in a general set up. What one should try to prove is that the event $\{Z_{\Omega}^{\zeta, \hat{\beta}} = 0\}$ is actually trivial, *i.e.* it has probability 0 or 1, so that since $\mathbb{E}[Z_{\Omega}^{\zeta, \hat{\beta}}] = 1$ it necessarily has probability 1. This is what is done for instance in [BL22, Proposition 4.4]. We comment further on this issue in the context of the disordered pinning model and the long-range directed polymer model, in Section 3.

Remark 2.8 (Universality of the scaling limit). *It is remarkable that the limiting object constructed in the above theorems depends only on a few parameters. For instance, $P_{\Omega}^{\zeta, \hat{\beta}}$ does not depend on the specific discretization Ω_{δ} of Ω nor on the slowly-varying function φ that appears in the tail (2.11) of the discrete environment. In particular, the rescaling β_{δ} depends on these microscopic details but the scaling limit is universal.*

2.3.3. About disorder relevance. One could wonder if it was truly necessary to rescale β to have a scaling limit, or if the intermediate disorder scale $\beta_{\delta} \sim \hat{\beta} J_{\delta} / V_{\delta}$ is truly the relevant scale. First of all, let us give the following result, which is a direct corollary of Theorem 2.6 and shows that $\lim_{\delta \downarrow 0} \beta_{\delta} \frac{V_{\delta}}{J_{\delta}} = 0$ lies in the so-called *weak disorder* regime.

Corollary 2.9 (Weak disorder). *Assume that the conditions of Theorem 2.6 hold. If $\lim_{\delta \downarrow 0} \beta_{\delta} \frac{V_{\delta}}{J_{\delta}} = 0$, then $Z_{\Omega_{\delta}}^{\omega, \beta_{\delta}}$ converges to $Z_{\Omega}^{\zeta, \hat{\beta}=0} = 1$ in $\mathbb{L}^p$ for every $p \in (0, \gamma)$.*

On the other hand, one would like to show that if $\beta_{\delta} V_{\delta} / J_{\delta} \rightarrow \infty$, then the partition function $Z_{\Omega_{\delta}}^{\omega, \beta_{\delta}}$ converges to 0 in probability. This property is a simplified instance of disorder relevance. As in the case of the strict positivity of the continuum partition function, it actually depends on finer properties of the homogeneous system than simply the scaling form of correlation functions and it is delicate to answer positively with such generality to this question. We have decided to include a sufficient condition (verified in practice in many models, including the pinning and directed polymer model, see Section 2.3 below), to illustrate that for subcritical systems, this property is generic.

Theorem 2.10 (Strong disorder). *Assume that $J_{\delta} v_{\delta} \sum_{x \in \Omega_{\delta}} \sigma_x$ converges in distribution under P_{δ}^{ref} as $\delta \downarrow 0$, to a strictly positive random variable. Then, if $\lim_{\delta \downarrow 0} \frac{\beta_{\delta} V_{\delta}}{J_{\delta}} = +\infty$, the partition function $Z_{\Omega_{\delta}}^{\omega, \beta_{\delta}}$ goes to 0 in probability.*

Note that one can check that

$$\mathbb{E}_{\delta}^{\text{ref}} \left[J_{\delta} v_{\delta} \sum_{x \in \Omega_{\delta}} \sigma_x \right] = v_{\delta} \sum_{x \in \Omega_{\delta}} \psi_{\Omega_{\delta}}^{(1)}(x) \xrightarrow{\delta \downarrow 0} \int_{\Omega} \psi_{\Omega}^{(1)}(x) dx.$$

This at least indicates that $J_{\delta} v_{\delta}$ is the correct normalization for the random variable $\sum_{x \in \Omega_{\delta}} \sigma_x$.

2.4. Further comments.

About the infinite mean case. In contrast with the approach of [BL21; BL22] for the directed polymer model, our approach does not cover the case of a γ -stable noise with $\gamma \in (0, 1]$. This is due to the fact that polynomial chaos with respect to such noises would not have any finite $\mathbb{L}^p$ moments for $p \geq 1$. We leave open the question of finding a general and necessary condition for the analogues of Theorems 2.3 and 2.5 in this case.

The case of a general Lévy noise. Another restriction compared to [BL21; BL22] is that we only consider stable noises instead of general Lévy noises. Given a measure λ on $\mathbb{R}$ satisfying $\int_{\mathbb{R}} (x^2 \wedge 1) \lambda(dx) < +\infty$, one could define the Lévy white noise ζ on Ω with Lévy measure λ as the almost sure limit as $a \downarrow 0$ of the truncated noise

$$\zeta^{(a)} = \sum_{(x,z) \in \Lambda} z \mathbb{1}_{|z| > a} \delta - \kappa(a) dx,$$

where Λ is the Poisson point process on $\Omega \times \mathbb{R}$ of intensity $dx \lambda(dz)$, and $\kappa(a) = \int_{\mathbb{R}} z \mathbb{1}_{|z| > a} \lambda(dz)$. In this case, relevant quantities are the truncated moments of the Lévy measure given by

$$m_p(a, b) = \int_{\mathbb{R}} |z|^p \mathbb{1}_{a < |z| \leq b} \lambda(dz),$$

and our proofs provide the following generalization of Theorem 2.3 (with the same notations):

Theorem 2.11. *Let ζ be a Lévy white noise with Lévy measure λ such that there exist $1 < p < q < 2$ such that $m_p(1, +\infty) < +\infty$ and $m_q(0, 1) < +\infty$. Suppose the family of functions $(\psi^{(k)})_{k \geq 0}$ verifies the following: for every constant $C \geq 0$,*

$$(2.18) \quad \sum_{k \geq 0} C^k \|\psi^{(k)}\|_q < +\infty.$$

Then $\Phi^{(a)}$ is well-defined for every $a \in (0, 1]$ and $(\Phi^{(a)})_{a \in (0, 1]}$ is a time-reversed martingale with respect to the filtration $\mathcal{F} = (\mathcal{F}_a)_{a > 0}$.

Additionally, $(\Phi^{(a)})_{a \in (0, 1]}$ is bounded in $\mathbb{L}^p$ and thus admits a limit $\Phi := \lim_{a \downarrow 0} \Phi^{(a)}$ a.s. and in $\mathbb{L}^p$. Furthermore, there exist a universal constant C'_0 such that

$$(2.19) \quad \|\Phi\|_p \leq \sum_{k \geq 0} \left(\frac{C'_0}{p-1} (m_p(1, +\infty) + m_q(0, 1)) \right)^k \|\psi^{(k)}\|_q.$$

We refer to Proposition 4.11 for further details.

A general Lévy noise ζ does not satisfy any scale invariance property, and consequently a polynomial chaos with respect to ζ cannot arise as the scaling limit of a discrete polynomial chaos with fixed disorder distribution. This obstruction can be circumvented by allowing the law of the discrete disorder random variables ω to depend on δ . For instance, for every $\delta > 0$ and every $x \in \Omega_\delta$, define

$$\omega_x^{(\delta)} = V_\delta \int_{\mathcal{C}_\delta(x)} \zeta(dy),$$

where the constant V_δ is chosen so that $\mathbb{P}[|\omega_x^{(\delta)}| > 1] = 1/2$ (the particular choice of the constants 1 and $1/2$ is immaterial). For every fixed δ , the random variables $(\omega_x^{(\delta)})$ are i.i.d. and centered. It is therefore natural to conjecture that an analogue of Theorem 2.5 holds in this setting for Lévy noises satisfying the assumptions of Theorem 2.11.

Relation to Stochastic PDEs. An important motivation behind the study of continuum partition functions is the construction of solutions to stochastic partial differential equations. Suppose the homogeneous system possesses a time direction (with finite horizon) and is Markovian i.e. $\Omega = (0, T) \times \mathbb{R}^d$ for some $T > 0$ (when $d = 0$, $\Omega = (0, T)$) and there is a function $\psi : [0, T] \times \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}_+$ (interpreted as the density of the transition kernel) such that the continuum correlation functions are given by

$$\psi^{(k)}((t_1, x_1), \dots, (t_k, x_k)) = \psi(t_1, 0, x_1) \psi(t_2 - t_1, x_1, x_2) \dots \psi(t_k - t_{k-1}, x_{k-1}, x_k),$$

for every $0 < t_1 < \dots < t_k < T$ and $x_1, \dots, x_k \in \mathbb{R}^d$. Then the point-to-point partition functions defined by the chaos expansions

$$\begin{aligned} \mathcal{Z}_{st}^{\hat{\beta}}(x, y) &= \psi(t-s, y-x) \sum_{k \geq 1} \hat{\beta}^k \int_{s < t_1 < \dots < t_k < t} \int_{x_1, \dots, x_k \in \mathbb{R}^d} \psi(t_1 - s, x, x_1) \psi(t_2 - t_1, x_1, x_2) \\ &\quad \dots \psi(t_k - t_{k-1}, x_{k-1}, x_k) \psi(t - t_k, x_k, y) \prod_{j=1}^k \zeta(dt_j dx_j), \end{aligned}$$

formally satisfy the flow equation

$$(2.20) \quad \mathcal{Z}_{st}^{\hat{\beta}}(x, y) = \psi(t - s, x, y) + \hat{\beta} \int_s^t \int_{\mathbb{R}^d} \mathcal{Z}_{su}^{\hat{\beta}}(x, z) \psi(t - u, z, y) \zeta(du dz).$$

If ψ is the Green function of a (time-translation invariant) differential operator $\mathcal{L}$, then formally this is the flow equation associated to

$$(2.21) \quad \mathcal{L}u(t, x) = \hat{\beta} u(t, x) \cdot \zeta(t, x), \quad t \geq 0, x \in \mathbb{R}^d.$$

Note that this equation is linear in u , but involves multiplicative noise, which may lead to singular behavior and ill-posedness.

Previous works ([Sai98; Bal14; BN17; Bal23; BJ24]) have addressed equations of the form (2.21), primarily in the context of two classical operators: the heat operator $\partial_t - \Delta_x$ and the wave operator $\partial_{tt} - \Delta_x$. Of particular relevance to the present work is the recent study [BJ24], which focuses on symmetric noises ($c_+ = c_-$ in (2.1)) and relies on the Lepage representation. Assumptions 1.2 and 1.3 in that work are closely related in spirit to the condition (2.6) of Theorem 2.3, as both impose integrability requirements slightly above L^γ .

2.5. Organisation of the rest of the article. The remainder of the article is organized as follows. In Section 3, we apply our general results to the two specific systems already introduced in Examples 3-4. Section 4 is devoted to the proof of the Theorems 4.1 and 4.2, which provide general L^p estimates for discrete and continuous chaos. These results are then applied in Section 5 to the proof of our main results (Theorems 2.3 and 2.5). Section 6 is devoted to the proof of the results stated in Section 2.3. The proofs of the results of Section 3 are presented in Sections 7-8. Finally, the Appendix collects some further technical results. In Appendix A, we collect results on the discrete noise ζ_δ and its convergence towards the continuum noise ζ . In Appendix B, we prove a useful inequality that compares a multivariate series with its continuous integral version.

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3. APPLICATIONS TO SPECIFIC STATISTICAL MECHANICS MODELS

Let us now provide more details on how our results of Section 2.1 apply in our two guideline examples: the disordered pinning model (from Example 1 and 3) and the long-range directed polymer model (from Example 2 and 4).

3.1. The (disordered) pinning model. Before we introduce our results for the pinning model, let us review the definition of the pinning model, which generalizes the setting of Example 1.

3.1.1. The homogeneous pinning model and its scaling limit. Let $\tau = (\tau_k)_{k \geq 1}$ be a recurrent and aperiodic renewal process on $\mathbb{N}$ with inter-arrival law given by

$$(3.1) \quad \mathbb{P}[\tau_1 = n] := \frac{\ell(n)}{n^{1+\alpha}},$$

with $\alpha \in (0, 1)$ and $\ell(\cdot)$ a slowly-varying function⁵. We may also interpret τ as a random subset of $\mathbb{N}$. Let us now introduce the renewal mass function $u(N) := \mathbb{P}[N \in \tau]$, which plays a crucial role. In [Don97, Theorem B], it is proved that

$$(3.2) \quad u(N) := \mathbb{P}[N \in \tau] \underset{N \rightarrow +\infty}{\sim} \frac{C_\alpha}{\ell(N)N^{1-\alpha}}, \quad \text{with} \quad C_\alpha = \frac{\alpha \sin(\pi\alpha)}{\pi}.$$

Let us stress that in [Gia07, Section A.5.4], it is shown that the rescaled subset $\frac{1}{N}\tau \subset \mathbb{R}_+$ converges in distribution as $N \rightarrow +\infty$ to $\mathcal{A}_\alpha$, the regenerative set of index α . The convergence takes place in the space $\mathcal{C}_\infty$ of closed subsets of $\mathbb{R}$, endowed with the Matheron topology, which makes it a Polish

⁵A function ℓ is said to be slowly-varying if $\lim_{t \rightarrow \infty} \frac{\ell(at)}{\ell(t)} = 1$ for any $a > 0$, see [BGT87].

space. The regenerative set $\mathcal{A}_\alpha$ can be defined as the range of the α -stable subordinator $(\sigma_t^{(\alpha)})_{t \geq 0}$, characterized by its Laplace transform

$$\mathbf{E}[e^{-\lambda \sigma_t^{(\alpha)}}] = e^{-t|\lambda|^\alpha}, \quad \text{for } \lambda \geq 0, t \geq 0,$$

that is $\mathcal{A}_\alpha := \overline{\{\sigma_t^{(\alpha)}, t \geq 0\}}$.

The *homogeneous pinning model* is the probability measure on subsets of $\mathbb{N}$ defined for every $h \in \mathbb{R}$ (the pinning parameter) and $N \geq 1$ by

$$(3.3) \quad \frac{dP_{N,h}}{dP}(\tau) := \frac{1}{Z_{N,h}} \exp\left(h \sum_{n=1}^N \mathbf{1}_{n \in \tau}\right) \quad \text{with} \quad Z_{N,h} := \mathbf{E}\left[\exp\left(h \sum_{n=1}^N \mathbf{1}_{n \in \tau}\right)\right].$$

Intuitively, the parameter h gives a reward (if $h > 0$) or a penalty (if $h < 0$) to renewal points in τ . A classical reference for this model is [Gia07, Chapter 2].

As far as the scaling limit of the homogeneous model is concerned, it is shown in [Soh09] that one needs to rescale the pinning parameter h as

$$(3.4) \quad h_N := \frac{\hat{h}}{Nu(N)}, \quad \hat{h} \in \mathbb{R}.$$

Then, [Soh09, Thm 3.1] states that when the closed subset $\frac{1}{N}\tau \cap \llbracket 1, N \rrbracket$ is sampled according to P_{N,h_N} with h_N as in (3.4), it converges in distribution as $N \rightarrow +\infty$ (for the Matheron topology) to a random closed subset $\mathcal{B}_{\alpha,\hat{h}} \subset [0, 1]$. The law $\mathbf{P}_{\alpha,\hat{h}}$ of $\mathcal{B}_{\alpha,\hat{h}}$ is absolutely continuous with respect to the law of $\mathcal{A}_\alpha$, with Radon-Nikodym derivative equal to $e^{\hat{h}L_1^{(\alpha)}} / \mathbf{E}[e^{\hat{h}L_1^{(\alpha)}}]$ where $L_t^{(\alpha)} = \inf\{s \geq 0, \sigma_s^{(\alpha)} \geq t\}$ is the local time of the subordinator $\sigma^{(\alpha)}$. In particular, this shows that Assumption (a) in Theorem 2.7 holds for the homogeneous reference measure P_{N,h_N} .

In order to define the limiting correlation function $\psi_{\hat{h}}^{(k)}$ of the continuum homogeneous pinning model, let us define the *continuum homogeneous partition function* and its *conditioned* analogue. For every $t \geq 0$, let

$$(3.5) \quad \mathcal{Z}_{t,\hat{h}} := \mathbf{E}[e^{\hat{h}L_t^{(\alpha)}}] \quad \text{and} \quad \mathcal{Z}_{t,\hat{h}}^c := \alpha t^{\alpha-1} \mathbf{E}[e^{\hat{h}L_t^{(\alpha)}} \mid t \in \mathcal{A}_\alpha].$$

Let us stress that the second definition is only formal since $\mathbf{P}[t \in \mathcal{A}_\alpha] = 0$. A consequence of [Soh09] is that for every $t \geq 0$, $(Z_{Nt,h_N})_{N \geq 1}$ converges as $N \rightarrow +\infty$ to $\mathcal{Z}_{t,\hat{h}}$. On the other hand, if one considers the discrete conditioned partition function $Z_{N,h}^c := \mathbf{E}[\exp(h \sum_{n=1}^N \mathbf{1}_{n \in \tau}) \mathbf{1}_{N \in \tau}]$, then $\mathcal{Z}_{t,\hat{h}}^c$ is rigorously defined as the limit of $P[N \in \tau]^{-1} Z_{Nt,h_N}^c$ as $N \rightarrow +\infty$. We provide a proof of the existence of this limit in Lemma 7.1, as well as *explicit* expressions of the conditioned and unconditioned continuum homogeneous partition functions in terms of Mittag-Leffler functions.

With the definition (3.5) at hand, the *continuum k -point correlation function* (for $k \geq 1$) is the symmetric function $\psi_{\hat{h}}^{(k)} : (0, 1)^k \rightarrow \mathbb{R}$ defined for $t_0 := 0 < t_1 < \dots < t_k < 1$ by

$$(3.6) \quad \psi_{\hat{h}}^{(k)}(t_1, \dots, t_k) = \frac{1}{\mathcal{Z}_{1,\hat{h}}} \prod_{j=1}^k (\mathcal{Z}_{t_j - t_{j-1}, \hat{h}}^c) \mathcal{Z}_{1-t_k, \hat{h}}.$$

3.1.2. The disordered pinning model and its scaling limit. Let $(\omega_n)_{n \geq 1}$ be i.i.d. random variables such that $\omega > -1$ a.s. Then, for $N \in \mathbb{N}$, $\beta \in [0, 1]$ and $h \in \mathbb{R}$, let us define the disordered pinning measure as follow:

$$(3.7) \quad \frac{dP_{N,h}^{\omega,\beta}}{dP_{N,h}^{\omega,\beta}}(\tau) := \frac{1}{Z_{N,h}^{\omega,\beta}} \prod_{n=1}^N (1 + \beta \omega_n \mathbf{1}_{n \in \tau}), \quad \text{with} \quad Z_{N,h}^{\omega,\beta} := \mathbf{E}_{N,h} \left[\prod_{n=1}^N (1 + \beta \omega_n \mathbf{1}_{n \in \tau}) \right].$$

We can then apply Theorem 2.6 to the partition function $Z_{N,h}^{\omega,\beta}$ of the pinning model to obtain the following. We assume that $\mathbf{E}[\omega] = 0$ and that $\mathbf{P}[\omega > t] \sim \varphi(t)t^{-\gamma}$ as $t \rightarrow \infty$ for some $\gamma \in (1, 2)$, as in (2.11). Also, as in (2.12), let V_N be such that

$$(3.8) \quad \mathbf{P}[\omega > V_N] \sim N^{-1} \quad \text{as} \quad N \rightarrow +\infty,$$

and note that V_N is regularly varying with index $1/\gamma$.

Theorem 3.1 (Intermediate disorder for the partition function). *Let the renewal process τ satisfy (3.1) and assume that $1 - \alpha < \frac{1}{\gamma}$. Note in particular that this guarantees that $u(N)V_N \rightarrow \infty$ as $N \rightarrow \infty$. Assume also the following scaling for the parameters β_N, h_N :*

$$(3.9) \quad \lim_{N \rightarrow \infty} \beta_N u(N)V_N = \hat{\beta} \in (0, +\infty), \quad \lim_{N \rightarrow \infty} h_N u(N)N = \hat{h} \in \mathbb{R}.$$

Then we have the following convergence in distribution:

$$(3.10) \quad Z_{N, h_N}^{\omega, \beta_N} \xrightarrow[N \rightarrow \infty]{(d)} Z_{\hat{h}}^{\zeta, \hat{\beta}} = 1 + \sum_{k \geq 1} \frac{\hat{\beta}^k}{k!} \int_{(0,1)^k} \psi_{\hat{h}}^{(k)}(t_1, \dots, t_k) \prod_{j=1}^k \zeta(dt_j),$$

where ζ is a γ -stable noise on $(0, 1)$ and for every $k \geq 1$, and $\psi_{\hat{h}}^{(k)}$ is defined in (3.6). This convergence holds jointly with the convergence of the discrete noise to ζ . Furthermore, for every $p \in [1, \gamma)$, $Z_{\hat{h}}^{\zeta, \hat{\beta}}$ has a finite moment of order p , and $\lim_{N \rightarrow +\infty} \mathbb{E}[(Z_{N, h_N}^{\omega, \beta_N})^p] = \mathbb{E}[(Z_{\hat{h}}^{\zeta, \hat{\beta}})^p]$.

Note that the multiple integrals appearing in (3.10) can be expressed as iterated stochastic integrals as follows. Define the (spectrally positive) γ -stable Lévy process $(\mathbf{L}_t)_{t \geq 0}$ by setting $\mathbf{L}_t = \int_0^t \zeta(dt)$ for every $t \geq 0$. Then, we can write

$$Z_{\hat{h}}^{\zeta, \hat{\beta}} = 1 + \sum_{k \geq 1} \hat{\beta}^k \int_{0 < t_1 < \dots < t_k < 1} \psi_{\hat{h}}^{(k)}(t_1, \dots, t_k) \prod_{j=1}^k d\mathbf{L}_{t_j},$$

where the integrals over the simplex $\{0 < t_1 < \dots < t_k < 1\}$ can be viewed as iterated Itô integrals with respect to the process $\mathbf{L}$.

Remark 3.2. *Let us stress that there is an alternative representation of the continuum partition function, closer to the limit obtained in [CSZ16a, Thm 3.1], given by:*

$$(3.11) \quad Z_{\hat{h}}^{\zeta, \hat{\beta}} = (Z_{1, \hat{h}})^{-1} \left(1 + \sum_{k \geq 1} \frac{1}{k!} \int_{(0,1)^k} \prod_{j=1}^k \phi^{(k)}(t_1, \dots, t_k) \prod_{j=1}^k (\hat{\beta} \zeta(dt_j) + \hat{h} dt_j) \right),$$

where for every $k \geq 1$, $\phi^{(k)}$ is a symmetric function (corresponding to the $hh = 0$ continuum homogeneous pinning model, i.e. $\phi^{(k)} = \psi_{\hat{h}=0}^{(k)}(t_1, \dots, t_k)$), defined for $0 =: t_0 < t_1 < \dots < t_k < 1$ by

$$\phi^{(k)}(t_1, \dots, t_k) := \prod_{i=1}^k (t_i - t_{i-1})^{-(1-\alpha)}.$$

The idea is simply to expand the product $\prod_{j=1}^k (\hat{\beta} \zeta(dt_j) + \hat{h} dt_j)$ in (3.11) and regroup the integration variables with respect to dt_j . Using the explicit expression of the continuum homogeneous partition functions (see Lemma 7.1), one reconstructs the correlation functions (3.6) and recovers (3.10). We refer to [BL24, §5.1], where this procedure is explained in detail in a very similar context.

We interpret here the discrete pinning measure $P_{N, h_N}^{\omega, \beta_N}$ as a measure on the space $\mathcal{C}_\infty$ of closed subsets of $\mathbb{R}$, endowed with the Matheron topology, by considering the closed subset $\frac{1}{N}\tau$. Let us denote $\mathcal{M}_1(\mathcal{C}_\infty)$ the set of probability measures on $\mathcal{C}_\infty$, endowed with the topology of the convergence in distribution.

Theorem 3.3 (Intermediate disorder for the disordered pinning). *Under the same assumptions as Theorem 3.1, for every $\hat{\beta} \in [0, \infty)$ and $\hat{h} \in \mathbb{R}$, there exists a random probability measure $\mathbf{Q}_{\hat{h}}^{\zeta, \hat{\beta}}$ on $\mathcal{C}_\infty$ such that the following convergence in distribution holds in $\mathcal{M}_1(\mathcal{C}_\infty)$,*

$$P_{N, h_N}^{\omega, \beta_N} \xrightarrow[N \rightarrow +\infty]{(d)} \mathbf{Q}_{\hat{h}}^{\zeta, \hat{\beta}}.$$

This convergence holds jointly with the convergence of the discrete noise to ζ . We refer to the measure $\mathbf{Q}_{\hat{h}}^{\zeta, \hat{\beta}}$ as the continuum disordered pinning model with γ -stable Lévy noise.

The proof of this result follows essentially from Theorem 3.1 and existing literature on the pinning model. The only additional ingredient is the strict positivity of the continuum partition function $\mathcal{Z}_{\hat{h}}^{\zeta, \hat{\beta}}$, which is necessary to apply our general Theorem 2.7. This delicate problem is solved in [FL].

Let us conclude by mentioning the following results in the cases where in (3.9) one has $\hat{\beta} = 0$ (weak disorder) or $\hat{\beta} = +\infty$ (strong disorder).

Theorem 3.4. *Let the aperiodic and recurrent renewal process τ satisfy (3.1) for some $\alpha \in (\frac{\gamma-1}{\gamma}, 1)$. Assume that $\lim_{N \rightarrow \infty} h_N u(N)N = \hat{h} \in \mathbb{R}$, as in (3.9).*

- (i) *If $\lim_{N \rightarrow \infty} \beta_N u(N) V_N = 0$, then $Z_{N, h_N}^{\omega, \beta_N}$ goes to 1 in probability (and in $\mathbb{L}^p$ for every $1 \leq p < \gamma$).*
- (ii) *If $\lim_{N \rightarrow \infty} \beta_N u(N) V_N = +\infty$, then $Z_{N, h_N}^{\omega, \beta_N}$ goes to 0 in probability.*

In particular, the above shows that the disorder relevance properties discussed in Section 2.3.3 hold for the disordered pinning model.

3.1.3. Further remarks and perspectives. The point-to-point disordered partition function is defined for every $1 \leq M \leq N$ by

$$Z_{M, N, h}^{\omega, \beta, c} := \mathbb{E}_{N, h} \left[\prod_{n=M+1}^N (1 + \beta \omega_n \mathbf{1}_{n \in \tau}) \mathbf{1}_{N \in \tau} \mid M \in \tau \right].$$

Our main result also applies to this case, and we can show that if $\lim_{N \rightarrow \infty} \beta_N u(N) V_N = \hat{\beta} \in \mathbb{R}_+$ and $\lim_{N \rightarrow \infty} h_N u(N)N = \hat{h} \in \mathbb{R}$, then for every $0 \leq s \leq t \leq 1$, $u(N)^{-1} Z_{Ns, Nt, h_N}^{\omega, \beta_N, c}$ converges in distribution to a random variable denoted $\mathcal{Z}_{s, t, \hat{h}}^{\hat{\beta}}$. In fact, one can show that

$$(u(N)^{-1} Z_{[Ns], [Nt], h_N}^{\omega, \beta_N, c})_{0 \leq s \leq t \leq 1} \xrightarrow[N \rightarrow \infty]{(d)} (\mathcal{Z}_{s, t, \hat{h}}^{\hat{\beta}})_{0 \leq s \leq t \leq 1},$$

in the sense of finite-dimensional distributions. It would be interesting to strengthen this convergence to convergence in distribution in a suitable functional space, as in [CSZ16b]. Since the limit is expressed as a polynomial chaos with respect to a Lévy white noise, a natural candidate is the space $\mathcal{D}([0, 1]^2, \leq, \mathbb{R}_+)$, consisting of functions defined on the simplex $\{0 \leq s \leq t \leq 1\}$ that are càdlàg in both variables.

This question is motivated by the fact that when $\hat{h} = 0$, the limiting random process is a candidate solution to the following *stochastic Volterra flow equation*:

$$\mathcal{Z}_{s, t}^{\hat{\beta}} = (t - s)^{-(1-\alpha)} + \hat{\beta} \int_s^t (t - u)^{-(1-\alpha)} \mathcal{Z}_{u, t}^{\hat{\beta}} \zeta(du),$$

where ζ is the γ -stable Lévy white noise. Inspired by the recent work [CSZ23], a major breakthrough was achieved for this equation by [WY24] in the case where ζ is Gaussian and in the critical regime $\alpha = 1/2$. We refer to their introduction for further background and references.

3.2. The long-range directed polymer. Before we introduce our results for the directed polymer model, let us review our assumptions on the underlying random walk, which slightly generalizes the setting of Example 2.

3.2.1. The long-range random walk and its scaling limit. We consider a random walk $S = (S_n)_{n \geq 0}$ on $\mathbb{Z}^d$, $d \geq 1$, that belongs to the domain of attraction of an α -stable law, with $\alpha \in (0, 2]$, without centering. In other words, we assume that there exists a sequence $(a_n)_{n \geq 1}$ such that $(\frac{1}{a_n} S_n)_{n \geq 0}$ converges in distribution as $n \rightarrow +\infty$ to a *non-degenerate* d -dimensional α -stable law $\mathbf{X}_1$. (We say that the law of $\mathbf{X}_1$ is non-degenerate if it cannot be supported on any proper subspace of $\mathbb{R}^d$.) For simplicity, we assume in the following that the random walk is aperiodic; this excludes for instance the case of the simple random walk (considered in [BL21]) but makes the exposition much easier.

The characterization of d -dimensional domains of attractions is very well-known in dimension $d = 1$, see the seminal reference [Fel66, p. IX.8], and in dimension $d \geq 2$, this dates back to [HOR84; RG79]; we refer to [Res04] for a general overview. In particular, the scaling sequence $(a_N)_{N \geq 1}$ is necessarily regularly varying with exponent $1/\alpha$; the case $\alpha = 2$ corresponds to the Gaussian case. Note that since

we consider a domain of attraction without centering, this imposes that $E[S_1] = 0$ when $\alpha \in (1, 2]$; the case $\alpha = 1$ is more delicate.

Let us stress that we also have a Donsker's invariance principle as well as a local limit theorem for $(S_N)_{N \geq 0}$. This will ensure in particular that the conditions ((a)) and ((b)) of Theorem 2.7 hold.

Let us start with a statement of the local limit theorem, see e.g. [Don91; Gri86]: under the aperiodicity condition, we have

$$\sup_{x \in \mathbb{Z}^d} \left| (a_n)^d P(S_n = x) - g_1\left(\frac{x}{a_n}\right) \right| \xrightarrow{n \rightarrow +\infty} 0,$$

where $g_1(\cdot)$ is the density of $\mathbf{X}_1$.

As far as the invariance principle is concerned, let us introduce $\mathcal{D} := \mathcal{D}([0, 1], \mathbb{R}^d)$ the space of càdlàg functions with values in $\mathbb{R}^d$, endowed with the Skorokhod topology (also called the J_1 topology). For every $N \geq 1$ and $t \in [0, 1]$, we consider the rescaled random walk $S_t^{(N)} = \frac{1}{a_N} S_{\lfloor Nt \rfloor}$, where $(S_t^{(N)})_{t \in [0, 1]}$ is interpreted as an element of $\mathcal{D}$. Then, the invariance principle (see e.g. [Res07, Thm. 7.1]) tells that we have the following convergence in distribution in $\mathcal{D}$:

$$(S_t^{(N)})_{t \in [0, 1]} \xrightarrow[N \rightarrow \infty]{(d)} (\mathbf{X}_t)_{t \in [0, 1]},$$

where $(\mathbf{X}_t)_{t \in [0, 1]}$ is a d -dimensional α -stable process. For every $t \in (0, 1]$, we denote by $g_t(\cdot)$ the density of $\mathbf{X}_t$: it is given by $g_t(x) = t^{-d/\alpha} g_1(x t^{-1/\alpha})$.

In particular, with this at hand, the *continuum k -point correlation function* (for $k \geq 1$) is the symmetric function $\psi^{(k)} : ((0, 1) \times \mathbb{R}^d)^k \rightarrow \mathbb{R}$ defined by

$$(3.12) \quad \psi^{(k)}((t_1, x_1), \dots, (t_k, x_k)) = \prod_{j=1}^k g_{t_j - t_{j-1}}(x_j - x_{j-1})$$

for $t_0 := 0 < t_1 < \dots < t_k < 1$ and $x_0 := 0, x_1, \dots, x_k \in \mathbb{R}^d$.

3.2.2. The long-range directed polymer model and its scaling limit. We are now ready to introduce the disordered version of the model. Let $(\omega_{n,x})_{n \in \mathbb{N}, x \in \mathbb{Z}^d}$ be i.i.d. and centered heavy-tailed random variables, satisfying (2.11) for some $\gamma \in (1, 2)$; assume also that $\omega > -1$ a.s. Then, the (long-range) directed polymer model is the probability measure defined as a Gibbs modification of the law of the random walk $(S_n)_{n \geq 1}$ defined above: for every $N \in \mathbb{N}$ and $\beta \in [0, 1]$, we let

$$(3.13) \quad \frac{dP_N^{\omega, \beta}}{dP}(S) = \frac{1}{Z_N^{\omega, \beta}} \prod_{n=1}^N (1 + \beta \omega_{n, S_n}), \quad \text{with} \quad Z_N^{\omega, \beta} = E \left[\prod_{n=1}^N (1 + \beta \omega_{n, S_n}) \right].$$

Let us define V_N up to asymptotic equivalence by

$$(3.14) \quad \mathbb{P}[|\omega| > V_N] \sim \frac{1}{N(a_N)^d} \quad \text{as } N \rightarrow +\infty,$$

and note that V_N is regularly varying with index $\frac{1}{\gamma}(1 + \frac{d}{\alpha})$.

Let us mention that we cannot apply directly Theorems 2.6 and 2.7 to obtain results on the directed polymer model, since the ambient space is here $\Omega = (0, 1) \times \mathbb{R}^d$, which is not bounded. We are however able to prove the following results.

Theorem 3.5 (Intermediate disorder for the partition function). *Let $S = (S_n)_{n \geq 0}$ be a random walk on $\mathbb{Z}^d$ satisfying the assumptions of Section 3.2.1, and assume that $\gamma < 1 + \frac{\alpha}{d}$. Note in particular that it guarantees that $V_N(a_N)^{-d} \rightarrow \infty$ as $N \rightarrow \infty$. Assume also the following scaling for the parameter β_N :*

$$(3.15) \quad \lim_{N \rightarrow \infty} \beta_N V_N(a_N)^{-d} = \hat{\beta} \in [0, \infty).$$

Then, we have the following convergence in distribution:

$$(3.16) \quad Z_N^{\omega, \beta_N} \xrightarrow[N \rightarrow \infty]{(d)} \mathcal{Z}^{\zeta, \hat{\beta}} = 1 + \sum_{k \geq 1} \frac{\hat{\beta}^k}{k!} \int_{(0, 1)^k \times (\mathbb{R}^d)^k} \prod_{j=1}^k \psi^{(k)}((t_1, x_1), \dots, (t_k, x_k)) \prod_{j=1}^k \zeta(dt_j dx_j),$$

where ζ is a γ -stable noise on $(0, 1) \times \mathbb{R}^d$ and $\psi^{(k)}$ is defined in (3.12). This convergence holds jointly with the convergence of the discrete noise to ζ . Furthermore, for every $p \in [1, \gamma)$, $\mathcal{Z}^{\zeta, \hat{\beta}}$ has a finite moment of order p , and $\lim_{N \rightarrow +\infty} \mathbb{E}[(Z_N^{\omega, \beta_N})^p] = \mathbb{E}[(\mathcal{Z}^{\zeta, \hat{\beta}})^p]$.

Note that the polymer measure can be viewed as a probability measure on $\mathcal{D} := \mathcal{D}([0, 1], \mathbb{R}^d)$ by setting: for every continuous and bounded function $G : \mathcal{D} \rightarrow \mathbb{R}$,

$$\mathbb{E}_N^{\omega, \beta}[G] := \frac{1}{Z_N^{\omega, \beta}} \mathbb{E} \left[G(S^{(N)}) \prod_{n=1}^N (1 + \beta \omega_{n, S_n}) \right].$$

We then have the following convergence for the directed polymer measure towards a continuum object.

Theorem 3.6 (Intermediate disorder for the long-range directed polymer). *Under the same assumptions as Theorem 3.5, for every $\hat{\beta} \in [0, \infty)$, there exists a random probability measure $\mathbf{Q}^{\zeta, \hat{\beta}}$ on $\mathcal{D}$ such that the following convergence in distribution holds in $\mathcal{M}_1(\mathcal{D})$,*

$$\mathbb{P}_N^{\omega, \beta_N} \xrightarrow[N \rightarrow +\infty]{(d)} \mathbf{Q}^{\zeta, \hat{\beta}}.$$

This convergence holds jointly with the convergence of the discrete noise to ζ . We refer to the measure $\mathbf{Q}^{\zeta, \hat{\beta}}$ as the continuum long-range directed polymer model with γ -stable Lévy noise.

Let us mention that Theorems 3.5-3.6 are the analogous of Theorems 2.1 and 2.4 in [BL21], where the results are obtained in the case of the simple random walk (which corresponds to $\alpha = 2$). Our results are therefore more general, and answer completely the questions of [BL21, Sec. 2.2] (note that the roles of γ and α are reversed).

Finally, let us conclude by mentioning the following results in the cases where in (3.15) one has $\hat{\beta} = 0$ (weak disorder) or $\hat{\beta} = +\infty$ (strong disorder).

Theorem 3.7. *Let the random walk $(S_n)_{n \geq 0}$ satisfy the assumptions of Section 3.2.1 and assume that $1 + \frac{\alpha}{d} > \gamma - 1$.*

- (i) *If $\lim_{N \rightarrow +\infty} \beta_N V_N(a_N)^{-d} = 0$, then Z_N^{ω, β_N} goes to 1 in probability (and in L^1).*
- (ii) *If $\lim_{N \rightarrow +\infty} \beta_N V_N(a_N)^{-d} = +\infty$, then Z_N^{ω, β_N} goes to 0 in probability.*

This shows that the disorder relevance properties discussed in Section 2.3.3 hold for the long-range directed polymer model.

3.2.3. Further remarks and perspectives. For this model, the point-to-point partition function is defined for every $1 \leq M \leq N$ and $x, y \in \mathbb{Z}^d$ by

$$Z_{M, N}^{\omega, \beta, c}(x, y) := \mathbb{E} \left[\prod_{n=M+1}^N (1 + \beta \omega_{n, S_n}) \mathbf{1}_{S_N=y} \mid S_M = x \right].$$

For $x \in \mathbb{R}^d$, we denote by $\lfloor x \rfloor$ a nearest neighbor of x in $\mathbb{Z}^d$. Our techniques adapt to prove that there exist a process $(\mathcal{Z}_{s, t}^{\zeta, \hat{\beta}}(x, y))_{0 \leq s \leq t \leq 1; x, y \in \mathbb{R}^d}$ such that

$$(3.17) \quad (a_N)^{-1} Z_{\lfloor Ns \rfloor, \lfloor Nt \rfloor}^{\omega, \beta_N, c}(\lfloor a_N x \rfloor, \lfloor a_N y \rfloor) \xrightarrow[N \rightarrow +\infty]{(d)} \mathcal{Z}_{s, t}^{\zeta, \hat{\beta}}(x, y),$$

in the sense of finite-dimensional distributions. The limiting process is a candidate solution for the flow associated with the following *stochastic fractional heat equation*

$$(3.18) \quad \partial_t u(t, x) = \mathcal{L}_\alpha u(t, x) + \hat{\beta} \zeta(t, x) \cdot u(t, x), \quad t \geq 0, x \in \mathbb{R}^d.$$

Here, $\mathcal{L}_\alpha$ denotes the fractional Laplacian of index α , and ζ is a γ -stable noise. When $\alpha = 2$, $\mathcal{L}_\alpha$ reduces to the standard Laplacian $\Delta = \sum_{j=1}^d \partial_{x_j}^2$, and (3.18) becomes the stochastic heat equation (SHE) with multiplicative noise. This equation has been extensively studied when the noise ζ is Gaussian. In this case, the Cole-Hopf transform of the solution of the SHE is expected to solve the KPZ equation, a singular SPDE which modeling the evolution of random growth fronts; recent breakthroughs were achieved in this direction [CSZ20; CSZ23]. Another important motivation for the study of the SHE

is the phenomenon of *localization*, or *intermittency* [Zel+87; CM94]. Informally, this refers to the appearance of sharp peaks at large times in the random field $(u(t, x))_{t \geq 0, x \in \mathbb{R}^d}$, where u is a solution of (3.18).

Early results for the SHE driven by Lévy noise were obtained in [Sai98], but relied on assumptions on the Lévy measure that excluded γ -stable noise. In the γ -stable case, existence, uniqueness and intermittency were established in [BCL23] (see also [CK23]).

As already mentioned for the stochastic Volterra equation, it would be interesting to strengthen the convergence (3.17) to a convergence in a suitable functional space. We again expect càdlàg regularity in time. Spatial regularity is a more delicate and less explored issue, particularly in view of intermittency. We refer to [CDH18] for related results in this direction.

4. IMPORTANT $\mathbb{L}^p$ ESTIMATES FOR CHAOS

In this section we prove useful and general moment inequalities for discrete and continuous chaos. They illustrate in particular why we needed to introduce symmetric norms in Section 2.4. All the setting and notations are the same as in Section 2.

A function $\mathbf{f} : \Omega^k \mapsto \mathbb{R}$ is said to vanish on the diagonals if $\mathbf{f}(x_1, \dots, x_k) = 0$ whenever there are two indices $i \neq j$ such that $x_i = x_j$. We now state two results for chaos based on symmetric functions that vanish on the diagonals: one for the discrete chaos and one for the continuous chaos. In both cases, the meaning of the inequalities we state is that the left-hand side is well-defined if the right-hand side is finite, and the inequality holds when this is the case. We will then apply these inequalities to k -point correlation functions $\frac{1}{k!} \psi^{(k)}$.

Theorem 4.1. *Assume that ω is centered and verifies (2.11) with $\gamma \in (1, 2)$. For every $p \in (1, \gamma)$ and $q \in (\gamma, 2]$, there exists a constant $C_0 = C_0(p, q, \gamma, \Omega)$ such that, for every $k \geq 1$, $\delta \in (0, 1)$ and $f : \Omega_\delta^k \rightarrow \mathbb{R}$ which is symmetric and vanishes on the diagonal:*

$$(4.1) \quad \left\| \frac{1}{k!} \sum_{x_1, \dots, x_k \in \Omega_\delta} f(x_1, \dots, x_k) \prod_{j=1}^k \frac{\omega_{x_j}}{V_\delta} \right\|_p \leq (C_0)^k \|\bar{f}\|_q.$$

Here, we recall that V_δ is defined (up to asymptotic equivalence) by the relation (2.12) and that $\bar{f}$ is the piecewise extension of f as introduced in (2.9).

We refer to Remark 2.4 (see (2.8)) for the way that the constant C_0 depends on p, q, γ, Ω .

Theorem 4.2. *Assume that ζ is a γ -stable Lévy white noise, with $\gamma \in (1, 2)$. For every $p \in (1, \gamma)$ and $q \in (\gamma, 2]$, the constant C_0 of Theorem 4.1 is such that for every $k \geq 1$, $a \in (0, 1]$ and $\mathbf{f} : \Omega^k \mapsto \mathbb{R}$ which is symmetric and vanishes on the diagonals,*

$$(4.2) \quad \left\| \frac{1}{k!} \int_{\Omega^k} \mathbf{f}(x_1, \dots, x_k) \prod_{j=1}^k \zeta^{(a)}(dx_j) \right\|_p \leq (C_0)^k \|\mathbf{f}\|_q.$$

4.1. Decoupling inequalities and some consequences. A crucial tool for the proof of Theorems 4.1 and 4.2 is so-called *decoupling inequalities*. We refer to [PG00] for an overview of these techniques. For pedagogical reasons, we recall the main definitions and inequalities that we use in our setting. In this section only, instead of working on a general summation index Ω_δ , we sum over sets of the form $\llbracket 1, N \rrbracket = \{1, \dots, N\}$, where N is the number of points in Ω_δ . Also, instead of working with symmetric functions null on the diagonals, we use tetrahedral functions, *i.e.* functions defined on $(n_1, \dots, n_k)$ with $n_1 < \dots < n_k$. The main result of this section (Theorem 4.3 below) can then easily be transferred to the setting of Theorem 4.1, the estimates being uniform in N — this is done in Section 4.2. The adaptation to continuous chaos is carried out in Section 4.3.

4.1.1. Statement of the result. Let $k \geq 1$ and for every $1 \leq j \leq k$, let $\omega^{(j)} = (\omega_n^{(j)})_{n \geq 1}$ be a sequence of independent and centered random variables. We do not assume that the sequences $\omega^{(j)}$ are independent or identically distributed, but we require that they are jointly *level independent*. This means that, if for $n \in \mathbb{N}$ one lets $\mathcal{L}_n = \sigma(\omega_n^{(j)}, 1 \leq j \leq k)$ be the “level n ” σ -algebra, then $(\mathcal{L}_n)_{n \geq 1}$ forms an independent

family. For example, let $\omega = (\omega_n)_{n \geq 1}$ be a sequence of independent random variables and define $\omega^{(j)} = \omega$ for every $1 \leq j \leq k$: then, the sequences $\omega^{(j)}$ are identical, but level independent nevertheless.

For any $k \geq 1$ and $N \geq 1$, let $\mathcal{T}_N^k$ denote the set of functions defined on the tetrahedra

$$\{(n_1, \dots, n_k) \in \llbracket 1, N \rrbracket, 1 \leq n_1 < \dots < n_k \leq N\}.$$

For $f \in \mathcal{T}_N^k$, define

$$X_N(f; \omega^{(1)}, \dots, \omega^{(k)}) := \sum_{1 \leq n_1 < \dots < n_k \leq N} f(n_1, \dots, n_k) \prod_{j=1}^k \omega_{n_j}^{(j)}.$$

Decoupling inequalities compare the law of the random variable $X_N(f; \omega^{(1)}, \dots, \omega^{(k)})$ with the law of the same random variable where the random sequences $\omega^{(1)}, \dots, \omega^{(k)}$ are replaced by a version where they are now independent.

Theorem 4.3 ($\mathbb{L}^p$ decoupling inequality). *There exists a constant C_1 such that for every $p \in (1, 2]$, for any $k \geq 1$, $N \geq 1$ and any tetrahedral function $f \in \mathcal{T}_N^k$,*

$$(4.3) \quad \left\| X_N(f; \omega^{(1)}, \dots, \omega^{(k)}) \right\|_p \leq \left(\frac{C_1}{p-1} \right)^k \left\| X_N(f; \tilde{\omega}^{(1)}, \dots, \tilde{\omega}^{(k)}) \right\|_p,$$

where the random sequences $\tilde{\omega}^{(1)}, \dots, \tilde{\omega}^{(k)}$ are independent and for each $1 \leq j \leq k$, $\tilde{\omega}^{(j)}$ has the same distribution as $\omega^{(j)}$.

Let us note that the random variable $X_N(f; \omega^{(1)}, \dots, \omega^{(k)})$ is an example of U-statistics, for which decoupling inequalities have a long history; we refer to [PG00] for an introduction to this topic. Let us stress however that the decoupling inequality of Theorem 4.3 has the remarkable feature that the dependence in k is of the form C_1^k with a universal constant C_1 . This is not true for general U-statistics and the method used here to derive this inequality is quite different from the techniques usually employed to obtain decoupling inequalities. For instance, in [PM95], the authors obtain more general decoupling inequalities but the dependence in k is of the type C^{k^2} .

We postpone the proof of Theorem 4.3 to Section 4.1.3 below, and first derive some corollaries.

4.1.2. Some useful corollaries. From Theorem 4.3 we can deduce the following corollary, which is the most useful result of this section for our purposes.

Corollary 4.4. *Assume that $\omega^{(j)} = (\omega_n^{(j)})_{n \in \mathbb{N}}$ are sequences of independent and centered random variables. There exist a constant C_2 such that for every $p \in (1, 2]$, for every $k \geq 1$, $N \geq 1$ and any tetrahedral function $f \in \mathcal{T}_N^k$,*

$$\left\| X_N(f; \omega^{(1)}, \dots, \omega^{(k)}) \right\|_p \leq \left(\frac{C_2}{p-1} \right)^k \left(\sum_{1 \leq n_1 < \dots < n_k \leq N} |f(n_1, \dots, n_k)|^p \prod_{j=1}^k \left\| \omega_{n_j}^{(j)} \right\|_p^p \right)^{1/p}.$$

Proof. First, apply Theorem 4.3: we obtain

$$(4.4) \quad \left\| X_N(f; \omega^{(1)}, \dots, \omega^{(k)}) \right\|_p \leq \left(\frac{C_1}{p-1} \right)^k \left\| X_N(f; \tilde{\omega}^{(1)}, \dots, \tilde{\omega}^{(k)}) \right\|_p.$$

Let us now freeze the $k-1$ first variables $\tilde{\omega}^{(1)}, \dots, \tilde{\omega}^{(k-1)}$ in $X_N(f; \tilde{\omega}^{(1)}, \dots, \tilde{\omega}^{(k)})$ and denote $\mathcal{F}_n^{(k)} = \sigma(\tilde{\omega}_i^{(k)}, i \leq n)$ for $n \geq 1$. Then, we consider the $\mathcal{F}_n^{(k)}$ -martingale

$$X_N(f; \tilde{\omega}^{(1)}, \dots, \tilde{\omega}^{(k)}) = \sum_{n=1}^N F(n) \tilde{\omega}_n^{(k)}$$

with for $n \geq 1$,

$$F(n) := \sum_{1 \leq n_1 < \dots < n_{k-1} < n} f(n_1, \dots, n_{k-1}, n) \prod_{j=1}^{k-1} \tilde{\omega}_{n_j}^{(j)},$$

and note that the $F(n)$ are random variables independent of $(\mathcal{F}_n^{(k)})_{n \geq 1}$. Applying the Burkholder–Davis–Gundy (BDG) inequality [BDG72] (see also Proposition 4.7 below), this yields that

$$\|X_N(f; \tilde{\omega}^{(1)}, \dots, \tilde{\omega}^{(k)})\|_p \leq C_p \mathbb{E} \left[\left(\sum_{n=1}^N F(n)^2 (\tilde{\omega}_n^{(k)})^2 \right)^{p/2} \right]^{1/p},$$

for some constant C_p (which is bounded by a universal constant uniformly for $p \in [1, 2]$). Since $p/2 \in [0, 1]$, we can use the inequality $(\sum_i a_i)^{p/2} \leq \sum_i a_i^{p/2}$ for any non-negative a_i 's to get that the above is bounded by

$$C_p \left(\sum_{n=1}^N \mathbb{E}[|F(n)|^p] \mathbb{E}[|\omega_n^{(k)}|^p] \right)^{1/p},$$

using also that $(\tilde{\omega}_i^{(k)})_{i \geq 1}$ has the same distribution as $(\omega_i^{(k)})_{i \geq 1}$.

All together, if for any $1 \leq n_k \leq N$ we denote $f_{n_k}(n_1, \dots, n_{k-1}) = f(n_1, \dots, n_k)$, then $f_{n_k} \in \mathcal{T}_{n_k-1}^{(k-1)}$ and recalling the definition of $F(n)$ we have that $F(n_k) = X_{n_k-1}(f_{n_k}; \tilde{\omega}^{(1)}, \dots, \tilde{\omega}^{(k-1)})$, so that

$$\left\| X_N(f; \tilde{\omega}^{(1)}, \dots, \tilde{\omega}^{(k)}) \right\|_p^p \leq (C_p)^p \sum_{n_k=1}^N \left\| X_{n_k-1}(f_{n_k}; \tilde{\omega}^{(1)}, \dots, \tilde{\omega}^{(k-1)}) \right\|_p^p \|\omega_{n_k}^{(k)}\|_p^p.$$

We can then iterate this inequality to finally obtain

$$(4.5) \quad \left\| X_N(f; \tilde{\omega}^{(1)}, \dots, \tilde{\omega}^{(k)}) \right\|_p^p \leq (C_p)^{pk} \sum_{1 \leq n_1 < \dots < n_k \leq N} |f(n_1, \dots, n_k)|^p \prod_{j=1}^k \|\omega_{n_j}^{(j)}\|_p^p.$$

This concludes the proof of Corollary 4.4 follows, recalling also (4.4). $\square$

Remark 4.5. *Let us stress that Corollary 4.4 will be very useful in what follows but it is not completely sharp, contrary to Theorem 4.1. For instance, consider the case $k = 1$ and i.i.d. random variables ω as in (2.11) with a pure polynomial tail, i.e. $\mathbb{P}(|\omega| > t) \sim t^{-\gamma}$ as $t \rightarrow \infty$. It is a classical fact, see [Fel66, p. XVII.7], that*

$$\sum_{1 \leq n \leq N} f(n) \omega_n \propto \left(\sum_{1 \leq n \leq N} |f(n)|^\gamma \right)^{1/\gamma},$$

in the sense that the ratio of the left-hand side by the right-hand side converges in distribution to a non-degenerate (γ -stable) random variable; this corresponds to the heavy-tailed analogue of the central limit theorem. Then, let us observe that Corollary 4.4 does not capture this correct scaling, since one cannot apply the inequality with $p = \gamma$. The limitation of Corollary 4.4 is that it does not exploit the full heavy-tail structure of the ω 's and in fact holds for any random variable ω with a finite $\mathbb{L}^p$ moment.

Before turning to the proof of Theorem 4.3, in view of the above remark, let us therefore state a slightly stronger version of Corollary 4.4. In order to fully take advantage of the heavy tail structure, we will need to perform $\mathbb{L}^p$ estimates for some families of random variables $\omega^{(j)}$, and $\mathbb{L}^q$ estimates with $p \neq q$ for the others: in the statement below, the set $I \subset \llbracket 1, k \rrbracket$ is thought as the set of indices for which $\mathbb{L}^p$ estimates are to be used.

Corollary 4.6. *There exist a constant C_2 such that for every $p \in (1, 2]$ and $p \leq q \leq 2$, for every $k \geq 1$, $N \geq 1$ and $I \subset \llbracket 1, k \rrbracket$, for any tetrahedral function $f \in \mathcal{T}_N^k$,*

$$(4.6) \quad \begin{aligned} & \left\| X_N(f; \omega^{(1)}, \dots, \omega^{(k)}) \right\|_p \\ & \leq \left(\frac{C_2}{p-1} \right)^k \left(\sum_{n_i, i \in I} \prod_{i \in I} \|\omega_{n_i}^{(i)}\|_p^p \left(\sum_{\substack{n_i, i \notin I \\ 1 \leq n_1 < \dots < n_k \leq N}} |f(n_1, \dots, n_k)|^q \prod_{i \notin I} \|\omega_{n_i}^{(i)}\|_q^q \right)^{p/q} \right)^{1/p}. \end{aligned}$$

Additionally, applying Jensen's inequality (to the first sum) shows that the above is bounded by

$$(4.7) \quad \left(\frac{C_2}{p-1}\right)^k N^{|I|(\frac{1}{p}-\frac{1}{q})} \left(\sum_{1 \leq n_1 < \dots < n_k \leq N} |f(n_1, \dots, n_k)|^q \prod_{i \in I} \|\omega_{n_i}^{(i)}\|_p^q \prod_{i \notin I} \|\omega_{n_i}^{(i)}\|_q^q \right)^{1/q}.$$

Proof. By the decoupling inequality of Theorem 4.3, we only need to treat $X_N(f; \tilde{\omega}^{(1)}, \dots, \tilde{\omega}^{(k)})$, that is when the sequences $\tilde{\omega}^{(j)}$ are independent.

Let $I \subset \llbracket 1, k \rrbracket$ and define, for every indices $(n_i)_{i \in I}$, the random variable

$$g((n_i)_{i \in I}) = \sum_{\substack{n_i, i \notin I \\ 1 \leq n_1 < \dots < n_k \leq N}} f(n_1, \dots, n_k) \prod_{i \notin I} \tilde{\omega}_{n_i}^{(i)}.$$

Note that random variables $g((n_i)_{i \in I})$ are independent of the random sequences $\tilde{\omega}^{(i)}$ for $i \in I$. Rewriting

$$X_N(f; \tilde{\omega}^{(1)}, \dots, \tilde{\omega}^{(k)}) = \sum_{n_i, i \in I} g((n_i)_{i \in I}) \prod_{i \in I} \tilde{\omega}_{n_i}^{(i)},$$

we can therefore apply Corollary 4.4, or more precisely (4.5), conditionally to the random variables $\tilde{\omega}^{(i)}$ for $i \notin I$. We obtain

$$\left\| X_N(f; \tilde{\omega}^{(1)}, \dots, \tilde{\omega}^{(k)}) \right\|_p \leq (C_p)^{|I|} \left(\sum_{n_i, i \in I} \prod_{i \in I} \|\omega_{n_i}^{(i)}\|_p^p \mathbb{E} \left[|g((n_i)_{i \in I})|^p \right] \right)^{1/p}.$$

Now, using Jensen's inequality, since $p \leq q$, we get that

$$\mathbb{E} \left[|g((n_i)_{i \in I})|^p \right] \leq \mathbb{E} \left[|g((n_i)_{i \in I})|^q \right]^{p/q}.$$

We now apply again Corollary 4.4 (more precisely (4.5)), to get that

$$\mathbb{E} \left[|g((n_i)_{i \in I})|^q \right]^{1/q} \leq (C_q)^{k-|I|} \left(\sum_{\substack{n_i, i \notin I \\ 1 \leq n_1 < \dots < n_k \leq N}} |f(n_1, \dots, n_k)|^q \prod_{i \notin I} \|\omega_{n_i}^{(i)}\|_q^q \right)^{1/q}.$$

Putting everything together, recalling (4.4) and the fact that C_p, C_q are bounded by a universal constant, this concludes the proof. $\square$

4.1.3. Proof of Theorem 4.3. The proof is based on two elementary results from the theory of martingales and is a direct adaptation of the proof for the continuous version which can be found in [BCL23].

Let us state two results that we use in the following. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $(\mathcal{F}_n)_{n \geq 1}$ be a filtration. The first result is the BDG inequality for discrete martingales, see [BDG72, Theorem 1.1].

Proposition 4.7 (BDG inequality). *Let $(M_n)_{n \geq 1}$ be a martingale with respect to $(\mathcal{F}_n)_{n \geq 1}$ such that $M_0 = 0$. Denote $\langle M \rangle$ its quadratic variation, i.e. $\langle M \rangle^2 = \sum_{n \geq 1} (M_n - M_{n-1})^2$ and $M^* = \sup_{n \geq 1} |M_n|$. Then, for any $p \geq 1$, there is a constant C_p which depends only on p (and is bounded by a constant uniformly in $p \in [1, 2]$) such that*

$$(4.8) \quad \frac{1}{C_p} \mathbb{E}[\langle M \rangle^{p/2}] \leq \mathbb{E}[(M^*)^p] \leq C_p \mathbb{E}[\langle M \rangle^{p/2}].$$

The second result deals with tangent processes. Two adapted processes $(d_n)_{n \geq 1}$ and $(e_n)_{n \geq 1}$ are said to be *tangent* if for all $n \geq 1$ and all $t \in \mathbb{R}$, a.s. $\mathbb{P}[e_n \geq t | \mathcal{F}_{n-1}] = \mathbb{P}[d_n \geq t | \mathcal{F}_{n-1}]$. The proof of the following result can be found in [Hit88, Theorem 2].

Proposition 4.8 (Comparing $\mathbb{L}^p$ norms of tangent processes). *Let $(e_n)_{n \geq 1}$ and $(d_n)_{n \geq 1}$ be two non-negative processes which are tangent. Then for any $p > 0$, there is a constant C'_p such that*

$$\mathbb{E} \left[\left(\sum_{n \geq 1} e_n \right)^p \right] \leq C'_p \mathbb{E} \left[\left(\sum_{n \geq 1} d_n \right)^p \right].$$

The remarkable feature of this last proposition is that it is also valid for $p \in (0, 1]$ and in particular does not use convexity; in fact, for $p \in (0, 1]$ (more generally for concave functions) one can take $C_p = 6$, see [Hit88, p.228]. This is crucial for us since we will apply it for $p/2 \leq 1$.

Remark 4.9. *Both of the above results are actually valid in a more general setting which strictly includes $\mathbb{L}^p$ norms. One could thus expect that the result of Theorem 4.1 holds for more general functionals. However we will ultimately use Doob's inequality and this is where we have to restrict ourselves to $\mathbb{L}^p$ norms.*

Proof of Theorem 4.3. Our main goal is to show that there exists a constant C_1 such that for every $1 \leq i \leq k$,

$$\left\| X_N(f; \omega^{(1)}, \dots, \omega^{(i)}, \tilde{\omega}^{(i+1)}, \dots, \tilde{\omega}^{(k)}) \right\|_p \leq C_1 \left\| X_N(f; \omega^{(1)}, \dots, \omega^{(i-1)}, \tilde{\omega}^{(i)}, \tilde{\omega}^{(i+1)}, \dots, \tilde{\omega}^{(k)}) \right\|_p.$$

The proof then follows easily by iteration. Additionally, by conditioning with respect to the random variables $\tilde{\omega}^{(i+1)}, \dots, \tilde{\omega}^{(k)}$ it is actually enough to prove this for $i = k$.

For $N \geq 1$, define the σ -algebra $\mathcal{F}_N = \sigma(\omega_n^{(j)}, 1 \leq j \leq k, 1 \leq n \leq N; \tilde{\omega}_n^{(k)}, 1 \leq n \leq N)$. Then the processes $(X_N(f; \omega^{(1)}, \dots, \omega^{(k)}))_{N \geq 1}$ and $(X_N(f; \omega^{(1)}, \dots, \omega^{(k-1)}, \tilde{\omega}^{(k)}))_{N \geq 1}$ are martingales with respect to the filtration $(\mathcal{F}_N)_{N \geq 1}$. Therefore, by the BDG inequality (Proposition 4.7), we have

$$\left\| X_N(f; \omega^{(1)}, \dots, \omega^{(k)}) \right\|_p \leq C_p \mathbb{E} \left[\langle X(f; \omega^{(1)}, \dots, \omega^{(k)}) \rangle_N^{p/2} \right]^{1/p}.$$

The increments of the process $(\langle X(f; \omega^{(1)}, \dots, \omega^{(k)}) \rangle_N)_{N \geq 1}$ are given by

$$e_N = \left(\sum_{1 \leq n_1 < \dots < n_{k-1} < N} f(n_1, \dots, n_{k-1}, N) \prod_{j=1}^{k-1} \omega_{n_j}^{(j)} \omega_N^{(k)} \right)^2,$$

which are non-negative. Since $\tilde{\omega}^{(k)}$ is identically distributed to $\omega^{(k)}$ and thanks to the joint level independence property, $(e_N)_{N \geq 1}$ is tangent to $(d_N)_{N \geq 1}$, where

$$d_N = \left(\sum_{1 \leq n_1 < \dots < n_{k-1} < N} f(n_1, \dots, n_{k-1}, N) \prod_{j=1}^{k-1} \omega_{n_j}^{(j)} \tilde{\omega}_N^{(k)} \right)^2.$$

Let us now observe that $(d_N)_{N \geq 1}$ is exactly the process of the increments of the quadratic variation of $(X_N(f; \omega^{(1)}, \dots, \omega^{(k-1)}, \tilde{\omega}^{(k)}))_{N \geq 1}$. Therefore, by Proposition 4.8 we have

$$\begin{aligned} \left\| X_N(f; \omega^{(1)}, \dots, \omega^{(k)}) \right\|_p &\leq C_p C'_{p/2} \mathbb{E} \left[\langle X(f; \omega^{(1)}, \dots, \tilde{\omega}^{(k)}) \rangle_N^{p/2} \right]^{1/p} \\ &\leq C_p^2 C'_{p/2} \left\| \sup_{1 \leq n \leq N} X_n(f; \omega^{(1)}, \dots, \tilde{\omega}^{(k)}) \right\|_p \\ &\leq C_p^2 C'_{p/2} \frac{p}{p-1} \left\| X_N(f; \omega^{(1)}, \dots, \tilde{\omega}^{(k)}) \right\|_p, \end{aligned}$$

where we have applied the reversed BDG for the second inequality and finally Doob's inequality for the last one. This concludes the proof of Theorem 4.3, with $C_1 = \sup_{p \in [1, 2]} \{p C_p^2 C'_{p/2}\}$. $\square$

4.2. Application to discrete chaos: proof of Theorem 4.1. First of all, let us reformulate Corollary 4.6 in the context of polynomial chaos. For $k \geq 1$, let $\omega^{(j)} = (\omega_x^{(j)})_{x \in \Omega_\delta}$, $1 \leq j \leq k$, be sequences of i.i.d. and centered random variables jointly level independent (recall the beginning of Section 4.1 for a definition). We obtain the following, as an easy consequence of Corollary 4.6.

Proposition 4.10. *There exist a constant C_2 such that for every $p \in (1, 2]$ and $p \leq q \leq 2$, for every $k \geq 1$, $\delta > 0$ and $J \subset \llbracket 1, k \rrbracket$, for any symmetric function $f : \Omega_\delta^k \rightarrow \mathbb{R}$ that vanishes on the diagonals,*

$$\left\| \frac{1}{k!} \sum_{x_1, \dots, x_k \in \Omega_\delta} f(x_1, \dots, x_k) \prod_{j=1}^k \omega_{x_j}^{(j)} \right\|_p \leq \left(\frac{C_2}{p-1} \right)^k |\Omega|^{J|(\frac{1}{p} - \frac{1}{q})|} \|\bar{f}\|_q \prod_{j \in J} \frac{\|\omega^{(j)}\|_p}{v_\delta^{1/p}} \prod_{j \notin J} \frac{\|\omega^{(j)}\|_q}{v_\delta^{1/q}}.$$

Recall here that $\bar{f}$ is the piecewise constant extension of f and v_δ is the volume of any cell of Ω_δ .

Note that we may also bound $|\Omega|^{J|(\frac{1}{p}-\frac{1}{q})|} \leq (|\Omega| \vee 1)^{k(\frac{1}{p}-\frac{1}{q})}$ since $\frac{1}{p} - \frac{1}{q} \geq 0$ in the above.

Proof. Since the set Ω_δ is finite, it is enough to show the result for $\Omega_\delta = \llbracket 1, N \rrbracket$ with N the number of elements of Ω_δ (which is also the number of cells). Then, letting $\mathfrak{S}_k$ be the symmetric group of $\llbracket 1, k \rrbracket$, we can write

$$\frac{1}{k!} \sum_{n_1, \dots, n_k \in \llbracket 1, N \rrbracket} f(n_1, \dots, n_k) \prod_{j=1}^k \omega_{n_j}^{(j)} = \frac{1}{k!} \sum_{\sigma \in \mathfrak{S}_k} \sum_{1 \leq n_1 < \dots < n_k \leq N} f(n_1, \dots, n_k) \prod_{j=1}^k \omega_{n_j}^{(\sigma(j))}.$$

We can then apply Corollary 4.6-(4.7) for every fixed $\sigma \in \mathfrak{S}_k$ and $I = \sigma^{-1}(J)$, to obtain that

$$\begin{aligned} & \left\| \frac{1}{k!} \sum_{x_1, \dots, x_k \in \Omega_\delta} f(x_1, \dots, x_k) \prod_{j=1}^k \omega_{x_j}^{(j)} \right\|_p \\ & \leq \left(\frac{C_2}{p-1} \right)^k \prod_{j \in J} \|\omega^{(j)}\|_p \prod_{j \notin J} \|\omega^{(j)}\|_q \times N^{|J|(\frac{1}{p}-\frac{1}{q})} \left(\sum_{1 \leq n_1 < \dots < n_k \leq N} |f(n_1, \dots, n_k)|^q \right)^{1/q}, \end{aligned}$$

where we have used the fact that for every fixed $j \in \llbracket 1, k \rrbracket$, the random variables $(\omega_n^{(j)})$ are identically distributed. Notice now that, by symmetry of f and recalling the formula (2.10) of $\|\bar{f}\|_q$, we have

$$N^{|J|(\frac{1}{p}-\frac{1}{q})} \left(\sum_{1 \leq n_1 < \dots < n_k \leq N} |f(n_1, \dots, n_k)|^q \right)^{1/q} = |\Omega|^{J|(\frac{1}{p}-\frac{1}{q})|} (v_\delta)^{-|J|/p} (v_\delta)^{-(k-|J|)/q} \|\bar{f}\|_q,$$

having also used the fact that $|\Omega| = Nv_\delta$. This concludes the proof. $\square$

We are now ready to complete the proof of Theorem 4.1.

Proof of Theorem 4.1. Define, for every $x \in \Omega_\delta$, the truncated random variables

$$(4.9) \quad \hat{\omega}_x = \omega_x \mathbb{1}_{|\omega_x| > V_\delta} - \mathbb{E}[\omega_x \mathbb{1}_{|\omega_x| > V_\delta}],$$

and

$$(4.10) \quad \check{\omega}_x = \omega_x \mathbb{1}_{|\omega_x| \leq V_\delta} - \mathbb{E}[\omega_x \mathbb{1}_{|\omega_x| \leq V_\delta}].$$

In particular, we have $\omega_x = \hat{\omega}_x + \check{\omega}_x$ and $\mathbb{E}[\hat{\omega}_x] = \mathbb{E}[\check{\omega}_x] = 0$. Expanding the product $\prod_{j=1}^k (\hat{\omega}_{x_j} + \check{\omega}_{x_j})$ and applying Minkowski's inequality, we obtain

$$\left\| \frac{1}{k!} \sum_{x_1, \dots, x_k \in \Omega_\delta} f(x_1, \dots, x_k) \prod_{j=1}^k \omega_{x_j} \right\|_p \leq \sum_{J \subset \llbracket 1, k \rrbracket} \left\| \frac{1}{k!} \sum_{x_1, \dots, x_k \in \Omega_\delta} f(x_1, \dots, x_k) \prod_{j \in J} \hat{\omega}_{x_j} \prod_{j \notin J} \check{\omega}_{x_j} \right\|_p.$$

For every subset $J \subset \llbracket 1, k \rrbracket$, we apply Proposition 4.10, using $\mathbb{L}^p$ estimates for the variables $\hat{\omega}$ and $\mathbb{L}^q$ estimates for the variables $\check{\omega}$ with $p \in (1, \gamma)$ and $q \in (\gamma, 2]$. This gives

$$\left\| \frac{1}{k!} \sum_{x_1, \dots, x_k \in \Omega_\delta} f(x_1, \dots, x_k) \prod_{j \in J} \hat{\omega}_{x_j} \prod_{j \notin J} \check{\omega}_{x_j} \right\|_p \leq \left(\frac{C_2}{p-1} (|\Omega| \vee 1)^{\frac{1}{p}-\frac{1}{q}} \right)^k \|\bar{f}\|_q \left(\frac{\|\hat{\omega}\|_p}{v_\delta^{1/p}} \right)^{|J|} \left(\frac{\|\check{\omega}\|_q}{v_\delta^{1/q}} \right)^{k-|J|}.$$

Then, summing over $J \subset \llbracket 1, k \rrbracket$ yields

$$(4.11) \quad \left\| \frac{1}{k!} \sum_{x_1, \dots, x_k \in \Omega_\delta} f(x_1, \dots, x_k) \prod_{j=1}^k \omega_{x_j} \right\|_p \leq \left(\frac{C_2}{p-1} (|\Omega| \vee 1)^{\frac{1}{p}-\frac{1}{q}} \right)^k \|\bar{f}\|_q \left(\frac{\|\hat{\omega}\|_p}{v_\delta^{1/p}} + \frac{\|\check{\omega}\|_q}{v_\delta^{1/q}} \right)^k.$$

Using the fact that the random variable $|\omega|$ has a tail given by (2.11), one can easily check by properties of regularly varying functions that for any fixed $p < \gamma$ and $q > \gamma$, as $t \rightarrow \infty$ we have

$$\mathbb{E}[|\omega|^p \mathbb{1}_{|\omega| > t}] \sim \frac{\gamma}{\gamma-p} t^{p-\gamma} \varphi(t) \quad \text{and} \quad \mathbb{E}[|\omega|^q \mathbb{1}_{|\omega| \leq t}] \sim \frac{\gamma}{q-\gamma} t^{\gamma-q} \varphi(t).$$

In particular, recalling the definition (2.12) of V_δ , we get that for any $a > 0$

$$(4.12) \quad \begin{aligned} \text{for } 0 < p < \gamma, \quad \mathbb{E}[|\omega|^p \mathbb{1}_{|\omega| > aV_\delta}] &\underset{\delta \rightarrow 0}{=} (1 + o(1)) \frac{\gamma}{\gamma - p} a^{p-\gamma} (V_\delta)^p v_\delta, \\ \text{for } q > \gamma \quad \mathbb{E}[|\omega|^q \mathbb{1}_{|\omega| \leq aV_\delta}] &\underset{\delta \rightarrow 0}{=} (1 + o(1)) \frac{\gamma}{q - \gamma} a^{q-\gamma} (V_\delta)^q v_\delta. \end{aligned}$$

(We have included a parameter $a > 0$ for future use, but in this proof we only use it for $a = 1$.) In particular, we get that there is some constant C (which does not depend on p, q , or γ) such that, for every $\delta > 0$,

$$(4.13) \quad \|\hat{\omega}\|_p \leq \frac{C}{(\gamma - p)^{1/p}} v_\delta^{1/p} V_\delta \quad \text{and} \quad \|\check{\omega}\|_q \leq \frac{C}{(q - \gamma)^{1/q}} v_\delta^{1/q} V_\delta.$$

Plugging these estimates into (4.11) yields the upper bound $(C_0)^k (V_\delta)^k \|\bar{f}\|_q$, with the constant

$$C_0 = \frac{2CC_2}{p-1} \max\{(\gamma - p)^{-1/p}, (q - \gamma)^{-1/q}\} (|\Omega| \vee 1)^{\frac{1}{p} - \frac{1}{q}},$$

as claimed in Remark 2.4. This concludes the proof. $\square$

4.3. Application to continuous chaos: proof of Theorem 4.2. In this section we prove Theorem 4.2. Our strategy is to first establish 4.2 for simple functions and then conclude by a density argument. Fix $k \geq 1$. A simple symmetric function $\mathbf{f} : \Omega^k \rightarrow \mathbb{R}$ is a measurable function of the form

$$(4.14) \quad \mathbf{f} = \sum_{1 \leq i_1, \dots, i_k \leq M} \alpha_{i_1, \dots, i_k} \mathbb{1}_{A_{i_1} \times \dots \times A_{i_k}},$$

where the measurable sets $(A_i)_{1 \leq i \leq M}$ form a partition of Ω , and the coefficients $\alpha_{i_1, \dots, i_k}$ are symmetric and such that $\alpha_{i_1, \dots, i_k} = 0$, if for some $l \neq m$, $i_l = i_m$. In particular, a symmetric simple function vanishes on the diagonals. We set for every $a \in (0, 1)$,

$$\mathbf{X}^{(a)}(\mathbf{f}) = \frac{1}{k!} \int_{\Omega^k} \mathbf{f}(x_1, \dots, x_k) \prod_{j=1}^k \zeta^{(a)}(dx_j).$$

Define, for every $0 \leq a \leq b \leq +\infty$ and $p \geq 1$, the truncated moments of the noise ζ ,

$$(4.15) \quad m_p(a, b) = \int_{\mathbb{R}} |z|^p \mathbb{1}_{a < |z| \leq b} \lambda(dz).$$

In our case, the Lévy measure λ is explicit, and we can compute the truncated moments but the following result is more general (see Theorem 2.11 and the discussion before).

Proposition 4.11. *There exists a universal constant C_3 such that, for every $1 < p < q \leq 2$, for every $a \in (0, 1)$ and every simple symmetric function $\mathbf{f} : \Omega^k \rightarrow \mathbb{R}$,*

$$\left\| \mathbf{X}^{(a)}(\mathbf{f}) \right\|_p \leq \left(\frac{C_3}{p-1} (|\Omega| \vee 1)^{1/p-1/q} \right)^k \left(m_p(1, +\infty)^{1/p} + m_q(a, 1)^{1/q} \right)^k \|\mathbf{f}\|_q.$$

Proof of Theorem 4.2. If the Lévy measure is $\lambda(dz) = (c_+ \mathbb{1}_{z>0} + c_- \mathbb{1}_{z<0}) \gamma |z|^{-1-\gamma} dz$, then for every $p \in (1, \gamma)$ and $q \in (\gamma, 2)$,

$$m_p(1, +\infty) \leq \frac{2\gamma}{\gamma - p}, \quad \text{and} \quad m_q(a, 1) \leq m_q(0, 1) \leq \frac{2\gamma}{q - \gamma}.$$

This proves Theorem 4.2 for simple functions and with a constant which is bounded by C_0 .

It remains to deal with a general function $\mathbf{f} \in L_s^q(\Omega^k)$, using a density argument that we now explain. For any $\varepsilon \in (0, 1)$, let $\mathbf{f}^{(\varepsilon)}$ be a simple symmetric function which approximates $\mathbf{f}$ in $L_s^q(\Omega^k)$ as $\varepsilon \downarrow 0$. Then, applying (4.2) to $\mathbf{f}^{(\varepsilon)} - \mathbf{f}^{(\varepsilon')}$ with $\varepsilon, \varepsilon' \in (0, 1)$, we obtain

$$\left\| \mathbf{X}^{(a)}(\mathbf{f}^{(\varepsilon)}) - \mathbf{X}^{(a)}(\mathbf{f}^{(\varepsilon')}) \right\|_p \leq (C_0)^k \|\mathbf{f}^{(\varepsilon)} - \mathbf{f}^{(\varepsilon')}\|_q.$$

Since $\mathbf{f}^{(\varepsilon)}$ converges in $L_s^q(\Omega^k)$ as $\varepsilon \downarrow 0$, the family of random variables $\mathbf{X}^{(a)}(\mathbf{f}^{(\varepsilon)})$ is a Cauchy sequence in $\mathbb{L}^p$. Therefore, it converges to a random variable $\mathbf{X}^{(a)}(\mathbf{f}) := \frac{1}{k!} \int_{\Omega^k} \mathbf{f}(x_1, \dots, x_k) \prod_{j=1}^k \zeta^{(a)}(dx_j)$ for

which (4.2) holds. The definition of $\mathbf{X}^{(a)}(\mathbf{f})$ does not depend on the choice of the approximations $\mathbf{f}^{(\varepsilon)}$ nor on p . $\square$

Proof of Proposition 4.11. Denote for every $i \in \llbracket 1, k \rrbracket$, $\omega_i = \zeta^{(a)}(A_i)$. The properties of the coefficients $\alpha_{i_1, \dots, i_k}$ imply that

$$\begin{aligned} \frac{1}{k!} \int_{\Omega^k} \mathbf{f}(x_1, \dots, x_k) \prod_{j=1}^k \zeta^{(a)}(dx_j) &= \frac{1}{k!} \sum_{1 \leq i_1, \dots, i_k \leq M} \alpha_{i_1, \dots, i_k} \prod_{j=1}^k \zeta^{(a)}(A_{i_j}) \\ &= \sum_{1 \leq i_1 < \dots < i_k \leq M} \alpha_{i_1, \dots, i_k} \prod_{j=1}^k \omega_{i_j}. \end{aligned}$$

Define, for every $i \in \llbracket 1, k \rrbracket$, the truncated random variables

$$(4.16) \quad \hat{\omega}_i = \sum_{(x, z) \in \Lambda} \mathbb{1}_{A_i}(x) z \mathbb{1}_{|z| > 1} - |A_i| \int_{\mathbb{R}} z \mathbb{1}_{|z| > 1} \lambda(dz),$$

and

$$(4.17) \quad \check{\omega}_i = \sum_{(x, z) \in \Lambda} \mathbb{1}_{A_i}(x) z \mathbb{1}_{a < |z| \leq 1} - |A_i| \int_{\mathbb{R}} z \mathbb{1}_{a < |z| \leq 1} \lambda(dz).$$

In particular, we have $\omega_i = \hat{\omega}_i + \check{\omega}_i$ and $\mathbb{E}[\hat{\omega}_i] = \mathbb{E}[\check{\omega}_i] = 0$. Expanding the product $\prod_{j=1}^k (\hat{\omega}_{i_j} + \check{\omega}_{i_j})$ and applying Minkowski's inequality, we obtain

$$\left\| \sum_{1 \leq i_1 < \dots < i_k \leq M} \alpha_{i_1, \dots, i_k} \prod_{j=1}^k \omega_{i_j} \right\|_p \leq \sum_{J \subset \llbracket 1, k \rrbracket} \left\| \sum_{1 \leq i_1 < \dots < i_k \leq M} \alpha_{i_1, \dots, i_k} \prod_{j \in J} \hat{\omega}_{i_j} \prod_{j \notin J} \check{\omega}_{i_j} \right\|_p.$$

For every subset $J \subset \llbracket 1, k \rrbracket$, we apply Corollary 4.6, using $\mathbb{L}^p$ estimates for the variables $\hat{\omega}$ and $\mathbb{L}^q$ estimates for the variables $\check{\omega}$ with $p \in (1, \gamma)$ and $q \in (\gamma, 2]$. This gives

$$(4.18) \quad \left\| \sum_{1 \leq i_1 < \dots < i_k \leq M} \alpha_{i_1, \dots, i_k} \prod_{j \in J} \hat{\omega}_{i_j} \prod_{j \notin J} \check{\omega}_{i_j} \right\|_p \leq \left(\frac{C_2}{p-1} \right)^k \left(\sum_{i_j, j \in J} \prod_{j \in J} \|\hat{\omega}_{i_j}\|_p^p \left(\sum_{\substack{i_j, j \notin J \\ 1 \leq i_1 < \dots < i_k \leq M}} |\alpha_{i_1, \dots, i_k}|^q \prod_{j \notin J} \|\check{\omega}_{i_j}\|_q^q \right)^{p/q} \right)^{1/p}.$$

It is proved in [BCL23, Section 6.2] that there is a universal constant C such that for every measurable function $g : \Omega \rightarrow \mathbb{R}$, every $p \in (1, 2]$ and $0 < a \leq b \leq +\infty$,

$$\left\| \sum_{(x, z) \in \Lambda} g(x) z \mathbb{1}_{a < |z| \leq b} - \int_{\Omega \times \mathbb{R}} g(x) z \mathbb{1}_{a < |z| \leq b} dx \lambda(dz) \right\|_p \leq C \left(\int_{\Omega \times \mathbb{R}} |g(x)|^p |z|^p \mathbb{1}_{a < |z| \leq b} dx \lambda(dz) \right)^{1/p}.$$

This inequality can also be seen as the Poisson analogue of Corollary 4.4 (for an order 1 chaos). Therefore, for every $i \in \llbracket 1, k \rrbracket$,

$$\|\hat{\omega}_i\|_p \leq C |A_i|^{1/p} m_p(1, +\infty)^{1/p}, \quad \text{and} \quad \|\check{\omega}_i\|_q \leq C |A_i|^{1/q} m_q(a, 1)^{1/q}.$$

Plugging these estimates into (4.18) yields,

$$\begin{aligned} \left\| \sum_{1 \leq i_1 < \dots < i_k \leq M} \alpha_{i_1, \dots, i_k} \prod_{j \in J} \hat{\omega}_{i_j} \prod_{j \notin J} \check{\omega}_{i_j} \right\|_p &\leq \left(\frac{C'_2}{p-1} \right)^k m_p(1, +\infty)^{|J|/p} m_q(a, 1)^{k-|J|/q} \left(\sum_{i_j, j \in J} \prod_{j \in J} |A_{i_j}| \left(\sum_{\substack{i_j, j \notin J \\ 1 \leq i_1 < \dots < i_k \leq M}} |\alpha_{i_1, \dots, i_k}|^q \prod_{j \notin J} |A_{i_j}| \right)^{p/q} \right)^{1/p}. \end{aligned}$$

Applying Jensen's inequality to the first sum, we finally obtain,

$$\left\| \sum_{1 \leq i_1 < \dots < i_k \leq M} \alpha_{i_1, \dots, i_k} \prod_{j \in J} \hat{\omega}_{i_j} \prod_{j \notin J} \check{\omega}_{i_j} \right\|_p \leq \left(\frac{C'_2}{p-1} \right)^k m_p(1, +\infty)^{|J|/p} m_q(a, 1)^{k-|J|/q} |\Omega|^{|J|(1/p-1/q)} \left(\sum_{1 \leq i_1 < \dots < i_k \leq M} |\alpha_{i_1, \dots, i_k}|^q \prod_{j=1}^k |A_{i_j}| \right)^{1/q}.$$

Summing over $J \subset \llbracket 1, k \rrbracket$ concludes the proof since the symmetric L^q norm of $\mathbf{f}$ is given by

$$\|\mathbf{f}\|_q = \left(\frac{1}{k!} \sum_{1 \leq i_1, \dots, i_k \leq M} |\alpha_{i_1, \dots, i_k}|^q \prod_{j=1}^k |A_{i_j}| \right)^{1/q}.$$

□

5. MULTIPLE LÉVY INTEGRALS AND SCALING LIMITS OF POLYNOMIAL CHAOS

This section is dedicated to the proof of Theorems 2.3 and 2.5. One of the strengths of our method, compared with the proof of Berger and Lacoïn in [BL21] and [BL22], is that we can proceed chaos by chaos. This was true for the $\mathbb{L}^p$ estimates, as illustrated in Section 4, but it also holds for scaling limits. We therefore postpone the full proof of Theorems 2.3 and 2.5 to the last Section 5.3. Before that, in Sections 5.1 and 5.3, we work with a chaos of a fixed order $k \geq 1$.

5.1. Defining the continuous chaos of order k . Let $k \geq 1$, $q \in (\gamma, 2)$ and let $\mathbf{f} \in L_s^q(\Omega^k)$ be a symmetric function. Recall the definition (2.1) of the truncated noise $\zeta^{(a)}$, built thanks to a Poisson point process Λ on $\mathbb{R} \times \Omega$ of intensity $\mu(dx dz) = dx \lambda(dz)$ with $\lambda(dz) = (c_+ \mathbb{1}_{z>0} + c_- \mathbb{1}_{z<0}) \gamma |z|^{-1-\gamma} dz$.

Denote, for every $a \in (0, 1]$,

$$(5.1) \quad \mathbf{X}^{(a)}(\mathbf{f}) = \frac{1}{k!} \int_{\Omega^k} \mathbf{f}(x_1, \dots, x_k) \prod_{j=1}^k \zeta^{(a)}(dx_j),$$

which is measurable with respect to $\mathcal{F}^{(a)} := \sigma(\Lambda \cap (\Omega \times (-\infty, a] \cup [a, \infty))$.

Proposition 5.1. *The process $(\mathbf{X}^{(a)}(\mathbf{f}))_{a \in (0, 1]}$ is a time-reversed càdlàg martingale with respect to the filtration $\mathcal{F} = (\mathcal{F}^{(a)})_{a \in (0, 1]}$. Furthermore it is bounded in $\mathbb{L}^p$ for every $p \in (1, \gamma)$ and, for every $a \in (0, 1]$ we have,*

$$(5.2) \quad \|\mathbf{X}^{(a)}(\mathbf{f})\|_p \leq (C_0)^k \|\mathbf{f}\|_q,$$

where the constant $C_0 = C_0(p, q, \gamma, \Omega)$ is given by Theorem 4.1. In particular, we have the following limit $\mathbf{X}(\mathbf{f}) := \lim_{a \downarrow 0} \mathbf{X}^{(a)}(\mathbf{f})$ $\mathbb{P}$ -a.s. and in $\mathbb{L}^p$.

Proof. The function $\mathbf{f}$ has a finite L_s^q norm therefore, according to Theorem 4.2, the random variable $\mathbf{X}^{(a)}(\mathbf{f})$ is well-defined and the family $(\mathbf{X}^{(a)}(\mathbf{f}))_{a \in (0, 1]}$ is bounded in $\mathbb{L}^p$, with $\|\mathbf{X}^{(a)}(\mathbf{f})\|_p \leq (C_0)^k \|\mathbf{f}\|_q$.

It only remains to show that it is a martingale with respect to the filtration $(\mathcal{F}^{(a)})_{a \in (0, 1]}$. Recalling the definition (2.1), let us start by writing

$$\mathbf{X}^{(a)}(\mathbf{f}) = \frac{1}{k!} \int_{\Omega^k \times \mathbb{R}^k} \mathbf{f}(x_1, \dots, x_k) \prod_{j=1}^k z_j \mathbb{1}_{|z_j| \geq a} \prod_{j=1}^k (\delta_\Lambda - \mu)(dx_j dz_j).$$

Now, for $b \in (a, 1]$, write $z \mathbb{1}_{|z| \geq a} = z \mathbb{1}_{|z| \geq b} + z \mathbb{1}_{a \leq |z| < b}$ and expand the product. Note that the terms of the form $z_j \mathbb{1}_{|z_j| \geq b}$, when integrated against the point process Λ are measurable with respect to $\mathcal{F}^{(b)}$ and those of the form $z_j \mathbb{1}_{a \leq |z_j| < b}$ are independent of $\mathcal{F}^{(b)}$. Therefore, we have that

$$\begin{aligned} \mathbb{E}[\mathbf{X}^{(a)}(\mathbf{f}) \mid \mathcal{F}^{(b)}] &= \frac{1}{k!} \sum_{I \subset \llbracket 1, k \rrbracket} \int_{\Omega^{|I|} \times \mathbb{R}^{|I|}} \mathbb{E} \left[\int_{\Omega^{k-|I|} \times \mathbb{R}^{k-|I|}} \mathbf{f}(x_1, \dots, x_k) \prod_{i \notin I} z_i \mathbb{1}_{a \leq |z_i| < b} (\delta_\Lambda - \mu)(dx_i dz_i) \right] \\ &\quad \cdot \prod_{i \in I} z_i \mathbb{1}_{|z_i| \geq b} \prod_{i \in I} (\delta_\Lambda - \mu)(dx_i dz_i). \end{aligned}$$

By Mecke's multivariate formula [LP18, Theorem 4.4], when $I \neq \llbracket 1, k \rrbracket$ one has

$$\mathbb{E} \left[\int_{\Omega^{k-|I|} \times \mathbb{R}^{k-|I|}} \mathbf{f}(x_1, \dots, x_k) \prod_{i \notin I} z_i \mathbb{1}_{a \leq |z_i| < b} (\delta_\Lambda - \mu)(dx_i dz_i) \right] = 0.$$

Therefore only the term $I = \llbracket 1, k \rrbracket$ in the above sum remains, which yields $\mathbb{E}[\mathbf{X}^{(a)}(\mathbf{f}) \mid \mathcal{F}^{(b)}] = \mathbf{X}^{(b)}(\mathbf{f})$ and concludes the proof. $\square$

5.2. Scaling limit of a discrete chaos of order k . We now address the question of convergence of discrete multiple integrals to their continuous counterparts. Let us denote, for any function $f : \Omega_\delta^k \rightarrow \mathbb{R}$,

$$X_\delta(f) := \frac{1}{k!} \sum_{x_1, \dots, x_k \in \Omega_\delta} f(x_1, \dots, x_k) \prod_{j=1}^k \frac{\omega_{x_j}}{V_\delta} = \frac{1}{k!} \int_{x_1, \dots, x_k \in \Omega} \bar{f}(x_1, \dots, x_k) \prod_{j=1}^k \zeta_\delta(dx_j),$$

where for the second identity we recall the definition (1.12) of the discrete noise ζ_δ and $\bar{f}$ denotes the piecewise extension (constant on every cell) of f to Ω^k .

Proposition 5.2. *Assume that ω is centered and satisfies (2.11) with $\gamma \in (1, 2)$. Let $\mathbf{f} : \Omega^k \rightarrow \mathbb{R}$ be a symmetric function and let for every $\delta > 0$, $f_\delta : \Omega_\delta^k \rightarrow \mathbb{R}$ be a symmetric function that vanishes on the diagonals, and assume that there is some $q \in (\gamma, 2]$ such that*

$$\lim_{\delta \downarrow 0} \|\mathbf{f} - \bar{f}_\delta\|_q = 0.$$

Then for every $s > \frac{D}{2}$, we have the following joint convergence in distribution in $H_{\text{loc}}^{-s}(\Omega) \times \mathbb{R}$: as $\delta \downarrow 0$,

$$(\zeta_\delta, X_\delta(f_\delta)) \xrightarrow{(d)} (\zeta, \mathbf{X}(\mathbf{f})),$$

where ζ is a γ -stable Lévy noise and $\mathbf{X}(\mathbf{f}) = \int_{\Omega^k} \mathbf{f}(x_1, \dots, x_k) \prod_{j=1}^k \zeta(dx_j)$ is defined in Proposition 5.1 (and is ζ -measurable). The conclusion extends to the joint convergence of a finite collection of chaos $(X_\delta(f_{j,\delta}))_{1 \leq j \leq M}$, provided for each $j \in \llbracket 1, M \rrbracket$, $f_{j,\delta} : \Omega^{k_j} \rightarrow \mathbb{R}$ ($k_j \geq 1$) is symmetric, vanishes on the diagonals, and converges in $L_s^q(\Omega^{k_j})$ to some function $\mathbf{f}_j$.

Proof. Our strategy to prove Proposition 5.2 will rely on two approximations: (i) we approximate the function $\mathbf{f}$ by a smooth (and compactly supported) one; (ii) we approximate the discrete noise ζ_δ by a truncated one. The proof for a single chaos readily extends to the case of several chaos.

For technical reasons, we need to introduce the following notation: given $\mathbf{f} : \Omega^k \rightarrow \mathbb{R}$ and $\delta > 0$, let $\mathbf{f}_\delta$ denote the piecewise constant extension to Ω of $\mathbf{f}|_{\Omega_\delta}$, i.e. $\mathbf{f}_\delta(y) = \mathbf{f}(x)$ for every $y \in \mathcal{C}_\delta(x)$. If the function $\mathbf{f}$ is continuous and compactly supported in Ω^k , $\mathbf{f}_\delta$ converges uniformly to $\mathbf{f}$ as $\delta \downarrow 0$.

(i) *Reducing to smooth functions $\mathbf{f}$ with compact support.* Let $(\mathbf{f}^{(\varepsilon)})_{\varepsilon \in (0,1]}$ be infinitely differentiable functions compactly supported in $\Omega^{k,*} = \{(x_1, \dots, x_k) \in \Omega^k, \forall i \neq j, x_i \neq x_j\}$, which approximate $\mathbf{f}$ in $L_s^q(\Omega^k)$ as $\varepsilon \downarrow 0$ ⁶. Let F be a bounded and Lipschitz function on the space $H_{\text{loc}}^{-s}(\Omega) \times \mathbb{R}$, with Lipschitz constant $\|\nabla F\|_\infty$. Then, we have

$$\begin{aligned} |\mathbb{E}[F(\zeta, \mathbf{X}(\mathbf{f}))] - \mathbb{E}[F(\zeta_\delta, X_\delta(f_\delta))]| &\leq \|\nabla F\|_\infty \mathbb{E} \left[|\mathbf{X}(\mathbf{f}) - \mathbf{X}(\mathbf{f}^{(\varepsilon)})| \right] \\ &\quad + \left| \mathbb{E} \left[F(\zeta, \mathbf{X}(\mathbf{f}^{(\varepsilon)})) \right] - \mathbb{E} \left[F(\zeta_\delta, X_\delta(\mathbf{f}^{(\varepsilon)})) \right] \right| + \|\nabla F\|_\infty \mathbb{E} \left[|X_\delta(\mathbf{f}^{(\varepsilon)}) - X_\delta(f_\delta)| \right]. \end{aligned}$$

By Proposition 5.1, the first term is bounded by $(C_0)^k \|\mathbf{f} - \mathbf{f}^{(\varepsilon)}\|_q$ and by Theorem 4.1 the third term is bounded by $(C_0)^k \|\mathbf{f}_\delta^{(\varepsilon)} - \bar{f}_\delta\|_q$. But by the triangular inequality we have

$$\|\mathbf{f}_\delta^{(\varepsilon)} - \bar{f}_\delta\|_q \leq \|\mathbf{f}_\delta^{(\varepsilon)} - \mathbf{f}\|_q + \|\mathbf{f} - \bar{f}_\delta\|_q.$$

⁶For the existence of such approximating functions $\mathbf{f}^{(\varepsilon)}$, first approximate $\mathbf{f}(x_1, \dots, x_k)$ by $\mathbf{f}(x_1, \dots, x_k) \mathbb{1}_{\forall i \neq j, |x_i - x_j| > \varepsilon}$ and then approximate this last function in $L_s^q(\Omega^{k,\varepsilon})$ where $\Omega^{k,\varepsilon} = \{(x_1, \dots, x_k) \in \Omega^k, \forall i \neq j, |x_i - x_j| > \varepsilon\}$.

Therefore, since $\mathbf{f}_\delta^{(\varepsilon)}$ converges uniformly to $\mathbf{f}^{(\varepsilon)}$, and by assumption $\|\mathbf{f} - \overline{f_\delta}\|_q \rightarrow 0$ as $\delta \downarrow 0$, we have

$$\begin{aligned} \limsup_{\delta \downarrow 0} \left| \mathbb{E}[F(\zeta, \mathbf{X}(\mathbf{f}))] - \mathbb{E}[F(\zeta_\delta, X_\delta(f_\delta))] \right| &\leq 2\|\nabla F\|_\infty \|\mathbf{f} - \mathbf{f}^{(\varepsilon)}\|_q \\ &\quad + \limsup_{\delta \downarrow 0} \left| \mathbb{E}[F(\zeta, \mathbf{X}(\mathbf{f}^{(\varepsilon)}))] - \mathbb{E}[F(\zeta_\delta, X_\delta(\mathbf{f}^{(\varepsilon)}))] \right|. \end{aligned}$$

One therefore only needs to show that, for any fixed $\varepsilon \in (0, 1)$, the last term goes to 0 as $\delta \downarrow 0$. Then, letting $\varepsilon \downarrow 0$ afterwards gives the conclusion.

In other words, it is enough to show that for any infinitely differentiable and compactly supported function $\mathbf{f}$, we have the following convergence in distribution: as $\delta \downarrow 0$

$$(5.3) \quad (\zeta_\delta, X_\delta(\mathbf{f})) \xrightarrow{(d)} (\zeta, \mathbf{X}(\mathbf{f})).$$

(ii) *Truncating the noise.* In order to prove (5.3), let us introduce a cut-off on the random variables ω . For $a > 0$ and $x \in \Omega_\delta$, define

$$(5.4) \quad \omega_x^{(a)} = \omega_x \mathbf{1}_{|\omega_x| > aV_\delta} - \kappa_\delta(a),$$

with $\kappa_\delta(a) = \mathbb{E}[\omega \mathbf{1}_{|\omega| > aV_\delta}]$. The random variables $(\omega_x^{(a)})_{x \in \Omega_\delta}$ are i.i.d. and centered. Then define, for every $\delta > 0$ and $a \in (0, 1)$:

$$X_\delta^{(a)}(\mathbf{f}) = \frac{1}{k!} \sum_{x_1, \dots, x_k \in \Omega_\delta} \mathbf{f}(x_1, \dots, x_k) \prod_{j=1}^k \frac{\omega_{x_j}^{(a)}}{V_\delta},$$

Lemma 5.3. *If $\mathbf{f}$ is a smooth function with compact support in $\Omega^{k,*}$, the following uniform approximation holds:*

$$\lim_{a \downarrow 0} \limsup_{\delta \downarrow 0} \|X_\delta(\mathbf{f}) - X_\delta^{(a)}(\mathbf{f})\|_1 = 0.$$

Let us first see how this lemma implies (5.3), i.e. Proposition 5.2. Let F be a bounded and Lipschitz function on the space $H_{\text{loc}}^{-s}(\Omega) \times \mathbb{R}$, with Lipschitz constant $\|\nabla F\|_\infty$. Then

$$\begin{aligned} \left| \mathbb{E}[F(\zeta_\delta, X_\delta(\mathbf{f}))] - \mathbb{E}[F(\zeta, \mathbf{X}(\mathbf{f}))] \right| &\leq \|\nabla F\|_\infty \mathbb{E}[|X_\delta(\mathbf{f}) - X_\delta^{(a)}(\mathbf{f})|] \\ &\quad + \left| \mathbb{E}[F(\zeta_\delta, X_\delta^{(a)}(\mathbf{f}))] - \mathbb{E}[F(\zeta, \mathbf{X}^{(a)}(\mathbf{f}))] \right| + \|\nabla F\|_\infty \mathbb{E}[|\mathbf{X}^{(a)}(\mathbf{f}) - \mathbf{X}(\mathbf{f})|]. \end{aligned}$$

By Proposition A.3, the second term goes to 0 as $\delta \downarrow 0$, so we obtain

$$\begin{aligned} \limsup_{\delta \downarrow 0} \left| \mathbb{E}[F(\zeta_\delta, X_\delta(\mathbf{f}))] - \mathbb{E}[F(\zeta, \mathbf{X}(\mathbf{f}))] \right| \\ \leq \|\nabla F\|_\infty \limsup_{\delta \downarrow 0} \|X_\delta(\mathbf{f}) - X_\delta^{(a)}(\mathbf{f})\|_1 + \|\nabla F\|_\infty \|\mathbf{X}^{(a)}(\mathbf{f}) - \mathbf{X}(\mathbf{f})\|_1. \end{aligned}$$

Then, Lemma 5.3 shows that the first term vanishes as $a \downarrow 0$. Additionally, Proposition 5.1 shows that $\lim_{a \downarrow 0} \|\mathbf{X}^{(a)}(\mathbf{f}) - \mathbf{X}(\mathbf{f})\|_1 = 0$, which concludes the proof of (5.3) and thus of Proposition 5.2.

Proof of Lemma 5.3. Again, we need to decompose the random variables ω into two (centered) components: recalling the definition (5.4) of $\omega_x^{(a)}$, we have

$$\omega_x = \omega_x^{(a)} + \check{\omega}_x^{(a)},$$

with $\check{\omega}_x^{(a)} = \omega_x \mathbf{1}_{|\omega_x| \leq aV_\delta} + \kappa_\delta(a)$. Then, developing the product $\prod_{j=1}^k (\omega_{x_j}^{(a)} + \check{\omega}_{x_j}^{(a)})$ yields

$$X_\delta(\mathbf{f}) - X_\delta^{(a)}(\mathbf{f}) = \sum_{J \subsetneq \llbracket 1, k \rrbracket} \frac{1}{(V_\delta)^k} \sum_{x_1, \dots, x_k \in \Omega_\delta} f_\delta(x_1, \dots, x_k) \prod_{j \in J} \omega_{x_j}^{(a)} \prod_{j \notin J} \check{\omega}_{x_j}^{(a)}.$$

By Jensen's and Minkowski inequality, we obtain for $p > 1$

$$\|X_\delta(\mathbf{f}) - X_\delta^{(a)}(\mathbf{f})\|_1 \leq \sum_{J \subsetneq \llbracket 1, k \rrbracket} \frac{1}{(V_\delta)^k} \left\| \sum_{x_1, \dots, x_k \in \Omega_\delta} \mathbf{f}(x_1, \dots, x_k) \prod_{j \in J} \omega_{x_j}^{(a)} \prod_{j \notin J} \check{\omega}_{x_j}^{(a)} \right\|_p.$$

Choose $p \in (1, \gamma)$ and $q \in (\gamma, 2]$ such that $\frac{p(q-\gamma)}{\gamma(q-p)} > \frac{k-1}{k}$ (i.e. p sufficiently close to γ). For every $J \subset \llbracket 1, k \rrbracket$, we apply Proposition 4.10 with $\mathbb{L}^p$ estimates for the random variables $\omega^{(a)}$ and $\mathbb{L}^q$ estimates for the random variables $\check{\omega}^{(a)}$. Thus, the terms in the sum above are bounded by

$$\frac{1}{(V_\delta)^k} \left(\frac{C_2}{p-1} (|\Omega| \vee 1)^{\frac{1}{p}-\frac{1}{q}} \right)^k \|\mathbf{f}_\delta\|_q \left(\frac{\|\omega^{(a)}\|_p}{v_\delta^{1/p}} \right)^{|J|} \left(\frac{\|\check{\omega}^{(a)}\|_q}{v_\delta^{1/q}} \right)^{k-|J|},$$

where we recall that $\mathbf{f}_\delta$ denotes the piecewise constant extension to Ω of $\mathbf{f}|_{\Omega_\delta}$, i.e. $\mathbf{f}_\delta(y) = \mathbf{f}(x)$ for every $y \in \mathcal{C}_\delta(x)$.

Additionally, using (4.12), we get that there is a constant $C > 0$ such that for any $a \in (0, 1)$, for δ small enough we have

$$\frac{\|\omega^{(a)}\|_p}{v_\delta^{1/p} V_\delta} \leq \frac{C}{(\gamma-p)^{1/p}} a^{1-\frac{\gamma}{p}} \quad \text{and} \quad \frac{\|\check{\omega}^{(a)}\|_q}{v_\delta^{1/q} V_\delta} \leq \frac{C}{(q-\gamma)^{1/q}} a^{1-\frac{\gamma}{q}}.$$

All together, we obtain that

$$\begin{aligned} \|X_\delta(\mathbf{f}) - X_\delta^{(a)}(\mathbf{f})\|_1 &\leq (C')^k a^{(1-\frac{\gamma}{q})k} \|\mathbf{f}_\delta\|_q \sum_{J \subset \llbracket 1, k \rrbracket} a^{-(\frac{\gamma}{p}-\frac{\gamma}{q})|J|} \\ &\leq (C')^k a^{(1-\frac{\gamma}{q})k - (\frac{\gamma}{p}-\frac{\gamma}{q})(k-1)} \|\mathbf{f}^{(\delta)}\|_q. \end{aligned}$$

where for the second inequality we have used that $\frac{1}{p} - \frac{1}{q} > 0$ and that $|J| \leq k-1$ in the sum. This concludes the proof, since we took p sufficiently close to 1 so that $\frac{p(q-\gamma)}{\gamma(q-p)} > \frac{k-1}{k}$, i.e. $(1 - \frac{\gamma}{q})k > (\frac{\gamma}{p} - \frac{\gamma}{q})(k-1)$. $\square$

$\square$

5.3. Polynomial chaos: proof of Theorems 2.3 and 2.5. We conclude this section with the proof of Theorems 2.3 and 2.5. The proofs are straightforward and consist of assembling the results from Section 4 with those of this Section.

Proof of Theorem 2.3. Let $\psi^{(k)}$, $k \geq 0$, be symmetric functions defined respectively on Ω^k such that (2.6) holds. Then, by Theorem 4.2, for every $a \in (0, 1]$ the random variable $\Phi^{(a)}$ from (2.3) is well-defined and for every $p \in (1, \gamma)$ we have

$$\|\Phi^{(a)}\|_p \leq \sum_{k \geq 0} C_0^k \|\psi^{(k)}\|_q < +\infty.$$

(The fact that the sum is finite comes from the assumption (2.6).) Additionally, by Proposition 5.1, $(\Phi^{(a)})_{a \in (0, 1]}$ is a càdlàg time-reversed martingale with respect to the filtration $(\mathcal{F}_a)_{a \in (0, 1]}$. Since for $p \in (1, \gamma)$ it is bounded in $\mathbb{L}^p$, it converges a.s. and in $\mathbb{L}^p$ to some well-defined random variable Φ . $\square$

We now move to the proof of Theorem 2.5.

Proof of Theorem 2.5. Let $\psi_\delta^{(k)}$, $k \geq 0$, be symmetric functions defined on Ω_δ^k satisfying the assumptions of Theorem 2.5. For every $k \geq 0$, denote $\psi^{(k)}$ the limit of $\bar{\psi}_\delta^{(k)}$ as $\delta \downarrow 0$. Then Assumption (ii) tells that for every $C \geq 0$ we have

$$\sum_{k \geq 0} C^k \|\psi^{(k)}\|_q < +\infty.$$

Therefore, the family of functions $(\psi^{(k)})_{k \geq 0}$ satisfies the assumption of Theorem 2.3, which ensures that the following continuous chaos expansion is well-defined:

$$\Phi = \sum_{k \geq 0} \frac{1}{k!} \int_{\Omega^k} \psi^{(k)}(x_1, \dots, x_k) \prod_{j=1}^k \zeta(dx_j).$$

We are going to show that Φ_δ converges in law to Φ . The proof essentially consists in proving the term by term convergence of the chaos. First, let us truncate the expansion: for $M \in \mathbb{N}$, define

$$\Phi_\delta^{(M)} := \sum_{k=0}^M \frac{1}{k!} \int_{\Omega^k} \psi_\delta^{(k)}(x_1, \dots, x_k) \prod_{j=1}^k \zeta_\delta(dx_j), \quad \text{and} \quad \Phi^{(M)} := \sum_{k=0}^M \frac{1}{k!} \int_{\Omega^k} \psi^{(k)}(x_1, \dots, x_k) \prod_{j=1}^k \zeta(dx_j).$$

By Theorem 4.1, for $p \in (1, \gamma)$ we have

$$\|\Phi_\delta - \Phi_\delta^{(M)}\|_p \leq \sum_{k>M} (C_0)^k \|\psi_\delta^{(k)}\|_q.$$

Hence, using Assumption (ii), we obtain

$$(5.5) \quad \lim_{M \rightarrow +\infty} \limsup_{\delta \downarrow 0} \|\Phi_\delta - \Phi_\delta^{(M)}\|_p = 0.$$

Similarly, using Theorem 4.2, for the continuous counterpart and Assumption (ii), we obtain

$$(5.6) \quad \lim_{M \rightarrow +\infty} \|\Phi - \Phi^{(M)}\|_p = 0.$$

Finally, Proposition 5.2 shows that the following convergence in distribution holds: for any fixed M , as $\delta \downarrow 0$,

$$\begin{aligned} & \left(\zeta_\delta, \int_{\Omega} \psi_\delta^{(1)}(x) \zeta_\delta(dx), \dots, \int_{\Omega^M} \psi_\delta^{(M)}(x_1, \dots, x_M) \prod_{j=1}^M \zeta_\delta(dx_j) \right) \\ & \xrightarrow{(d)} \left(\zeta, \int_{\Omega} \psi^{(1)}(x) \zeta(dx), \dots, \int_{\Omega^M} \psi^{(M)}(x_1, \dots, x_M) \prod_{j=1}^M \zeta(dx_j) \right). \end{aligned}$$

Therefore $\Phi_\delta^{(M)}$ converges in distribution as $\delta \downarrow 0$ to $\Phi^{(M)}$, jointly with the noise. Together with (5.5) and (5.6), this concludes the proof of Theorem 2.5. $\square$

6. CONVERGENCE OF THE DISORDERED GIBBS MEASURE AND DISORDER RELEVANCE

In this section, we prove the convergence of the disordered Gibbs measure, *i.e.* Theorem 2.7, in Section 6.1. We also address the question of disorder relevance, *i.e.* Theorem 2.10, in Section 6.2

6.1. Proof of Theorem 2.7. For every bounded, continuous function G on E , define

$$Z_{\Omega_\delta}^{\omega, \beta_\delta}(G) := \mathbb{E}_\delta^{\text{ref}} \left[G(\sigma) \prod_{x \in \Omega_\delta} (1 + \beta_\delta \omega_x) \right],$$

so that with this notation we have $P_{\Omega_\delta}^{\omega, \beta_\delta}(G) = Z_{\Omega_\delta}^{\omega, \beta_\delta}(G) / Z_{\Omega_\delta}^{\omega, \beta_\delta}$. Given a distribution $T \in \mathcal{D}'(\Omega)$ and a test function $\mathbf{f} \in \mathcal{D}(\Omega)$ (infinitely differentiable with compact support), we denote by $\langle T, \mathbf{f} \rangle$ the action of T on $\mathbf{f}$.

The following lemma is adapted from [BL21] and stated in a slightly more general context. For the sake of completeness, we include the proof here. It provides sufficient conditions for the discrete probability measure $P_{\Omega_\delta}^{\omega, \beta_\delta}$ to admit a scaling limit. An important feature of the proof is that it does not require the state space Ω to be bounded; this allows us to apply it to the long-range polymer model in the proof of Theorem 3.6.

Lemma 6.1. *Suppose that there exist β_δ such that the following conditions are satisfied:*

- (a) $P_{\Omega_\delta}^{\text{ref}}$ converges in $\mathcal{M}_1(E)$ to a probability distribution $\mathbf{P}_\Omega$;
- (b) for every function $G \in \mathcal{C}_b(E)$, for every $\mathbf{f} \in \mathcal{D}(\Omega)$, there exist a random variable Z_Ω^ζ measurable with respect to ζ such that, as $\delta \downarrow 0$,

$$(\langle \zeta_\delta, \mathbf{f} \rangle, Z_{\Omega_\delta}^{\omega, \beta_\delta}(G)) \xrightarrow{(d)} (\langle \zeta, \mathbf{f} \rangle, Z_\Omega^\zeta(G)).$$

- (c) The random variable $Z_\Omega^\zeta(1)$ is almost surely strictly positive.

Then there exist a random probability measure $\mathbf{P}_\Omega^\zeta \in \mathcal{M}_1(E)$ such that we have the following joint convergence in distribution in $H_{\text{loc}}^{-s}(\Omega) \times \mathcal{M}_1(E)$, $s > \frac{D}{2}$,

$$(\zeta_\delta, \mathbf{P}_{\Omega_\delta}^{\omega, \beta_\delta}) \xrightarrow{(d)} (\zeta, \mathbf{P}_\Omega^\zeta) \quad \text{as } \delta \downarrow 0.$$

Proof. First we prove convergence of finite-dimensional marginals. Since ζ_δ and $\mathbf{P}_{\Omega_\delta}^{\omega, \beta_\delta}$ can be both viewed as linear forms, it is actually enough to prove convergence of one-dimensional marginals, which is exactly condition (b).

Second, we prove tightness of the couple $(\zeta_\delta, \mathbf{P}_{\Omega_\delta}^{\omega, \beta_\delta})$. The family of random distributions $\{\zeta_\delta\}_{\delta \in (0,1)}$ is tight in $H_{\text{loc}}^{-s}(\Omega)$, see Theorem A.2 in Appendix. Therefore, it only remains to prove that the family of random probability measures $(\mathbf{P}_{\Omega_\delta}^{\omega, \beta_\delta})_{\delta \in (0,1)}$ is tight. By conditions (b) and (c), $Z_{\Omega_\delta}^{\omega, \beta_\delta} \xrightarrow{(d)} Z_\Omega^{\zeta, \hat{\beta}}$ with $Z_\Omega^{\zeta, \hat{\beta}} > 0$ almost surely, hence there exists a sequence $(\varepsilon_m)_{m \geq 0}$ going to zero such that, for all $\delta > 0$ and $m \geq 0$

$$\mathbb{P}[Z_{\Omega_\delta}^{\omega, \beta_\delta} \leq \varepsilon_m] \leq 2^{-m}.$$

Since by assumption (a), the family of probability measures $\mathbf{P}_\delta^{\text{ref}}$ converge as $\delta \downarrow 0$ in the space $\mathcal{M}_1(E)$, it is a tight family of $\mathcal{M}_1(E)$. We can therefore find compact subsets $(K_m)_{m \geq 1}$ of E such that for all $\delta > 0$ and $m \geq 0$,

$$\mathbf{P}_\delta^{\text{ref}}(K_m) \geq 1 - \frac{\varepsilon_m}{4^m}.$$

Now let us define, for every $n \geq 1$

$$\mathcal{K}_n = \{\mu \in \mathcal{M}_1(E) ; \forall m > n, \mu(K_m^c) \leq 2^{-m}\}.$$

The set $\mathcal{K}_n \subset \mathcal{M}_1(E)$ is closed, and any sequence in $\mathcal{K}_n$ is tight, therefore it is a compact subset of $\mathcal{M}_1(E)$. Additionally, by subadditivity, we have

$$\mathbb{P}[\mathbf{P}_{\Omega_\delta}^{\omega, \beta_\delta} \notin \mathcal{K}_n] \leq \sum_{m > n} \mathbb{P}[\mathbf{P}_{\Omega_\delta}^{\omega, \beta_\delta}(K_m^c) > 2^{-m}].$$

Using Markov inequality and the fact that $\mathbb{E}[Z_{\Omega_\delta}^{\omega, \beta_\delta}(\mathbb{1}_{K_m^c})] = \mathbf{P}_\delta^{\text{ref}}(K_m^c) \leq \varepsilon_m 4^{-m}$, we get

$$\begin{aligned} \mathbb{P}[\mathbf{P}_{\Omega_\delta}^{\omega, \beta_\delta}(K_m^c) > 2^{-m}] &\leq \mathbb{P}[Z_{\Omega_\delta}^{\omega, \beta_\delta}(\mathbb{1}_{K_m^c}) > \varepsilon_m 2^{-m}] + \mathbb{P}[Z_{\Omega_\delta}^{\omega, \beta_\delta} \leq \varepsilon_m] \\ &\leq 2^m \varepsilon_m^{-1} \mathbf{P}_\delta^{\text{ref}}(K_m^c) + 2^{-m} \leq 2 \cdot 2^{-m}. \end{aligned}$$

This shows that $\sup_{\delta \in (0,1)} \mathbb{P}[\mathbf{P}_{\Omega_\delta}^{\omega, \beta_\delta} \notin \mathcal{K}_n] \leq 2 \cdot 2^{-n}$, which can be made arbitrarily small by taking n large. This concludes the proof. $\square$

We now apply Lemma 6.1 to the proof of Theorem 2.7.

Proof of Theorem 2.7. All we need is to check that condition (b) of Lemma 6.1 is satisfied.

Let $G \in \mathcal{C}_b(E)$. Arguing as in (1.7) and (1.13), the random variable $Z_{\Omega_\delta}^{\omega, \beta_\delta}(G)$ has the following polynomial chaos expansion:

$$(6.1) \quad Z_{\Omega_\delta}^{\omega, \beta_\delta}(G) = \mathbf{E}_\delta^{\text{ref}}[G(\sigma)] + \sum_{k \geq 1} \frac{\hat{\beta}^k}{k!} \sum_{x_1, \dots, x_k \in \Omega_\delta} \psi_\delta^{(k)}(x_1, \dots, x_k; G) \prod_{j=1}^k \frac{\omega_{x_j}}{V_\delta},$$

recalling the definition (2.17) of the G -weighted k -point correlation function $\psi_\delta^{(k)}(x_1, \dots, x_k; G)$.

The function G being bounded, we therefore have, for every $k \geq 1$ and $x_1, \dots, x_k \in \Omega_\delta$,

$$|\psi_\delta^{(k)}(x_1, \dots, x_k; G)| \leq \|G\|_\infty \psi_\delta^{(k)}(x_1, \dots, x_k).$$

This implies that the functions $\psi_\delta^{(k)}(\cdot, G)$ satisfy the assumption (ii) of Theorem 2.5. By condition (b), they also satisfy the assumption (i) of Theorem 2.7. Note that the convergence also holds for $k = 0$ thanks to condition (a).

Therefore, Theorem 2.5 applies and yields condition (b), using also that the evaluation against a test function is a continuous functional from $H_{\text{loc}}^{-s}(\Omega)$ to $\mathbb{R}$. $\square$

6.2. Disorder relevance: proof of Theorem 2.10. The proof follows a standard change-of-measure argument. Recall that in Theorem 2.10, we assume that the random variable $J_\delta v_\delta \sum_{x \in \Omega_\delta} \sigma_x$ converges as $\delta \rightarrow 0$ to a strictly positive random variable.

Lemma 6.2. *Assume that $\hat{\beta}_\delta := \beta_\delta V_\delta J_\delta^{-1}$ verifies $\lim_{\delta \downarrow 0} \hat{\beta}_\delta = +\infty$, and define the following event (for the environment ω):*

$$A_\delta = \{\omega : \forall x \in \Omega_\delta, |\omega_x| < a_\delta V_\delta\}, \quad \text{with} \quad a_\delta := (\hat{\beta}_\delta)^{4/(\gamma-1)}.$$

Then, we have that

- (i) $\lim_{\delta \rightarrow 0} \mathbb{P}[(A_\delta)^c] = 0;$
- (ii) $\lim_{\delta \rightarrow 0} \mathbb{E}[Z_{\Omega_\delta}^{\omega, \beta_\delta} \mathbf{1}_{A_\delta}] = 0.$

This implies that $Z_{\Omega_\delta}^{\omega, \beta_\delta} \rightarrow 0$ in probability.

Proof. First of all, notice that by definition we have $\lim_{\delta \downarrow 0} a_\delta = +\infty$ and also $\lim_{\delta \downarrow 0} a_\delta^{-(\gamma-1)/2} \hat{\beta}_\delta = +\infty$. Notice also that by Potter's bound (see [BGT87, Thm 1.5.4]) for any $\varepsilon > 0$, there are constants C_ε and c_ε such that for every $a \geq 1$ and $\delta \in (0, 1)$,

$$(6.2) \quad c(\varepsilon) a^{-\varepsilon} \varphi(V_\delta) \leq \varphi(a V_\delta) \leq C(\varepsilon) a^\varepsilon \varphi(V_\delta).$$

Let us start with (i). Using subadditivity then Potter's bound (6.2) (say with $\varepsilon = \gamma/2$), we have that

$$\mathbb{P}[(A_\delta)^c] \leq \text{Card}(\Omega_\delta) \mathbb{P}[|\omega| \geq a_\delta V_\delta] \leq C a_\delta^{-\gamma/2} |\Omega| \frac{1}{v_\delta} \mathbb{P}[|\omega| \geq V_\delta],$$

where we have also used that $|\Omega| = v_\delta \text{Card}(\Omega_\delta)$. Recalling (2.12) we have that $\frac{1}{v_\delta} \mathbb{P}[|\omega| \geq V_\delta] \rightarrow 1$, so the above goes to 0 as $\delta \downarrow 0$, since $\lim_{\delta \downarrow 0} a_\delta = +\infty$.

For (ii), by Fubini's theorem we have

$$(6.3) \quad \mathbb{E}[Z_{\Omega_\delta}^{\omega, \beta_\delta} \mathbf{1}_{A_\delta}] = \mathbb{E}_\delta^{\text{ref}} \left[\mathbb{E} \left[\mathbf{1}_{A_\delta} \prod_{x \in \Omega_\delta} (1 + \beta_\delta \omega_x \sigma_x) \right] \right] = \mathbb{E}_\delta^{\text{ref}} \left[\mathbb{P}_\delta^\sigma[A_\delta] \right],$$

where, given a realization of the field σ , $\mathbb{P}_\delta^\sigma$ is the size-biased measure defined by

$$\frac{d\mathbb{P}_\delta^\sigma}{d\mathbb{P}}(\omega) = \prod_{x \in \Omega_\delta} (1 + \beta_\delta \omega_x \sigma_x).$$

Note that $\mathbb{P}_\delta^\sigma$ is a probability distribution since $\mathbb{E}[\prod_{x \in \Omega_\delta} (1 + \beta_\delta \omega_x \sigma_x)] = 1$ and that under $\mathbb{P}_\delta^\sigma$, the random variables $(\omega_x)_{x \in \Omega_\delta}$ are independent. Furthermore, if $\sigma_x = 0$, the law of ω_x is the same as under $\mathbb{P}$, and if $\sigma_x = 1$, the law of ω_x is $\tilde{\mathbb{P}}_\delta$, which is defined by $d\tilde{\mathbb{P}}_\delta(\omega) = (1 + \beta_\delta \omega) d\mathbb{P}(\omega)$.

Now, we bound 6.3 by observing that $A_\delta \subset \{\forall x \in \sigma, |\omega_x| \leq a_\delta V_\delta\}$: we get that

$$\begin{aligned} \mathbb{E}[Z_{\Omega_\delta}^{\omega, \beta_\delta} \mathbf{1}_{A_\delta}] &\leq \mathbb{E}_\delta^{\text{ref}} \left[\mathbb{P}_\delta^\sigma[\forall x \in \sigma, |\omega_x| \leq a_\delta V_\delta] \right] = \mathbb{E}_\delta^{\text{ref}} \left[\prod_{x \in \sigma} (1 - \tilde{\mathbb{P}}_\delta(|\omega_x| > a_\delta V_\delta)) \right] \\ &\leq \mathbb{E}_\delta^{\text{ref}} \left[\exp \left(- \tilde{\mathbb{P}}_\delta(|\omega| > a_\delta V_\delta) \sum_{x \in \Omega_\delta} \sigma_x \right) \right], \end{aligned}$$

where, for the last inequality, we have simply used that $1 - x \leq e^{-x}$ for any x . Now, using the definition of $\tilde{\mathbb{P}}_\delta$, we obtain that

$$\tilde{\mathbb{P}}_\delta(|\omega| > a_\delta V_\delta) = \mathbb{P}(|\omega| > a_\delta V_\delta) + \beta_\delta \mathbb{E}[\omega \mathbf{1}_{\{|\omega| > a_\delta V_\delta\}}] \sim \frac{\gamma}{\gamma-1} \beta_\delta \varphi(a_\delta V_\delta) (a_\delta V_\delta)^{-(\gamma-1)} \quad \text{as } \delta \downarrow 0.$$

Together with Potter's lower bound (6.2) (say with $\varepsilon = (\gamma-1)/2$), this yields

$$\tilde{\mathbb{P}}_\delta(|\omega| > a_\delta V_\delta) \geq c' \beta_\delta a_\delta^{-(\gamma-1)/2} \mathbb{P}(|\omega| > V_\delta) \geq c'' \hat{\beta}_\delta a_\delta^{-(\gamma-1)/2} J_\delta v_\delta,$$

recalling the definition of $\hat{\beta}_\delta$ and the fact that $\mathbb{P}[|\omega| \geq V_\delta] \sim v_\delta$ (see (2.12)). We end up with

$$\mathbb{E}[Z_{\Omega_\delta}^{\omega, \beta_\delta} \mathbf{1}_{A_\delta}] \leq \mathbb{E}_\delta^{\text{ref}} \left[\exp \left(- c'' \hat{\beta}_\delta a_\delta^{-(\gamma-1)/2} J_\delta v_\delta \sum_{x \in \Omega_\delta} \sigma_x \right) \right],$$

which goes to 0 as $\delta \downarrow 0$ since $\lim_{\delta \downarrow 0} \hat{\beta}_\delta a_\delta^{-(\gamma-1)/2} = +\infty$ and $J_\delta v_\delta \sum_{x \in \Omega_\delta} \sigma_x$ converges in distribution to a *positive* random variable. This concludes the proof. $\square$

7. PROOF FOR DISORDERED PINNING MODEL

In this section we prove Theorem 3.1 and Theorem 3.4. We recall that we have assumed that $1 - \alpha < \frac{1}{\gamma}$, which is equivalent to $\gamma(1 - \alpha) < 1$.

Consider for $N \in \mathbb{N}$ the discretization $\Omega_N := \{\frac{n}{N}, 1 \leq n \leq N\}$ of $\Omega := (0, 1)$. In this section, instead of working with a continuous discretization parameter δ as in Sections 1 and 2, we prefer the discrete parameter N (with $\delta = N^{-1}$). The volume of each cell is $v_N = N^{-1}$, and recall that V_N is defined by the asymptotic relation (3.8). We also set $\hat{\beta}_N := \beta_N u(N) V_N$, and recall that we assume that $\lim_{N \rightarrow \infty} \hat{\beta}_N = \hat{\beta} \in [0, \infty)$.

7.1. Proof of Theorem 3.1. A consequence of the strong renewal theorem (3.2) is that there exists a constant C such that for every $N \geq 1$,

$$(7.1) \quad \frac{1}{C} \leq u(N) \ell(N) N^{1-\alpha} \leq C.$$

Furthermore, denoting for every $N \geq 1$ and $t \in [0, 1]$,

$$(7.2) \quad U_N(t) := \frac{u(\lfloor Nt \rfloor)}{u(N)},$$

the functions $(U_N)_{N \geq 1}$ converge uniformly on every compact subset of $(0, 1]$ to $U(t) := t^{\alpha-1}$.

Arguing as in (1.4) and (1.13), we rewrite the partition function $Z_{N, h_N}^{\omega, \beta_N}$ as a polynomial chaos with respect to the variables ω :

$$\begin{aligned} Z_{N, h_N}^{\omega, \beta_N} &= 1 + \sum_{k \geq 1} \frac{(\beta_N)^k}{k!} \sum_{t_1, \dots, t_k \in \Omega_N} \mathbb{E}_{N, h_N} \left[\prod_{j=1}^k \mathbb{1}_{Nt_j \in \tau} \right] \prod_{j=1}^k \omega_{Nt_j} \\ &= 1 + \sum_{k \geq 1} \frac{(\hat{\beta}_N)^k}{k!} \sum_{t_1, \dots, t_k \in \Omega_N^k} \psi_N^{(k)}(t_1, \dots, t_k) \prod_{j=1}^k \frac{\omega_{Nt_j}}{V_N}. \end{aligned}$$

Here, for every $k \geq 1$, $\psi_N^{(k)} : (\Omega_N)^k \rightarrow \mathbb{R}$ is the symmetric function which vanishes on the diagonals, defined for $0 = t_0 < t_1 < \dots < t_k \leq 1$ (in Ω_N) by

$$\psi_N^{(k)}(t_1, \dots, t_k) := u(N)^{-k} \mathbb{P}_{N, h_N} [\{Nt_1, \dots, Nt_k\} \subset \tau].$$

Its piecewise constant extension to Ω^k is such that for every $t_0 := 0 < t_1 < \dots < t_k \leq 1$,

$$\bar{\psi}_N^{(k)}(t_1, \dots, t_k) = u(N)^{-k} \mathbb{P}_{N, h_N} [\{\lfloor Nt_1 \rfloor, \dots, \lfloor Nt_k \rfloor\} \subset \tau].$$

We now verify that the family of functions $(\psi_N^{(k)})_{k \geq 1}$ satisfy the assumptions of Theorem 2.5 and identify their scaling limit.

We recall the definition of the discrete homogeneous constrained partition function:

$$Z_{M, h}^c := \mathbb{E} \left[\exp \left(h \sum_{n=1}^M \mathbb{1}_{n \in \tau} \right) \mathbb{1}_{M \in \tau} \right].$$

By the Markov property of the renewal process τ , we obtain that for every $t_0 := 0 < t_1 < \dots < t_k \leq 1$

$$\bar{\psi}_N^{(k)}(t_1, \dots, t_k) = \frac{1}{Z_{N, h_N}} \prod_{j=1}^k \left(\frac{Z_{\lfloor Nt_j \rfloor - \lfloor Nt_{j-1} \rfloor, h_N}^c}{u(N)} \right) Z_{N - \lfloor Nt_k \rfloor, h_N}.$$

For every $\alpha \in (0, 1)$, the (modified) Mittag-Leffler function is defined by

$$E_\alpha(z) := \sum_{k=0}^{+\infty} \frac{z^k \Gamma(\alpha)^k}{\Gamma(\alpha k + 1)}.$$

The proof of the following lemma is essentially technical and is therefore postponed to Appendix B.

Lemma 7.1. *Recall the definitions (3.5) of the (free and conditioned) continuum partition functions $\mathcal{Z}_{t,\hat{h}}$ and $\mathcal{Z}_{t,\hat{h}}^c$. If $\lim_{N \rightarrow +\infty} \frac{M_N}{N} = t \in (0, 1]$, then*

$$\lim_{N \rightarrow +\infty} Z_{M_N, h_N} = \mathcal{Z}_{t,\hat{h}} := E_\alpha(\hat{h}t^\alpha), \quad \text{and} \quad \lim_{N \rightarrow +\infty} \frac{Z_{M_N, h_N}^c}{u(N)} = \mathcal{Z}_{t,\hat{h}}^c := \alpha t^{\alpha-1} E'_\alpha(\hat{h}t^\alpha),$$

These convergences hold uniformly in t on every compact subset of $(0, 1]$. Furthermore there exists a constant $C = C_{\hat{h}}$ such that for every $1 \leq M \leq N$,

$$(7.3) \quad Z_{M, h_N} \leq C \quad \text{and} \quad Z_{M, h_N}^c \leq Cu(M).$$

Therefore, for every $k \geq 1$, $(\bar{\psi}_N^{(k)})_{N \geq 1}$ converges to $\psi_{\hat{h}}^{(k)}$ (defined in (3.6)) uniformly on every compact subset of $\{(t_1, \dots, t_k) \in (0, 1)^k, \forall i \neq j, t_i \neq t_j\}$. Furthermore, the bound (7.3) implies that for every $t_0 := 0 < t_1 < \dots < t_k \leq 1$,

$$\psi_N^{(k)}(t_1, \dots, t_k) \leq C^k \prod_{j=1}^k \frac{u(\lfloor Nt_j \rfloor - \lfloor Nt_{j-1} \rfloor)}{u(N)},$$

which thanks to (7.1) gives

$$(7.4) \quad \psi_N^{(k)}(t_1, \dots, t_k) \leq (C')^k u(N)^{-k} \prod_{j=1}^k \frac{(\lfloor Nt_j \rfloor - \lfloor Nt_{j-1} \rfloor)^{-(1-\alpha)}}{\ell(\lfloor Nt_j \rfloor - \lfloor Nt_{j-1} \rfloor)}.$$

Taking the limit $N \rightarrow +\infty$ yields, for every $k \geq 1$,

$$\psi_{\hat{h}}^{(k)}(t_1, \dots, t_k) \leq (C')^k \prod_{j=1}^k \frac{1}{(t_j - t_{j-1})^{1-\alpha}}.$$

Let $q \in (\gamma, \frac{1}{1-\alpha})$. We now use these inequalities to verify that

1. for every $k \geq 1$, $\psi_{\hat{h}}^{(k)} \in L_s^q((0, 1)^k)$.
 2. the functions $\psi_N^{(k)}$ satisfy hypothesis (ii).
 3. for every $k \geq 1$, $\psi_N^{(k)}$ converges in $L_s^q((0, 1)^k)$ to $\psi_{\hat{h}}^{(k)}$.
1. Denoting $\xi := 1 - q(1 - \alpha) > 0$, we obtain

$$\begin{aligned} \frac{1}{k!} \int_{(0,1)^k} \psi_{\hat{h}}^{(k)}(t_1, \dots, t_k)^q \prod_{j=1}^k dt_j &= \int_{0 < t_1 < \dots < t_k < 1} \psi_{\hat{h}}^{(k)}(t_1, \dots, t_k)^q \prod_{j=1}^k dt_j \\ &\leq (C')^{kq} \int_{0 < t_1 < \dots < t_k < 1} \prod_{j=1}^k (t_j - t_{j-1})^{\xi-1} dt_j. \end{aligned}$$

Therefore, using the standard computation (see [BL22, Lemma A.3]), for $\xi > 0$ we have

$$(7.5) \quad \int_{t_0 := 0 < t_1 < \dots < t_k < t_{k+1} := t} \prod_{j=1}^{k+1} (t_j - t_{j-1})^{\xi-1} \prod_{j=1}^k dt_j = t^{(k+1)\xi-1} \frac{\Gamma(\xi)^{k+1}}{\Gamma((k+1)\xi)},$$

from which we deduce that

$$(7.6) \quad \left(\frac{1}{k!} \int_{(0,1)^k} \psi_{\hat{h}}^{(k)}(t_1, \dots, t_k)^q \prod_{j=1}^k dt_j \right)^{1/q} \leq (C')^k \left(\frac{\Gamma(\xi)^k}{\Gamma(\xi k + 1)} \right)^{1/q}.$$

2. Let us now estimate the L^q moment of $\bar{\psi}_N^{(k)}$ (we recall that the function $\bar{\psi}_N^{(k)}$ vanishes on the diagonals): we have

$$\begin{aligned} \frac{1}{k!} \int_{(0,1)^k} \bar{\psi}_N^{(k)}(t_1, \dots, t_k)^q \prod_{j=1}^k dt_j &= \int_{0 < t_1 < \dots < t_k < 1} \bar{\psi}_N^{(k)}(t_1, \dots, t_k)^q \prod_{j=1}^k dt_j \\ &= \frac{1}{N^k} \sum_{1 \leq n_1 < \dots < n_k \leq N} \psi_N^{(k)}\left(\frac{n_1}{N}, \dots, \frac{n_k - n_{k-1}}{N}\right)^q. \end{aligned}$$

By (7.4), this is bounded by

$$\frac{(C')^k}{N^k u(N)^{qk}} \sum_{1 \leq n_1 < \dots < n_k \leq N} \prod_{j=1}^k \frac{(n_j - n_{j-1})^{\xi-1}}{\ell(n_j - n_{j-1})^q},$$

with $\xi = 1 - q(1 - \alpha) > 0$ as above. Let $\varepsilon > 0$ be such that $\xi > \varepsilon$. Applying Proposition B.1 (say with $\varepsilon = \xi/2$), there exist a constant C such that for every $k \geq 1$ and $N \geq 1$,

$$\sum_{1 \leq n_1 < \dots < n_k \leq N} \prod_{j=1}^k \frac{(n_j - n_{j-1})^{\xi-1}}{\ell(n_j - n_{j-1})^q} \leq \frac{C^k}{\ell(N)^{kq}} N^{k\xi} \int_{0 < t_1 < \dots < t_k < 1} \prod_{j=1}^k (t_j - t_{j-1})^{\frac{1}{2}\xi-1} dt_j.$$

Therefore, using the lower bound of (7.1), we finally obtain

$$\begin{aligned} (7.7) \quad \left(\frac{1}{k!} \int_{(0,1)^k} (\bar{\psi}_N^{(k)}(t_1, \dots, t_k))^q \prod_{j=1}^k dt_j \right)^{1/q} &\leq (C'')^k \left(\int_{0 < t_1 < \dots < t_k < 1} \prod_{j=1}^k (t_j - t_{j-1})^{\frac{1}{2}\xi-1} dt_j \right)^{1/q} \\ &\leq (C'')^k \left(\frac{\Gamma(\frac{1}{2}\xi)^k}{\Gamma(\frac{1}{2}\xi k + 1)} \right)^{1/q}, \end{aligned}$$

with the same calculation as in (7.6). This decreases faster than exponentially as $k \rightarrow +\infty$, so the hypothesis (ii) of Theorem 2.5 is verified.

3. For every $k \geq 1$, the functions $(\psi_N^{(k)})_{N \geq 1}$ converge uniformly on every compact subset of $\{(t_1, \dots, t_k) \in (0, 1)^k, \forall i \neq j, t_i \neq t_j\}$. Furthermore, they are uniformly bounded in $L_s^q((0, 1)^q)$ for every $q \in (\gamma, \frac{1}{1-\alpha})$. By dominated convergence, this shows the convergence in $L_s^q((0, 1)^q)$ for every $q \in (\gamma, \frac{1}{1-\alpha})$. $\square$

7.2. Proof of Theorem 3.3.

We apply the general Theorem 2.7.

It is proved in [Gia07, Theorem A.8] that the rescaled renewal process

$$\tau^{(N)} := \frac{1}{N} \tau \subset \mathbb{R}_+$$

converges in distribution with respect to the Matheron topology, to the α -stable regenerative set $\mathcal{A}_\alpha$. Hence the assumption (a) is verified.

We now verify assumption (b). Let G be a bounded continuous function on the space $\mathcal{C}_\infty$. Adapting the definition 2.15 to the pinning model, we define for every fixed $k \geq 1$, for every $0 < t_1 < \dots < t_k < 1$,

$$\psi_N^{(k)}(t_1, \dots, t_k; G) := u(N)^{-k} \mathbb{E}_{N, h_N} \left[G\left(\frac{\tau}{N}\right) \mathbf{1}_{\lfloor Nt_1 \rfloor, \dots, \lfloor Nt_k \rfloor \in \tau} \right].$$

which can be rewritten as

$$\psi_N^{(k)}(t_1, \dots, t_k; G) = \psi_N^{(k)}(t_1, \dots, t_k) \mathbb{E}_{N, h_N} \left[G\left(\frac{\tau}{N}\right) \mid \lfloor Nt_1 \rfloor, \dots, \lfloor Nt_k \rfloor \in \tau \right].$$

We must show that the symmetric function $\psi_N^{(k)}(\cdot; G)$ converges in $L_s^q((0, 1)^k)$ for every $q \in (\gamma, \frac{1}{1-\alpha})$. Since G is bounded,

$$\psi_N^{(k)}(t_1, \dots, t_k; G) \leq \|G\|_\infty \psi_N^{(k)}(t_1, \dots, t_k),$$

which implies that $\psi_N^{(k)}(\cdot; G)$ is uniformly bounded in $L_s^q((0, 1)^k)$ by (7.7). Moreover, by [CSZ16b, Proposition A.8], for every fixed $T > 0$, the random closed set $\tau^{(N)} \cap [0, T]$ conditioned on $\lfloor NT \rfloor \in \tau$,

converges in distribution (for the Matheron topology) to the α -stable regenerative set $\mathcal{A}_\alpha$, conditioned on $T \in \mathcal{A}_\alpha$. Therefore, for every fixed $0 < t_1 < \dots < t_k < 1$,

$$\mathbf{E}_{N,h_N} \left[G\left(\frac{\tau}{N}\right) \mid \lfloor Nt_1 \rfloor, \dots, \lfloor Nt_k \rfloor \in \tau \right] \xrightarrow{N \rightarrow +\infty} \mathbf{E}_{\hat{h}} \left[G(\mathcal{A}_\alpha) \mid t_1, \dots, t_k \in \mathcal{A}_\alpha \right].$$

Here, the conditional expectation $\mathbf{E}_{\hat{h}}[\cdot \mid t_1, \dots, t_k \in \mathcal{A}_\alpha]$ is understood as the law of the concatenation of k independent α -stable regenerative sets $(\mathcal{A}_\alpha^{(i)} \cap [0, t_i - t_{i-1}])_{1 \leq i \leq k}$ respectively conditioned to $t_i - t_{i-1} \in \mathcal{A}_\alpha^{(i)}$ for every $i \in \llbracket 1, k \rrbracket$ and one unconditioned α -stable regenerative set $\mathcal{A}_\alpha^{(k+1)} \cap [0, 1 - t_k]$.

Finally, it is proved in [FL] that the continuum partition function $\mathcal{Z}_{\hat{h}}^{\zeta, \hat{\beta}}$ is almost surely strictly positive, which concludes the proof.

7.3. Proof of Theorem 3.4. By Theorem 2.10, it is enough to show that the random variable

$$D_N := \frac{1}{Nu(N)} \sum_{n=1}^N \mathbf{1}_{n \in \tau}$$

converges in distribution as $N \rightarrow +\infty$ to a strictly positive random variable. The Laplace transform of D_N is given by

$$\mathbf{E}_{N,h_N} [e^{\lambda D_N}] = \mathbf{E}_{N,h_N} \left[\exp \left(\lambda_N \sum_{n=1}^N \mathbf{1}_{n \in \tau} \right) \right] = \frac{Z_{N,h_N+\lambda_N}}{Z_{N,h_N}},$$

with $\lambda_N := \frac{\lambda}{Nu(N)}$. By Lemma 7.1 (or [Soh09, Thm. 3.1]), we have that

$$\lim_{N \rightarrow \infty} \mathbf{E}_{N,h_N} [e^{\lambda D_N}] = \frac{\mathcal{Z}_{1,\hat{h}+\lambda}}{\mathcal{Z}_{1,\hat{h}}} = \mathbf{E}_{\alpha,\hat{h}} [e^{\lambda L_1^{(\alpha)}}],$$

where $d\mathbf{P}_{\alpha,\hat{h}} = \frac{1}{\mathcal{Z}_{1,\hat{h}}} e^{\hat{h}L_1^{(\alpha)}} d\mathbf{P}$ is the law of the continuum homogeneous pinning model, with $\mathbf{P}$ the law of an α -stable subordinator $\sigma^{(\alpha)}$ and $L_1^{(\alpha)} = \inf\{s \geq 0, \sigma_s^{(\alpha)} \geq t\}$ its local time (see Section 3.1.1 for an overview). The right-hand side is the Laplace transform of the local time under $\mathbf{P}_{\alpha,\hat{h}}$, which is a.s. strictly positive. This concludes the proof. $\square$

8. PROOFS FOR LONG-RANGE DIRECTED POLYMER

In this section, we prove Theorems 3.5, Theorem 3.6 and 3.7. Let us stress that in this case, one cannot directly apply Theorems 2.6-2.7 since $\Omega = (0, 1) \times \mathbb{R}^d$ is not bounded. We therefore introduce some truncation argument, considering the random walk only in a bounded strip.

Denote for every $N \geq 1$, $S_N^* = \max_{1 \leq n \leq N} \frac{|S_n|}{a_N}$. Let $f : \mathbb{R}_+ \rightarrow [0, 1]$ be a smooth non-increasing function such that $f(x) = 1$ if $x \leq 1$ and $f(x) = 0$ if $x > 2$, and for every $A > 0$, let $f_A(x) = f(x/A)$. Then, define the truncated partition function

$$Z_{N,A}^{\omega,\beta_N} := \mathbf{E} \left[f_A(S_N^*) \prod_{n=1}^N (1 + \beta_N \omega_{n,S_n}) \right].$$

Let us recall that $(\mathbf{X}_t)_{t \geq 0}$ is the multivariate α -stable Lévy process that arises as the scaling limit of the random walk, with density $g_t(x)$ which is continuous and bounded (see e.g. [Wat07], which also give their rate of decay). We also denote by $\mathbf{X}^* = \sup_{t \in [0,1]} |\mathbf{X}_t|$, and we introduce the formal notation

$$\mathbf{P}(\cdot \mid \mathbf{X}_{t_1} = x_1, \dots, \mathbf{X}_{t_k} = x_k)$$

for the law of the concatenation of bridges of $\mathbf{X}$ from x_{i-1} to x_i between times t_{i-1} and t_i (this can be defined as the scaling limit of a random walk $(\frac{1}{a_N} S_{\lfloor t_N \rfloor})_{t \geq 0}$ conditioned to have $S_{\lfloor t_i N \rfloor} = \lfloor x_i a_N \rfloor$ for $1 \leq i \leq k$, see [Lig70]).

For a bounded continuous function $G : \mathcal{D} \rightarrow \mathbb{R}$, define

$$Z_N^{\omega,\beta_N} := \mathbf{E} \left[G(S^{(N)}) \prod_{n=1}^N (1 + \beta_N \omega_{n,S_n}) \right].$$

In section 8.2, we explain how the proof of Theorem 3.5 can be adapted to prove the convergence in law of $Z_N^{\omega, \beta_N}(G)$.

8.1. Proof of Theorem 3.5. The core of the proof is the convergence of the truncated partition function $Z_{N,A}^{\omega, \beta_N}$.

Lemma 8.1. *Assume that $\lim_{N \rightarrow \infty} \beta_N V_N(a_N)^{-d} = \hat{\beta} \in [0, \infty)$. Then, for every $A > 0$, we have the following convergence in distribution:*

$$(8.1) \quad Z_{N,A}^{\omega, \beta_N} \xrightarrow[N \rightarrow \infty]{(d)} \mathcal{Z}_A^{\zeta, \hat{\beta}} = \sum_{k \geq 0} \hat{\beta}^k \int_{0 < t_1 < \dots < t_k < 1} \int_{(\mathbb{R}^d)^k} \psi_A^{(k)}((t_1, x_1), \dots, (t_k, x_k)) \prod_{j=1}^k \zeta(dt_j dx_j),$$

where $\psi_A^{(k)}((t_1, x_1), \dots, (t_k, x_k)) := \prod_{j=1}^k g_{t_j - t_{j-1}}(x_j - x_{j-1}) \mathbb{E}[f_A(\mathbf{X}^*) \mid \mathbf{X}_{t_1} = x_1, \dots, \mathbf{X}_{t_k} = x_k]$.

The convergence also holds for the finite-dimensional marginal distributions of the process $(Z_{N,A}^{\omega, \beta_N})_{A>0}$. Furthermore, for every $p \in [1, \gamma)$, $\lim_{N \rightarrow +\infty} \mathbb{E}[|Z_{N,A}^{\omega, \beta_N}|^p] = \mathbb{E}[|\mathcal{Z}_A^{\zeta, \hat{\beta}}|^p] < +\infty$.

Before proving this lemma, let us first see how it implies Theorem 3.5.

Proof of Theorem 3.5. Since $f_A \leq 1$, we have

$$(8.2) \quad \mathbb{E}[|Z_N^{\omega, \beta_N} - Z_{N,A}^{\omega, \beta_N}|] = \mathbb{E}[Z_N^{\omega, \beta_N}] - \mathbb{E}[Z_{N,A}^{\omega, \beta_N}] = \mathbb{E}[1 - f_A(S_N^*)] \leq \mathbb{P}[S_N^* \geq A].$$

Now, by the invariance principle (see e.g. [Res07, Thm. 7.1]), we have that S_N^* converges in distribution to $\mathbf{X}^*$ as $N \rightarrow +\infty$, so we get

$$(8.3) \quad \limsup_{N \rightarrow +\infty} \mathbb{E}[|Z_N^{\omega, \beta_N} - Z_{N,A}^{\omega, \beta_N}|] \leq \mathbb{P}(\mathbf{X}^* \geq A) \xrightarrow[A \rightarrow \infty]{} 0.$$

For $A \leq B$, notice that we have $0 \leq Z_{N,A}^{\omega, \beta_N} \leq Z_{N,B}^{\omega, \beta_N}$. Hence, Lemma 8.1 implies that $0 \leq \mathcal{Z}_A^{\zeta, \hat{\beta}} \leq \mathcal{Z}_B^{\zeta, \hat{\beta}}$ almost surely. In particular, we have that $\mathcal{Z}^{\zeta, \hat{\beta}} := \lim_{A \rightarrow \infty} \mathcal{Z}_A^{\zeta, \hat{\beta}}$ exists a.s. Let us now show that $\mathcal{Z}^{\zeta, \hat{\beta}} < +\infty$ a.s. For this, notice that since $\mathcal{Z}_A^{\zeta, \hat{\beta}} \leq \mathcal{Z}_B^{\zeta, \hat{\beta}}$ we have

$$\mathbb{E}[|\mathcal{Z}_B^{\zeta, \hat{\beta}} - \mathcal{Z}_A^{\zeta, \hat{\beta}}|] = \mathbb{E}[\mathcal{Z}_B^{\zeta, \hat{\beta}}] - \mathbb{E}[\mathcal{Z}_A^{\zeta, \hat{\beta}}] = \lim_{N \rightarrow +\infty} (\mathbb{E}[Z_{N,B}^{\omega, \beta_N}] - \mathbb{E}[Z_{N,A}^{\omega, \beta_N}]).$$

By the invariance principle ([Res07, Thm. 7.1]), S_N^* converges in distribution to $\mathbf{X}^*$, so that

$$(8.4) \quad \lim_{N \rightarrow +\infty} \mathbb{E}[Z_{N,A}^{\omega, \beta_N}] = \lim_{N \rightarrow +\infty} \mathbb{E}[f_A(S_N^*)] = \mathbb{E}[f_A(\mathbf{X}^*)],$$

and therefore

$$\mathbb{E}[|\mathcal{Z}_B^{\zeta, \hat{\beta}} - \mathcal{Z}_A^{\zeta, \hat{\beta}}|] = \mathbb{E}[f_B(\mathbf{X}^*) - f_A(\mathbf{X}^*)] \leq \mathbb{P}(\mathbf{X}^* \geq A) \xrightarrow[A \rightarrow \infty]{} 0.$$

This shows that the family of random variables $(\mathcal{Z}_A^{\zeta, \hat{\beta}})_{A \geq 1}$ is a Cauchy family in $\mathbb{L}^1$, so the limit $\mathcal{Z}^{\zeta, \hat{\beta}} := \lim_{A \rightarrow \infty} \mathcal{Z}_A^{\zeta, \hat{\beta}}$ also holds in $\mathbb{L}^1$. Therefore, Lemma 8.1, together with (8.3), concludes the proof of Theorem 3.5. $\square$

Proof of Lemma 8.1. A fundamental tool that we will need is the local limit theorem, see e.g. [Don91; Gri86], that we now recall for convenience:

$$(8.5) \quad \sup_{x \in \mathbb{Z}^d} \left| (a_n)^d \mathbb{P}(S_n = x) - g_1\left(\frac{x}{a_n}\right) \right| \xrightarrow[n \rightarrow +\infty]{} 0.$$

Since the density g_1 is continuous and bounded, there exists a constant C such that for every $n \in \mathbb{N}$ and $x \in \mathbb{Z}^d$,

$$(8.6) \quad \mathbb{P}(S_n = x) \leq \frac{C}{(a_n)^d}.$$

We are now going to apply Theorem 2.5, and to this end, we need to specify how the convergence of $Z_{N,A}^{\omega,\beta_N}$ fits into its framework. First of all, the polynomial chaos expansion of $Z_{N,A}^{\omega,\beta_N}$ is given by:

$$Z_{N,A}^{\omega,\beta_N} = \mathbb{E}[f_A(S_N^*)] + \sum_{k \geq 1} (\beta_N)^k \sum_{\substack{1 \leq n_1 < \dots < n_k \leq N \\ x_1, \dots, x_k \in \mathbb{Z}^d}} \mathbb{E}[f_A(S_N^*) \mathbf{1}_{\{S_{n_1}=x_1, \dots, S_{n_k}=x_k\}}] \prod_{j=1}^k \omega_{n_j, x_j}.$$

Set $\Omega_A = (0, 1) \times \{x \in \mathbb{R}^d, |x| < 2A\}$, which is a bounded open set of $\mathbb{R}^{1+d}$ and for $N \geq 1$, let $\Omega_{N,A} = \{(\frac{n}{N}, \frac{x}{a_N}), 1 \leq n \leq N, x \in \mathbb{Z}^d \text{ with } |x| < 2Aa_N\}$. In this section, we also work with the discretization parameter N instead of δ (with $\delta = N^{-1}$). The discretization $\Omega_{N,A}$ naturally yields a tessellation of Ω with cells of volume $v_N = N^{-1}(a_N)^{-d}$; recall that V_N is defined by the asymptotic relation 3.14. For this model, $J_N = (a_N)^d$ by the local limit theorem (8.5). Let us also set $\hat{\beta}_N := \beta_N V_N (a_N)^{-d}$ and recall that we assume that $\lim_{N \rightarrow \infty} \hat{\beta}_N = \hat{\beta} \in [0, \infty)$.

We can therefore rewrite the polynomial chaos expansion as follows:

$$Z_{N,A}^{\omega,\beta_N} = \mathbb{E}[f_A(S_N^*)] + \sum_{k \geq 1} \frac{(\hat{\beta}_N)^k}{k!} \sum_{(t_1, x_1), \dots, (t_k, x_k) \in \Omega_{N,A}} \psi_{N,A}^{(k)}((t_1, x_1), \dots, (t_k, x_k)) \prod_{j=1}^k \frac{\omega_{Nt_j, a_N x_j}}{V_N}.$$

Here, for every $k \geq 1$, $\psi_{N,A}^{(k)} : \Omega_{N,A}^k \rightarrow \mathbb{R}_+$ is the symmetric function vanishing on the diagonals defined, for every $(t_1, x_1), \dots, (t_k, x_k) \in \Omega_{N,A}$ with $t_1 < \dots < t_k$, by

$$(8.7) \quad \psi_{N,A}^{(k)}((t_1, x_1), \dots, (t_k, x_k)) := (a_N)^{kd} \mathbb{E}[f_A(S_N^*) \mathbf{1}_{\{S_{Nt_1}=a_N x_1, \dots, S_{Nt_k}=a_N x_k\}}].$$

Its piecewise extension to Ω_A is denoted $\bar{\psi}_{N,A}^{(k)}$. Now, we need to identify the scaling limits of the functions $\bar{\psi}_{N,A}^{(k)}$ and show that they satisfy the assumptions of Theorem 2.5.

Let us introduce the notation $\psi_N^{(k)} := \psi_{N,+\infty}^{(k)}$, with by convention $f_{+\infty} := 1$: we can then write

$$\psi_{N,A}^{(k)}((t_1, x_1), \dots, (t_k, x_k)) = \psi_N^{(k)}((t_1, x_1), \dots, (t_k, x_k)) \mathbb{E}[f_A(S_N^*) \mid S_{Nt_j} = a_N x_j \forall j \in \llbracket 1, k \rrbracket].$$

By the Markov property of the random walk, when $t_1 < \dots < t_k$ we get

$$\psi_N^{(k)}((t_1, x_1), \dots, (t_k, x_k)) = (a_N)^{kd} \prod_{j=1}^k \mathbb{P}(S_{N(t_j - t_{j-1})} = a_N(x_j - x_{j-1})).$$

Therefore, by the local limit theorem (8.5) and the invariance principle for conditioned random walk [Lig70], for every $k \geq 1$, the functions $\bar{\psi}_N^{(k)}$ and $\bar{\psi}_{N,A}^{(k)}$ converge pointwise as $N \rightarrow +\infty$ respectively to the symmetric functions $\psi^{(k)}$ and $\psi_A^{(k)}$ given for every $0 < t_1 < \dots < t_k < 1$ and $x_1, \dots, x_k \in \mathbb{R}^d$ by

$$(8.8) \quad \begin{aligned} \psi^{(k)}((t_1, x_1), \dots, (t_k, x_k)) &= \prod_{j=1}^k g_{t_j - t_{j-1}}(x_j - x_{j-1}), \\ \psi_A^{(k)}((t_1, x_1), \dots, (t_k, x_k)) &= \prod_{j=1}^k g_{t_j - t_{j-1}}(x_j - x_{j-1}) \mathbb{E}[f_A(\mathbf{X}^*) \mid \mathbf{X}_{t_j} = x_j \forall j \in \llbracket 1, k \rrbracket], \end{aligned}$$

where we recall that $\mathbf{P}(\cdot \mid \mathbf{X}_{t_j} = x_j \forall j \in \llbracket 1, k \rrbracket)$ is a shorthand for the concatenation of k independent bridges connecting (t_{j-1}, x_{j-1}) to (t_j, x_j) for every $j \in \llbracket 1, k \rrbracket$.

In order to prove that this convergence actually holds in L_s^q and that $(\psi_{N,A}^{(k)})$ satisfy condition (ii) of Theorem 2.5, we show the following. For every $q \in (\gamma, 1 + \frac{\alpha}{d})$, denote $\xi = \xi_q := \frac{d}{\alpha}(1 + \frac{\alpha}{d} - q) > 0$. Then there exists a constant C such that the following hold:

(i) For every $k \geq 1$,

$$(8.9) \quad \frac{1}{k!} \int_{\Omega_A^k} \psi_A^{(k)}((t_1, x_1), \dots, (t_k, x_k))^q \prod_{j=1}^k dt_j dx_j \leq \frac{C^k}{\Gamma(\xi k + 1)}.$$

(ii) For every $k \geq 1$ and $N \geq 1$,

$$(8.10) \quad \frac{1}{k!} \int_{\Omega_A^k} \bar{\psi}_{N,A}^{(k)}((t_1, x_1), \dots, (t_k, x_k))^q \prod_{j=1}^k dt_j dx_j \leq \frac{C^k}{\Gamma(\frac{1}{2}\xi k + 1)}.$$

Notice that, since $\psi_{N,A}^{(k)} \leq \psi_N^{(k)}$ and $\psi_A^{(k)} \leq \psi^{(k)}$, it is enough to work with $\psi_N^{(k)}$ and $\psi^{(k)}$. Since these bounds hold for every $q \in (\gamma, 1 + \frac{\alpha}{d})$, the dominated convergence theorem implies that the pointwise convergence of $\bar{\psi}_{N,A}^{(k)}$ to $\psi_A^{(k)}$ extends to convergence in L_s^q . Hence the condition (i) of Theorem 2.5 is also satisfied.

Let us start with (8.9). For every $k \geq 1$, we have

$$\frac{1}{k!} \int_{\Omega_A^k} \psi((t_1, x_1), \dots, (t_k, x_k))^q \prod_{j=1}^k dt_j dx_j \leq \hat{\beta}^k \int_{0 < t_1 < \dots < t_k < 1} \int_{(\mathbb{R}^d)^k} \prod_{j=1}^k g_{t_j - t_{j-1}}(x_j - x_{j-1})^q \prod_{j=1}^k dt_j dx_j.$$

Now, the scale-invariance property of the function g gives that for every $t > 0$,

$$\int_{\mathbb{R}^d} g_t(x)^q dx = c_q t^{\xi-1},$$

where $\xi = \xi_q := \frac{d}{\alpha}(1 + \frac{\alpha}{d} - q) > 0$ and $c_q = \|g\|_{L^q(\mathbb{R}^d)}^q$ is finite since $g \in L^1(\mathbb{R}^d) \cap L^\infty(\mathbb{R}^d)$. Therefore, the left-hand side of (8.9) is bounded by

$$(c_q)^k \int_{0 < t_1 < \dots < t_k < 1} \prod_{j=1}^k (t_j - t_{j-1})^{\xi-1} \prod_{j=1}^k dt_j = \frac{(c_q \Gamma(\xi))^k}{\Gamma(\xi k + 1)},$$

having used (7.5). This proves 8.9.

We now turn to (8.10). For every $k \geq 1$, the left-hand side is equal to

$$\begin{aligned} & \int_{0 < t_1 < \dots < t_k} \int_{(\mathbb{R}^d)^k} \bar{\psi}_N^{(k)}((t_1, x_1), \dots, (t_k, x_k))^q \prod_{j=1}^k dt_j dx_j \\ &= (v_N(a_N)^{qd})^k \sum_{\substack{1 \leq n_1 < \dots < n_k \leq N \\ x_1, \dots, x_k \in \mathbb{Z}^d}} \prod_{j=1}^k \mathbb{P}(S_{n_j - n_{j-1}} = x_j - x_{j-1})^q. \end{aligned}$$

Applying the uniform bound (8.6) to $\mathbb{P}(S_{n_j - n_{j-1}} = x_j - x_{j-1})^{q-1}$ and using that $v_N = N^{-1}a_N^{-d}$, we get that this is bounded by

$$C^k (a_N)^{-k(q-1)d} N^{-k} \sum_{1 \leq n_1 < \dots < n_k \leq N} \prod_{j=1}^k \frac{1}{(a_{n_j - n_{j-1}})^{(q-1)d}} \sum_{x_1, \dots, x_k \in \mathbb{Z}^d} \prod_{j=1}^k \mathbb{P}(S_{n_j - n_{j-1}} = x_j - x_{j-1}).$$

Now we can use that $\sum_{x \in \mathbb{Z}^d} \mathbb{P}(S_n = x) = 1$ for every $n \in \mathbb{N}$ to sum over $x_1, \dots, x_k$. Then, writing that $a_n = \ell(n)n^{1/\alpha}$ for some slowly-varying function $\ell(\cdot)$ and recalling that $\xi := \frac{d}{\alpha}(1 + \frac{\alpha}{d} - q)$, we have that $(a_n)^{(q-1)d} = n^{-(\xi-1)}\ell(n)^{(q-1)d}$. We then obtain that the left-hand side of (8.10) is bounded by

$$\begin{aligned} & C^k (a_N)^{k(q-1)d} N^{-k} \sum_{1 \leq n_1 < \dots < n_k \leq N} \prod_{j=1}^k \frac{(n_j - n_{j-1})^{\xi-1}}{\ell(n_j - n_{j-1})^{(q-1)d}} \\ & \leq (C')^k (a_N)^{k(q-1)d} N^{-k} \frac{N^{k\xi}}{\ell(N)^{k(q-1)d}} \int_{0 < t_1 < \dots < t_k < 1} \prod_{j=1}^k (t_j - t_{j-1})^{\frac{1}{2}\xi-1} \prod_{j=1}^k dt_j = (C')^k \frac{\Gamma(\frac{1}{2}\xi)^k}{\Gamma(\frac{1}{2}\xi k + 1)}, \end{aligned}$$

where we have used Proposition B.1 (with $\varepsilon = \xi/2$) for the first inequality and then (7.5). This yields (8.10) and concludes the proof. $\square$

8.2. Adapting the proof for a general function G . Without loss of generality, we assume that the function G is positive. As we did in the case $G = 1$, we define a truncated version of $Z_N^{\omega, \beta_N}(G)$ for every $A > 0$ by

$$Z_{N,A}^{\omega, \beta_N}(G) := \mathbb{E} \left[f_A(S_N^*) G(S^{(N)}) \prod_{n=1}^N (1 + \beta_N \omega_{n, S_n}) \right].$$

Lemma 8.1 can be adapted as follows:

Lemma 8.2. *Assume that $\lim_{N \rightarrow \infty} \beta_N V_N(a_N)^{-d} = \hat{\beta} \in [0, \infty)$. Then, for every $A > 0$, we have the following convergence in distribution:*

$$Z_{N,A}^{\omega, \beta_N}(G) \xrightarrow[N \rightarrow \infty]{(d)} \mathcal{Z}_A^{\zeta, \hat{\beta}}(G) = \sum_{k \geq 0} \hat{\beta}^k \int_{0 < t_1 < \dots < t_k < 1} \int_{(\mathbb{R}^d)^k} \psi_A^{(k)}((t_1, x_1), \dots, (t_k, x_k); G) \prod_{j=1}^k \zeta(dt_j dx_j),$$

where $\psi_A^{(k)}((t_1, x_1), \dots, (t_k, x_k); G) := \prod_{j=1}^k g_{t_j - t_{j-1}}(x_j - x_{j-1}) \mathbb{E}[f_A(\mathbf{X}^*) G(\mathbf{X}) \mid \mathbf{X}_{t_1} = x_1, \dots, \mathbf{X}_{t_k} = x_k]$.

This lemma implies the convergence of $Z_N^{\omega, \beta_N}(G)$. Indeed, since we also have for every $0 \leq A \leq B$,

$$Z_{N,A}^{\omega, \beta_N}(G) \leq Z_{N,B}^{\omega, \beta_N}(G),$$

we get

$$\mathbb{E} \left[|Z_N^{\omega, \beta_N}(G) - Z_{N,A}^{\omega, \beta_N}(G)| \right] = \mathbb{E}[G(S^{(N)})(1 - f_A(S_N^*))] \leq \|G\|_\infty \mathbb{P}[S_N^* \geq A]$$

and similarly,

$$\mathbb{E} \left[|\mathcal{Z}_B^{\zeta, \hat{\beta}}(G) - \mathcal{Z}_A^{\zeta, \hat{\beta}}(G)| \right] \leq \|G\|_\infty \mathbb{P}[\mathbf{X}^* \geq A].$$

Therefore,

$$\lim_{A \rightarrow +\infty} \limsup_{N \rightarrow +\infty} \mathbb{E} \left[|Z_N^{\omega, \beta_N}(G) - Z_{N,A}^{\omega, \beta_N}(G)| \right] = 0,$$

and the family $(\mathcal{Z}_A^{\zeta, \hat{\beta}}(G))_{A \geq 1}$ is a Cauchy family in $\mathbb{L}^1$, so the limit $\mathcal{Z}^{\zeta, \hat{\beta}}(G) := \lim_{A \rightarrow +\infty} \mathcal{Z}_A^{\zeta, \hat{\beta}}(G)$ holds in $\mathbb{L}^1$.

The polynomial chaos expansion of $Z_{N,A}^{\omega, \beta_N}(G)$ is given by

$$Z_{N,A}^{\omega, \beta_N}(G) = \mathbb{E}[f_A(S_N^*) G(S^{(N)})] + \sum_{k \geq 1} \frac{(\hat{\beta}_N)^k}{k!} \sum_{(t_1, x_1), \dots, (t_k, x_k) \in \Omega_{N,A}} \psi_{N,A}^{(k)}((t_1, x_1), \dots, (t_k, x_k); G) \prod_{j=1}^k \frac{\omega_{N t_j, a_N x_j}}{V_N},$$

where, for every $k \geq 1$, $\psi_{N,A}^{(k)}(\cdot; G) : \Omega_{N,A}^k \rightarrow \mathbb{R}_+$ is the symmetric function vanishing on the diagonals defined, for every $(t_1, x_1), \dots, (t_k, x_k) \in \Omega_{N,A}$ with $t_1 < \dots < t_k$, by

$$\psi_{N,A}^{(k)}((t_1, x_1), \dots, (t_k, x_k); G) := (a_N)^{kd} \mathbb{E} \left[f_A(S_N^*) G(S^{(N)}) \mathbf{1}_{\{S_{N t_1} = a_N x_1, \dots, S_{N t_k} = a_N x_k\}} \right].$$

By the local limit theorem (8.5) and the invariance principle for conditioned random walk [Lig70], it converges pointwisely as $N \rightarrow +\infty$ to the symmetric function $\psi_A^{(k)}(\cdot; G)$ given for every $0 < t_1 < \dots < t_k < 1$ and $x_1, \dots, x_k \in \mathbb{R}^d$ by

$$\psi_A^{(k)}((t_1, x_1), \dots, (t_k, x_k); G) = \prod_{j=1}^k g_{t_j - t_{j-1}}(x_j - x_{j-1}) \mathbb{E} \left[f_A(\mathbf{X}^*) G(\mathbf{X}) \mid X_{t_1} = x_1, \dots, X_{t_k} = x_k \right].$$

Furthermore, for every $N \geq 1$,

$$\psi_{N,A}^{(k)}((t_1, x_1), \dots, (t_k, x_k); G) \leq \|G\|_\infty \psi_{N,A}^{(k)}((t_1, x_1), \dots, (t_k, x_k)),$$

and therefore, the integrability conditions proved for $G = 1$ in the proof of Lemma 8.1 apply to a general bounded function G .

8.3. Proof of Theorem 3.6. We apply Lemma 6.1 to prove Theorem 3.6.

Assumption (a) follows from the invariance principle, while assumption (b) is established in Section 8.2. It therefore remains to show that $\mathcal{Z}^{\zeta, \hat{\beta}}$ is a.s. strictly positive. This result follows by a straightforward adaptation of the proof in the case $\alpha = 2$ presented in [BL22, Section 4.7], so we have decided not to include it here.

8.4. Proof of Theorem 3.7. Let us stress that item (i) is a direct consequence of Theorem 3.5: if $\lim_{N \rightarrow +\infty} \beta_N V_N(a_N)^{-d} = \hat{\beta} = 0$ we have that Z_N^{ω, β_N} converges in distribution to $\mathcal{Z}^{\omega, \beta=0} = 1$.

For the proof of item (ii), this is also almost a consequence of Theorem 2.10, after a truncation argument.

Lemma 8.3. *Suppose $\lim_{N \rightarrow +\infty} \beta_N V_N(a_N)^{-d} = +\infty$, then for every fixed $A > 0$, $Z_{N,A}^{\omega, \beta_N}$ goes to 0 in probability.*

Proof. For a fixed $A > 0$, since $f_A(x) = 0$ for $x > 2$, we have

$$Z_{N,A}^{\omega, \beta_N} \leq \mathbb{E} \left[\mathbb{1}_{\forall 1 \leq n \leq N, |S_n| \leq 2Aa_N} \prod_{n=1}^N (1 + \beta_N \omega_{n, S_n}) \right] \leq \tilde{Z}_{N,A} := \mathbb{E}_{N,A} \left[\prod_{n=1}^N (1 + \beta_N \omega_{n, S_n}) \right],$$

where $\mathbb{P}_{N,A}(\cdot) = \mathbb{P}[\cdot \mid \forall 1 \leq n \leq N, |S_n| \leq 2Aa_N]$. In order to apply Theorem 2.10 to $\tilde{Z}_{N,A}$, we have to show that $(a_N)^d N^{-1} (a_N)^{-d} \sum_{n=1}^N \mathbb{1}_{|S_n| \leq 2Aa_N}$ converges in distribution under $\mathbb{P}$ to a strictly positive random variable (recall that for the long-range polymer, $J_N = (a_N)^d$ and $v_N = N^{-1} (a_N)^{-d}$). But this is a consequence of the invariance principle for long-range random walks (see e.g. [Res07, Thm. 7.1.]) since

$$\frac{1}{N} \sum_{n=1}^N \mathbb{1}_{|S_n| \leq 2Aa_N} \xrightarrow{(d)} \int_0^1 \mathbb{1}_{|\mathbf{X}_t| \leq 2A} dt,$$

which is almost surely strictly positive by stochastic continuity of the process $(\mathbf{X}_t)_{t \in [0,1]}$. $\square$

Now, we deduce from this lemma the proof of item (ii). Fix any $\varepsilon > 0$ and $\delta > 0$. For every $N \geq 1$ and $A > 0$, by (8.2), we have

$$\mathbb{E}[|Z_N^{\omega, \beta_N} - Z_{N,A}^{\omega, \beta_N}|] \leq \mathbb{P}[S_N^* \geq A].$$

Choose $A > 0$ large enough so that

$$(8.11) \quad \sup_{N \geq 1} \mathbb{E}[|Z_N^{\omega, \beta_N} - Z_{N,A}^{\omega, \beta_N}|] \leq \delta.$$

Then, for every $N \geq 1$,

$$\mathbb{P}[Z_N^{\omega, \beta_N} > 2\varepsilon] = \mathbb{P}[(Z_N^{\omega, \beta_N} - Z_{N,A}^{\omega, \beta_N}) + Z_{N,A}^{\omega, \beta_N} > 2\varepsilon] \leq \mathbb{P}[Z_N^{\omega, \beta_N} > \varepsilon] + \mathbb{P}[Z_N^{\omega, \beta_N} - Z_{N,A}^{\omega, \beta_N} > \varepsilon].$$

Therefore, by Markov's inequality and (8.11),

$$\mathbb{P}[Z_N^{\omega, \beta_N} > 2\varepsilon] \leq \mathbb{P}[Z_{N,A}^{\omega, \beta_N} > \varepsilon] + \delta \varepsilon^{-1},$$

which yields, applying the lemma, $\limsup_{N \rightarrow +\infty} \mathbb{P}[Z_N^{\omega, \beta_N} > 2\varepsilon] \leq \delta \varepsilon^{-1}$. Since δ is arbitrary, this concludes the proof.

APPENDIX A. LÉVY NOISES AND POISSON CONVERGENCE

After having extensively dissected, partitioned and truncated our polynomial chaos, it becomes necessary, at some point, to establish a result of convergence in distribution. This is the purpose of this appendix.

Let us recall our set up. The set Ω is bounded and open in $\mathbb{R}^D$, and for every $\delta \in (0, 1)$, Ω_δ is a finite subset of Ω associated to a tessellation $\mathcal{C}_\delta = \{\mathcal{C}_\delta(x)\}_{x \in \Omega_\delta}$ of Ω (see Section 2.2.1 for a definition). We are given for every $\delta \in (0, 1)$, i.i.d. and centered random variables $(\omega_x)_{x \in \Omega_\delta}$ such that for every $t > 0$

$$\mathbb{P}[|\omega_x| > t] = \varphi(t)t^{-\gamma},$$

where $\gamma \in (1, 2)$ and φ is a slowly-varying function. Furthermore, there are constants $c_+, c_- \in [0, 1]$ with $c_+ + c_- = 1$ such that $c_+ = \lim_{t \rightarrow +\infty} \frac{\mathbb{P}[\omega_x > t]}{\mathbb{P}[\omega_x > t]}$. The discrete noise ζ_δ is then defined by

$$\zeta_\delta := \frac{1}{V_\delta} \sum_{x \in \Omega_\delta} \omega_x \delta_x.$$

Let Λ be a Poisson point process on $\Omega \times \mathbb{R}$ of intensity $\mu(dx dz) = dx \lambda(dz)$ with $\lambda(dz) = \gamma(c_+ \mathbb{1}_{z>0} + c_- \mathbb{1}_{z<0})|z|^{-1-\gamma} dz$. Our first step is to define the γ -stable noise ζ . For every $a \in (0, 1]$, let $\Lambda^{(a)}$ denote the point process Λ restricted to $\{(x, z) \in \Omega \times \mathbb{R}, |z| > a\}$. This subset of $\Omega \times \mathbb{R}$ has a finite measure under μ , therefore the point process $\Lambda^{(a)}$ is almost surely finite. This allows to define

$$\zeta^{(a)} = \sum_{(x, z) \in \Lambda} z \mathbb{1}_{|z|>a} \delta_x - \kappa(a) dx,$$

with $\kappa(a) = \int_{\mathbb{R}} z \mathbb{1}_{|z|>a} \lambda(dz)$. The truncated noises $(\zeta^{(a)})_{a \in (0, 1]}$ converge almost surely as $a \downarrow 0$ in some negative Sobolev spaces. In order to formulate our result, let us briefly introduce these spaces.

The space of infinitely differentiable functions compactly supported in Ω is denoted by $\mathcal{D}_0(\Omega)$. Its dual space is the space of distributions $\mathcal{D}'(\Omega)$, and the action of a distribution $T \in \mathcal{D}'(\Omega)$ on an element $\mathbf{f} \in \mathcal{D}_0(\Omega)$ is denoted by $\langle T, \mathbf{f} \rangle$. Let $s \in \mathbb{R}$, $H^s(\mathbb{R}^D)$ is the completion of $\mathcal{D}_0(\Omega)$ for the Hilbertian norm

$$\|\theta\|_{H^s} = \left(\int_{\mathbb{R}^D} (1 + |z|^2)^s |\hat{\theta}(z)|^2 dz \right)^{1/2},$$

where $\hat{\theta}$ is the Fourier transform of θ . It is a linear subspace of $\mathcal{D}'(\mathbb{R}^D)$. The local version of this space is

$$H_{\text{loc}}^s(\Omega) = \{\theta \in \mathcal{D}'(\Omega); \forall \psi \in \mathcal{D}_0(\Omega), \psi \theta \in H^s(\mathbb{R}^D)\}.$$

It is a separable Fréchet space.

Theorem A.1. $\zeta^{(a)}$ converges almost surely in $H_{\text{loc}}^{-s}(\Omega)$ for every $s > \frac{D}{2}$. Its limit is denoted ζ and its law is characterized by

$$\mathbb{E}\left[e^{i\langle \zeta, \mathbf{f} \rangle}\right] = \exp\left(\int_{\Omega \times \mathbb{R}} \left(e^{i\mathbf{f}(x)z} - 1 - i\mathbf{f}(x)z\right) dx \lambda(dz)\right),$$

for every function $\mathbf{f} \in \mathcal{D}_0(\Omega)$.

The proof consists of showing that, for every function $\psi \in \mathcal{D}_0(\Omega)$, almost surely, the family $(\psi \zeta^{(a)})_{a>0}$ is Cauchy in H^s as $a \rightarrow 0$. Such a proof can be found in [BL22, appendix A]. Although the setting of their paper is slightly different - the authors consider more general measures λ , but supported only on $\mathbb{R}_+$ - their arguments extend to our framework without difficulty. The set of real numbers s such that $s > \frac{D}{2}$ coincides exactly with the set of s for which

$$\int_{\mathbb{R}^D} (1 + |z|^2)^{-s} dz < +\infty.$$

This illustrates the fact that Lévy noises have the same Sobolev regularity as the Dirac distribution.

Theorem A.2. Let $s > \frac{D}{2}$, then $(\zeta_\delta)_{\delta>0}$ converges in law as $\delta \rightarrow 0$ to ζ for the topology of $H_{\text{loc}}^{-s}(\Omega)$.

A proof of this result can be found in [BL21] (Theorem 2.4). Again the setting is slightly different but there is no difficulty to adapt their proof. One needs to show that the family of random variables $(\zeta_\delta)_{\delta>0}$ is tight in $H_{\text{loc}}^s(\Omega)$ (lemma 3.2), and then that for every $\mathbf{f} \in \mathcal{D}_0(\Omega)$, $\langle \zeta_\delta, \mathbf{f} \rangle$ converges in law as $\delta \rightarrow 0$ to $\langle \zeta, \mathbf{f} \rangle$ which is a consequence of the convergence of binomials to Poisson laws.

We now derive a useful consequence of this convergence in distribution. Let $k \geq 1$ and $a \in (0, 1]$. For every symmetric function $\mathbf{f} \in \mathcal{D}_0(\Omega^k)$, define

$$X_\delta^{(a)}(\mathbf{f}) = \frac{1}{k!} \sum_{x_1, \dots, x_k \in \Omega_\delta} \mathbf{f}(x_1, \dots, x_k) \prod_{j=1}^k \frac{\omega_{x_j}^{(a)}}{V_\delta},$$

where we recall (see (5.4)) that $\omega_x^{(a)} = \omega_x \mathbb{1}_{|\omega_x|>aV_\delta} - \mathbb{E}[\omega \mathbb{1}_{|\omega|>aV_\delta}]$.

Proposition A.3. *Let $\mathbf{f}$ be symmetric and infinitely differentiable function with compact support in $\Omega^{k,*} = \{(x_1, \dots, x_k) \in \Omega^k, \forall i \neq j, x_i \neq x_j\}$. Then for every $s > \frac{D}{2}$ the following joint convergence in distribution holds in $H_{\text{loc}}^{-s}(\Omega) \times \mathbb{R}$ as $\delta \downarrow 0$:*

$$(\zeta_\delta, X_\delta^{(a)}(\mathbf{f})) \xrightarrow{(d)} (\zeta, \mathbf{X}^{(a)}(\mathbf{f})),$$

where $\mathbf{X}^{(a)}(\mathbf{f}) = \frac{1}{k!} \int_{\Omega^k} \mathbf{f}(x_1, \dots, x_k) \prod_{j=1}^k \zeta^{(a)}(dx_j)$.

Proof. Define for $a \in (0, 1]$ and $\delta \in (0, 1)$,

$$\zeta_\delta^{(a)} := \frac{1}{V_\delta} \sum_{x \in \Omega_\delta} \omega_x^{(a)} \delta_x = \frac{1}{V_\delta} \sum_{x \in \Omega_\delta} \omega_x \mathbf{1}_{|\omega_x| > a V_\delta} \delta_x - \frac{1}{V_\delta} \mathbb{E}[\omega \mathbf{1}_{|\omega| > a V_\delta}] \sum_{x \in \Omega_\delta} \delta_x.$$

Using the fact that $|\Omega| = v_\delta \# \Omega_\delta$ and the definition of V_δ ((2.12)), the second term in the above expression (which is deterministic) converges (in any negative Sobolev space) to $\kappa(a) dx$ as $\delta \downarrow 0$. Additionally, the proof of Theorem A.2 adapts to the truncated random variables $\omega^{(a)}$, therefore for every fixed $a \in (0, 1]$, $\zeta_\delta^{(a)}$ converges in distribution in $H_{\text{loc}}^{-s}(\Omega)$ to $\zeta^{(a)}$. Now observe that

$$X_\delta^{(a)}(\mathbf{f}) = \frac{1}{k!} \int_{\Omega^k} \mathbf{f}(x_1, \dots, x_k) \prod_{j=1}^k \zeta_\delta^{(a)}(dx_j).$$

Therefore, the joint convergence of $(\zeta_\delta, X_\delta^{(a)}(\mathbf{f}))$ to $(\zeta, \mathbf{X}^{(a)}(\mathbf{f}))$ as $\delta \rightarrow 0$ follows from the continuity of the map on $H_{\text{loc}}^{-s}(\Omega)$ defined by

$$\theta \mapsto \int_{\Omega^k} \mathbf{f}(x_1, \dots, x_k) \prod_{j=1}^k \theta(dx_j).$$

In fact, we prove the slightly stronger statement: the multilinear map

$$T : \begin{cases} H_{\text{loc}}^{-s}(\Omega)^k \times H^{ks}(\mathbb{R}^{kd}) & \longrightarrow \mathbb{R} \\ (\theta_1, \dots, \theta_k; \mathbf{f}) & \longmapsto \int_{\Omega^k} \mathbf{f}(x_1, \dots, x_k) \prod_{j=1}^k \theta_j(dx_j), \end{cases}$$

is continuous. Let us first see why it is well-defined. Given $x_2, \dots, x_k \in \Omega$, the function $\mathbf{f}(\cdot, x_2, \dots, x_k)$ is an element of $\mathcal{D}_0(\Omega)$, therefore $\langle \theta_1, \mathbf{f}(\cdot, x_2, \dots, x_k) \rangle$ is well-defined and the map $(x_2, \dots, x_k) \mapsto \langle \theta_1, \mathbf{f}(\cdot, x_2, \dots, x_k) \rangle$ is in $\mathcal{D}_0(\Omega^{k-1})$. Therefore one can apply θ_2 with respect to the second variable x_2 as we did for the first variable. Following this procedure iteratively gives a definition of the integral of $\mathbf{f}$ with respect to $\theta_1, \dots, \theta_k$. This definition is not ambiguous since $\mathbf{f}$ is symmetric.

Assume that $\zeta_1, \dots, \zeta_k$ are in $H^{-s}(\mathbb{R}^D)$. Then by Plancherel's identity and the definition of the Sobolev norm,

$$|T(\theta_1, \dots, \theta_k; \mathbf{f})| \leq \prod_{j=1}^k \|\theta_j\|_{H^{-s}(\mathbb{R}^D)} \|\mathbf{f}\|_{H^{ks}(\mathbb{R}^{kD})}.$$

For the general case, let $K_1, \dots, K_k$ be compact sets of Ω such that the support of $\mathbf{f}$ is included in $\prod_{j=1}^k K_j$.

Take for every $1 \leq j \leq k$ a smooth and compactly supported function χ_j such that $\chi_j = 1$ on K_j . Define

$\chi(x_1, \dots, x_k) = \prod_{j=1}^k \chi_j(x_j)$. Then

$$T(\theta_1, \dots, \theta_k; \mathbf{f}) = T(\theta_1, \dots, \theta_k; \chi \mathbf{f}) = T(\chi_1 \theta_1, \dots, \chi_k \theta_k; \mathbf{f}) \leq \prod_{j=1}^k \|\chi_j \theta_j\|_{H^{-s}(\mathbb{R}^D)} \|\mathbf{f}\|_{H^{ks}(\mathbb{R}^{kD})},$$

which concludes the proof. $\square$

APPENDIX B. ABOUT HOMOGENEOUS PARTITION FUNCTIONS

This Appendix is devoted to the proof of Lemma 7.1. First, we need the following uniform bound, that we also use in Section 8.

Proposition B.1. *Let L be a strictly positive slowly-varying function and let $\xi > 0$. For every $\varepsilon > 0$, there exist a constant $C = C_\varepsilon$ such that for every $k, N \in \mathbb{N}^*$,*

$$\begin{aligned} \sum_{1 \leq n_1 < \dots < n_k \leq N} \prod_{j=1}^k L(n_j - n_{j-1}) (n_j - n_{j-1})^{\xi-1} \\ \leq C^k L(N)^k N^{k\xi} \int_{0 < t_1 < \dots < t_k < 1} \prod_{j=1}^k (t_j - t_{j-1})^{\xi-\varepsilon-1} dt_j, \end{aligned}$$

and

$$\begin{aligned} \sum_{1 \leq n_1 < \dots < n_k < n_{k+1} := N} \prod_{j=1}^{k+1} L(n_j - n_{j-1}) (n_j - n_{j-1})^{\xi-1} \\ \leq C^{k+1} L(N)^{k+1} N^{k\xi+\xi-1} \int_{0 < t_1 < \dots < t_k < t_{k+1} = 1} \prod_{j=1}^{k+1} L(t_j - t_{j-1}) (t_j - t_{j-1})^{\xi-\varepsilon-1} \prod_{j=1}^k dt_j. \end{aligned}$$

Proof. By standard comparison arguments, there exist a constant C such that for every $k, N \in \mathbb{N}^*$,

$$\begin{aligned} \sum_{1 \leq n_1 < \dots < n_k \leq N} \prod_{j=1}^k L(n_j - n_{j-1}) (n_j - n_{j-1})^{\xi-1} \\ \leq C^k \int_{0 < t_1 < \dots < t_k < N} \prod_{j=1}^k L(t_j - t_{j-1}) (t_j - t_{j-1})^{\xi-1} dt_j. \end{aligned}$$

With a change of variable $t_j = N s_j$, we get that the right-hand side is equal to

$$C^k N^{k\xi} L(N)^k \int_{0 < s_1 < \dots < s_k < 1} \prod_{j=1}^k \frac{L(N(s_j - s_{j-1}))}{L(N)} (s_j - s_{j-1})^{\xi-1} ds_j.$$

By Potter's bound (see [BGT87, Thm 1.5.6.]), for every $\varepsilon > 0$, there exist a constant $C = C(\varepsilon)$ such that for every $s \leq t$, $L(s) \leq L(t) \left(\frac{t}{s}\right)^\varepsilon$. This concludes the proof of the first inequality, and the proof of the second one follows the same lines. $\square$

Proof of Lemma 7.1. Introduce $\lambda = e^h - 1$. We can rewrite the conditioned partition function in terms of λ as follows:

$$Z_{M,h}^c = e^h \mathbb{E} \left[\prod_{n=1}^{M-1} (1 + \lambda \mathbb{1}_{n \in \tau}) \mathbb{1}_{M \in \tau} \right].$$

Expanding the product $\prod_{n=1}^N (1 + \lambda \mathbb{1}_{n \in \tau})$ and taking expectation yields

$$Z_{M,h}^c = e^h \sum_{k \geq 0} \lambda^k \sum_{n_0 := 0 < n_1 < \dots < n_k < n_{k+1} := M} \prod_{j=1}^{k+1} u(n_j - n_{j-1}).$$

First, let us prove the uniform bound (7.3). By (7.1), for every $k \geq 1$,

$$\sum_{n_0 := 0 < n_1 < \dots < n_k < n_{k+1} := M} \prod_{j=1}^{k+1} u(n_j - n_{j-1}) \leq C^{k+1} \sum_{n_0 := 0 < n_1 < \dots < n_k < n_{k+1} := M} \prod_{j=1}^{k+1} \frac{(n_j - n_{j-1})^{\alpha-1}}{\ell(n_j - n_{j-1})}.$$

Applying Proposition B.1 and (7.5), we obtain that the left-hand side is bounded by

$$C^{k+1} \frac{M^{k\alpha+\alpha-1}}{\ell(M)^{k+1}} \frac{\Gamma(\alpha)^{k+1}}{\Gamma((k+1)\alpha)}.$$

Since $\lambda = e^h - 1 \sim h$ as $h \rightarrow 0$, and $h_N \sim \hat{h} N^{-1} u(N)^{-1}$ as $N \rightarrow +\infty$, by (7.1), there exist a constant $C = C_{\hat{h}}$ such that

$$\lambda_N \leq C \frac{\ell(N)}{N^\alpha}.$$

Therefore,

$$Z_{M, h_N}^c \leq C \frac{M^{\alpha-1}}{\ell(M)} \sum_{k \geq 0} \left(C \frac{M^\alpha \ell(N)}{N^\alpha \ell(M)} \right)^k \frac{1}{\Gamma((k+1)\alpha)}.$$

By [BGT87, Thm 1.5.4.],

$$\sup_{N \geq 1} \sup_{1 \leq M \leq N} \frac{M^\alpha \ell(N)}{N^\alpha \ell(M)} < +\infty.$$

Since the sequence $(\Gamma((k+1)\alpha))_{k \geq 0}$ decreases faster than any exponential rate, this concludes the proof of (7.3).

Suppose now that $M_N/N \rightarrow t \in (0, 1]$. For every $\varepsilon > 0$ and $k \in \mathbb{N}$, we split the indexes of summations into

$$\mathcal{D}_1^{\varepsilon, N}(k) := \{n_0 := 0 < n_1 < \dots < n_k < n_{k+1} := M_N; \forall j \in \llbracket 1, k+1 \rrbracket, n_j - n_{j-1} \geq \varepsilon N\},$$

and

$$\mathcal{D}_2^{\varepsilon, N}(k) := \{n_0 := 0 < n_1 < \dots < n_k < n_{k+1} := M_N; \exists j \in \llbracket 1, k+1 \rrbracket, n_j - n_{j-1} < \varepsilon N\},$$

which gives a decomposition of Z_{M_N, h_N}^c into two pieces:

$$Z_{N, \varepsilon}^{c, 1} = e^{h_N} \sum_{k \geq 0} (\lambda_N)^k \sum_{n_1, \dots, n_k \in \mathcal{D}_1^{\varepsilon, N}(k)} \prod_{j=1}^{k+1} u(n_j - n_{j-1})$$

and

$$Z_{N, \varepsilon}^{c, 2} = e^{h_N} \sum_{k \geq 0} (\lambda_N)^k \sum_{n_1, \dots, n_k \in \mathcal{D}_2^{\varepsilon, N}(k)} \prod_{j=1}^{k+1} u(n_j - n_{j-1}).$$

Rewriting $Z_{N, \varepsilon}^{c, 1}$ with the function U_N (introduced in (7.2)), we get

$$Z_{N, \varepsilon}^{c, 1} = e^{h_N} u(N) \sum_{k \geq 0} (\lambda_N u(N) N)^k \int_{\substack{0 < t_1 < \dots < t_k < \frac{M_N}{N} \\ \forall j \in \llbracket 1, k+1 \rrbracket, t_j - t_{j-1} \geq \varepsilon}} \prod_{j=1}^{k+1} U_N\left(\frac{\lfloor N t_j \rfloor - \lfloor N t_{j-1} \rfloor}{N}\right) \prod_{j=1}^k dt_j.$$

Since the function U_N converges uniformly on $(0, 1]$ to the function $U(t) = t^{\alpha-1}$, we get that for every fixed $\varepsilon > 0$,

$$\frac{Z_{N, \varepsilon}^{c, 1}}{u(N)} \xrightarrow{N \rightarrow +\infty} \sum_{k \geq 0} \hat{h}^k \int_{\substack{0 < t_1 < \dots < t_k < t \\ \forall j \in \llbracket 1, k+1 \rrbracket, t_j - t_{j-1} \geq \varepsilon}} \prod_{j=1}^{k+1} (t_j - t_{j-1})^{\alpha-1} \prod_{j=1}^k dt_j,$$

and the right hand side converges as $\varepsilon \rightarrow 0$ to $Z_{t, \hat{h}}^c$ (recall 7.5). It only remains to show that $Z_{N, \varepsilon}^{c, 2}$ is asymptotically negligible as $\varepsilon \downarrow 0$. Using again the bound (7.1) and the fact that $M_N \leq N$, we get for every $k \geq 0$,

$$\sum_{n_1, \dots, n_k \in \mathcal{D}_2^{\varepsilon, N}(k)} \prod_{j=1}^{k+1} u(n_j - n_{j-1}) \leq C^{k+1} \sum_{\substack{1 \leq n_1 < \dots < n_k < N \\ \exists j \in \llbracket 1, k+1 \rrbracket, n_j - n_{j-1} \leq \varepsilon N}} \prod_{j=1}^k \frac{(n_j - n_{j-1})^{\alpha-1}}{\ell(n_j - n_{j-1})},$$

which is bounded (by Proposition B.1) by

$$C^{k+1} \frac{N^{k\alpha+\alpha-1}}{\ell(N)^{k+1}} \int_{\substack{0 < t_1 < \dots < t_k < 1 \\ \exists j \in \llbracket 1, k+1 \rrbracket, t_j - t_{j-1} < \varepsilon}} \prod_{j=1}^{k+1} (t_j - t_{j-1})^{\alpha-1} \prod_{j=1}^k dt_j.$$

Therefore,

$$\limsup_{N \rightarrow +\infty} \frac{Z_{N,\varepsilon}^{c,2}}{u(N)} \leq C \sum_{k \geq 0} C^k \int_{\substack{0 < t_1 < \dots < t_k < 1 \\ \exists j \in \llbracket 1, k+1 \rrbracket, t_j - t_{j-1} < \varepsilon}} \prod_{j=1}^{k+1} (t_j - t_{j-1})^{\alpha-1} \prod_{j=1}^k dt_j,$$

which goes to 0 as $\varepsilon \downarrow 0$. □

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